

ChainCert: A Permissioned Blockchain-Enabled Credential Verification and Management Platform

De La Salle University - Manila
Term 3, A.Y. 2024-2025

In Partial Fulfillment of the Course Requirements for
CAPIT0A

Submitted by CAP-2459-IT:
Kalinisan, Carl Adrian
Ramirez, Myles Clement
Tan, Kyle Justin
Tumulak, Ralph Lauren

Capstone Adviser:
Danny Cheng

Submitted to:
Bertumen, Estefanie
Tabora, Liandro

Submitted on:

August 5, 2025

Table of Contents

Chapter 1 ..... **10**

1.1 Background of the Study 10

1.2 Statement of the Problem 13

1.2.1 Specific Problem #1 - Reliance on Static Credential Formats Without Embedded Verifiability 13

1.2.1.1 5 Whys Analysis 13

1.2.1.2 Root Cause Evaluation 16

1.2.1.3 Opportunity 16

1.2.2 Specific Problem #2 - Lack of Credential Lifecycle Management (Issuance, Expiry, Revocation, and Updates) 16

1.2.2.1 5 Whys Analysis 16

1.2.2.2 Root Cause Evaluation 18

1.2.2.3 Opportunity 18

1.2.3 Specific Problem #3 - Absence of Standardized Credential Metadata and Format Structures 19

1.2.3.1 5 Whys Analysis 19

1.2.3.2 Root Cause Evaluation 20

1.2.3.3 Opportunity 20

1.2.4 Specific Problem #4 - Exclusion of Microcredentials and Informal Learning Proofs 21

1.2.4.1 5 Whys Analysis 21

1.2.4.2 Root Cause Evaluation 22

1.2.4.3 Opportunity 22

1.2.5 Specific Problem #5 - Decentralized and Paper-Based Storage Results in Credential Loss and Inaccessibility 22

1.2.5.1 5 Whys Analysis 23

1.2.5.2 Root Cause Evaluation 24

1.2.5.3 Opportunity 24

1.2.6 Specific Problem #6 - Manual Verification Delays Due to Lack of Credential Interoperability....25
1.2.6.1 5 Whys Analysis....25
1.2.6.2 Root Cause Evaluation....26
1.2.6.3 Opportunity....26
1.2.7 Specific Problem #7 - No Tamper-Evident Logs or Version History....26
1.2.7.1 5 Whys Analysis....27
1.2.7.2 Root Cause Evaluation....28
1.2.7.3 Opportunity....28
1.2.8 Specific Problem #8 - No Unified Registry or Searchable Credential Index....28
1.2.8.1 5 Why Analysis....28
1.2.8.2 Root Cause Evaluation....29
1.2.8.3 Opportunity....29
1.2.9 Specific Problem #9 - Credentials Are Not Interoperable Across Systems....29
1.2.9.1 5 Whys Analysis....29
1.2.9.2 Root Cause Evaluation....30
1.2.9.3 Opportunity....30
1.3 Solutions Framework (diagram and narrative)....31
1.3.1 Target Users....32
1.3.2 Problems....32
1.3.3 Inputs....32
1.3.4 Opportunities....32
1.3.5 Features....33
1.3.6 Modules and Functions....33
1.3.6.1 Credential Issuance Module....33
1.3.6.2 Credential Verification Module....34
1.3.6.3 Blockchain Integration Module....35
1.3.6.4 Credential Access and Notification Module....36

1.3.6.5 Certifying Body Sync Module....37
1.3.6.6 Employer/Regulator Interface Module....38
1.3.6.7 Lifelong Learning & Micro-Credentials Module....39
1.3.6.8 Smart Contract Automation Module....40
1.3.7 Technologies to be Integrated....41
1.3.8 Tools and Platforms to Be Used for Development....42
1.3.9 Integration with Existing Systems....42
1.3.10 Justification of Solution....43
1.4 Design Architecture....44
1.4.1 System Components....45
1.4.2 User Workflow and Process Flow....46
1.4.2.1 Credential Issuance Workflow....46
1.4.2.2 Credential Management by Credential Holders....48
1.4.2.3 Credential Verification Workflow....48
1.5 Study Objectives....51
1.5.1 General Objectives....51
1.5.2 Specific Objectives....51
1.5.2.1 Development of a Hyperledger Fabric-Based Credential System....51
1.5.2.2 Integration of Issuer Web Dashboard and Verifier Portal....52
1.5.2.3 Implementation of Credential Lifecycle Features....52
1.5.2.4 Privacy and Access Control Management....52
1.5.2.5 Wallet Creation and Management for Credential Ownership....52
1.5.2.6 System Testing and Performance Optimization....53
1.5.2.7 User Feedback Collection and Continuous Improvement....53
1.6 Significance of the Study....53
1.6.1 Theoretical Contributions....54
1.6.2 Practical Applications....54
1.6.3 Beneficiaries....55

1.7 Scope and Limitations..... 56
1.7.1 Users.....56
1.7.2 Features and Functions.....57
1.7.3 Technology..... 58
1.7.4 Credentials..... 58
1.7.4.1 Non-Expiring Credentials.....58
1.7.4.2 Expiring Credentials..... 59
Chapter 2..... **59**
2.1 Review of Related Concepts.....59
2.1.1 Prevalence, Causes, and Impact of Fake Academic Credentials..... 59
2.1.2 Legal Implications and Responsibilities in Credential Verification.....61
2.1.2.1 Labor Code of the Philippines | Republic Act No. 6715..... 62
2.1.2.2 PRC Modernization Act | Republic Act No. 8981..... 62
2.1.2.3 Revised Penal Code | Republic Act No. 3815..... 62
2.1.3 Existing Anti-Fraud Measures of Educational Institutions and its Challenges..... 63
2.1.4 Blockchain Technology and Its Role in Academic Verification..... 64
2.1.5 Barriers to Adopting Modern Technology for Academic Credential Verification and Anti-Fraud Measures.....65
2.1.6 Blockchain Technology vs Other Modern Technology Solutions..... 66
2.1.6.1 Blockchain vs. Centralized Databases.....66
2.1.6.2 Blockchain vs. Digital Signatures..... 67
2.1.6.3 Blockchain vs. Public Key Infrastructure (PKI).....68
2.1.6.4 Blockchain vs. Traditional Paper-Based Certificates..... 68
2.1.7 Digital Signatures and Public Key Infrastructure..... 70
2.1.8 Verifiable Credentials.....70
2.1.9 Zero-Knowledge Proofs (ZKPs)..... 70
2.1.10 Digital Wallets and Self-Sovereign Identity..... 71
2.1.11 Smart Contracts..... 71

2.1.12 Decentralized Identifiers (DIDs)....72
2.1.12.1 Architecture....72
2.1.12.2 Cryptography and Security Principles....73
2.1.12.3 Types and Implementation....73
2.1.13 Digital Credentials....74
2.1.14 Permissioned Blockchain, its Advantages, and Compatibility with Credential Management. 75
2.2 Review of Related Systems....77
2.2.1 Non-Blockchain Systems....77
2.2.1.1 Professional Regulation Commission (PRC)....77
2.2.1.2 DMW, POEA, POLO, and OWWA: Interconnected Agencies in OFW Deployment and Welfare....79
2.2.1.3 Corporate Sector....81
2.2.1.4 Government Sector (Public)....82
2.2.1.5 Credential Issuance Processes in Philippine Government Agencies....83
2.2.1.5.1 Department of Migrant Workers (DMW) / POEA: Overseas Employment Certificate (EOC)....84
2.2.1.5.2 Overseas Workers Welfare Administration (OWWA): Membership and OFW eCard....84
2.2.1.5.3 Technical Education and Skills Development Authority (TESDA): National Certificates (NC) and Digital Credentials....85
2.2.2 Blockchain Systems....85
2.2.2.1 Exonum: A Blockchain Framework for Secure and Transparent Systems....86
2.2.2.2 Akowe's Blockchain Solution for Academic Credential Verification in Africa....86
2.2.2.3 BCdiploma: A Blockchain-Enabled Solution for Secure Digital Credentialing....87
2.2.2.4 European Blockchain Services Infrastructure (EBSI) for Academic Credential Verification....87
2.2.2.5 NUS Green Bond Reporting Tokenisation Initiative....88
2.2.2.6 OpenCerts: Singapore's Blockchain-Powered Credential Verification System....89

2.2.2.7 TruScholar: Blockchain-Based Digital Credentialing Platform in India.....90
2.2.2.8 DiplomaSafe: A Digital Credentialing Solution for the Future.....91
2.2.2.9 Verifi-Chain: A Credentials Verifier using Blockchain and IPFS.....93
2.2.2.10 Cerberus: A Blockchain-Based Accreditation and Degree Verification System.....94
2.2.2.11 CredSec: A Blockchain-based Secure Credential Management System for University Adoption.....95
2.2.2.12 Blockchain for Academic Integrity: Developing the Blockchain Academic Credential Interoperability Protocol (BACIP).....96
2.2.2.13 TESDA's Blockchain-Based Credential Verification System Using Polygon.....97
2.2.2.14 Blockchain in Higher Education–Partido State University & University of the Cordilleras.....98
2.2.2.15 Hyperledger Fabric: Permissioned Blockchain Framework.....99
2.2.2.16 Credly by Pearson: Enterprise-Grade Digital Credentialing.....100
2.3 Review of Related Methodologies.....102
2.3.1 Agile Methodology.....102
2.3.1.1 Scrum.....102
2.3.1.2 Kanban.....103
2.3.2 Lean Software Development.....104
2.3.3 Waterfall Methodology.....105
2.3.4 DevOps Methodology.....105
2.3.5 5 Whys Analysis.....106
2.3.6 User Acceptance Testing (UAT).....107
Chapter 3.....**108**
3.1 Description.....108
3.2 Scrum Methodology Phase.....109
3.2.1 Initiation.....109
3.2.2 Planning.....109
3.2.3 Implementation Phase.....109
3.2.4 Review and Retrospective Phase.....110

3.2.5 Release..... 110
3.3 Phases..... 110
3.3.1 Product Backlog..... 110
3.3.1.1 Tools..... 111
3.3.1.2 Input..... 111
3.3.1.3 Output..... 112
3.3.2 Sprint Planning..... 112
3.3.2.1 Tools..... 112
3.3.2.2 Input..... 113
3.3.2.3 Output..... 113
3.3.3 System Deployment..... 114
3.3.3.1 Tools..... 114
3.3.3.2 Input..... 115
3.3.3.3 Output..... 116
3.3.4 Unit Testing..... 116
3.3.4.1 Tools..... 117
3.3.4.2 Input..... 117
3.3.4.3 Output..... 118
3.4 Work Plan/Gantt Chart..... 118
3.5 Capstone 1 and 2 Targets..... 119
3.5.1 Capstone 1 (80% of the development)..... 119
3.5.1.1 System Architecture, Requirements, and User Roles..... 119
3.5.1.2 Blockchain Backend and Smart Contract Implementation..... 119
3.5.1.3 Credential Verification System..... 120
3.5.1.4 Administrator Dashboard..... 120
3.5.2 Capstone 2 (20% of the development)..... 120
3.5.2.1 System Implementation & Environment Setup..... 120
3.5.2.2 User Acceptance Testing (UAT)..... 120

3.5.2.3 System Testing and User Acceptance Testing (UAT).....121
3.5.2.4 Deployment and Final Documentation.....121
References:.....122
Appendix A:Transcript of Interview with HR Recruiter of Avesco Marketing Corporation.....131
Appendix B: Transcript of Interview with HR Recruiter Intern of Sunlife Philippines..... 132
Appendix C: Transcript of Interview #2 with HR Recruiter of Avesco Marketing Corporation.....133
Appendix D: Survey Response From HR Recruiters Regarding Credential Verification Practices..... 134
Appendix D.1: Survey Response from Lagunilla, HR Recruiter Intern of Sunlife Philippines..... 134
Appendix D.2: Survey Response from Tan, HR Recruiter of Avesco Marketing Corporation..... 137
Appendix E: Survey Response from the Human Resources Development and Management Department Department Head 1 of the City Government of Bacoor, Cavite..... 138
Appendix F: Survey Response Summary – Measuring the Feasibility and Impact of Blockchain-Based Credential Verification Systems.....140
Appendix G: Survey Response from Roland Talagtag from Reliance Producers Cooperative..... 142

Chapter 1

1.1 Background of the Study

In an era where digital technology is ever growing, it is easier than ever to forge credentials, thus harder to distinguish between them being fake or real. The case of Marilee Jones, a former dean at MIT, who was found to have falsified her academic credentials for nearly 30 years, using fake degrees from Albany Medical College, Union College, and Rensselaer Polytechnic Institute (Hoover & Millman, 2007). Similarly, in January 2012, Yahoo's CEO, Scott Thompson, falsely claimed to have a degree in computer science from Stonehill College, alongside his actual degree in accounting. This was exposed through an investigation by activist shareholder Daniel Loeb (Luckerath-Rovers, 2025). These incidents are not isolated; in fact, even in the Philippines, a 2025 operation by the Manila Police District (MPD) resulted in the arrest of Mario Acantilado, who was caught producing counterfeit diplomas (Calucin, 2025). In addition, in Brazil, a large-scale investigation covered thousands of fake adult high school diplomas sold to individuals who had never completed their education. These fake adult high school diplomas resulted in wage gains of 5% to 8% for individuals compared to those without. These examples showcase the proliferation of fake academic credentials as a growing threat, not only undermining the integrity of education systems but also compromising the recruitment process and eroding trust between educational institutions, employers, and job seekers. Overall, the issue is driven by the desire for individuals to bypass educational qualifications and gain access to employment opportunities; almost all employers may perceive diplomas or credentials as a basis for productivity (Firmo, 2021). Moreover, in the Philippines and other developing countries, where education is a key factor in securing employment, individuals are encouraged to resort to making use of fraudulent credentials and certifications to gain either employment or admission to educational institutions.

Existing credential verification systems rely heavily on manual processes, such as contacting educational institutions or using third-party services to verify credentials. These systems are not only time-consuming but also vulnerable to errors and fraud. For example, when an applicant's credentials are in question, HR departments often rely on alumni networks or use personal references, both of which may not be reliable or standardized. This process can take up to five to ten working days, depending on the responsiveness of the institution being contacted. (Appendix D) Furthermore, there is a lack of standardization in verification methods across institutions, which further complicates the process. In addition, many educational institutions continue to use physical credentials, such as diplomas and

transcripts, which are prone to manipulation. Watermarks, holograms, and other anti-forgery techniques, while commonly employed, have proven insufficient in preventing fraud. The recent case involving two Filipino nationals shed light on vulnerabilities of the existing systems of issuance and verification, as both women submitted forged university certificates, one from De La Salle University and the other from Ateneo de Manila University, in order to secure positions at CL Enterprise, a retail company operating Mr. Fix outlets. Their deception remained undetected for over a year, until a random inspection by the Ministry of Manpower (MOM) uncovered the falsified documents. Moreover, in South Florida, thousands of fake nursing degrees were issued, leading to individuals passing national exams and entering the healthcare system without the necessary training (Putri, 2024). These cases highlight the significance of thorough document verification, as false educational qualifications or credentials are rampant.

The consequences of fraudulent credentials extend beyond recruitment issues; they have serious legal, financial, and reputational implications for both employers and educational institutions. Employers who hire individuals with fake credentials may face legal action, financial losses, and damage to their reputation. For instance, Mallory (2014) describes a case where a security guard company in Rhode Island was held liable for hiring a fraudster who misrepresented their qualifications, leading to serious legal consequences when the individual committed theft. Similarly, a Texas cab company was held accountable for hiring a driver with false credentials, who later committed a crime (Attewell & Domina, 2010). These cases illustrate the risks associated with negligent hiring practices, which can be mitigated by more reliable verification methods. Educational institutions also face significant risks when they fail to secure their credentials. Institutions that unknowingly issue authentic degrees alongside fraudulent ones may suffer from loss of credibility, affecting future enrollments and damaging their reputation in the academic community. The Philippine Labor Code (R.A 6715) mandates that employers exercise due diligence in verifying credentials, and failure to do so can lead to negligent hiring claims. Moreover, educational institutions and employers must ensure that their credentialing systems comply with data privacy laws, such as the Philippine Data Privacy Act of 2012, to avoid legal liabilities. Given these serious implications, there is an urgent need for a more secure, transparent, and efficient method of issuing. Outdated, manual verification and issuance systems are no longer adequate, as these are often slow, inefficient, and vulnerable to fraud, allowing fake documents to go undetected for long periods or, worst case, never at all.

According to Pulmano et al. (2023) the concept of self-sovereign identity (SSI) as a key framework was highlighted for empowering individuals to control their credentials. This concept is

increasingly seen as the future of digital identity management, allowing users to hold and share their data in a decentralized manner, without relying on centralized authorities for validation. In alignment with this concept and the challenges posed by academic fraud and inefficient verification methods, Hyperledger Fabric, a permissioned blockchain technology, offers a secure, transparent, and tamper-proof solution for credential management. Blockchain technology ensures that once credentials are recorded, they cannot be altered or forged, providing a reliable and immutable record for credential verification. Moreover, its model restricts access to trusted participants, ensuring that only authorized entities can validate and access sensitive data, thereby reducing the risk of fraudulent activities (Mitani & Otsuka, 2020).

This is made possible by the modular architecture of Hyperledger Fabric, which enhances scalability and efficiency by using a modular consensus mechanism that allows only specific nodes to validate transactions, reducing computational load and transaction fees. This feature is particularly useful for systems managing large datasets like credentials and ensuring cost-effective processing. Additionally, privacy controls such as data collection and channels allow for secure data sharing between authorized participants, ensuring confidentiality (Androulaki et al., 2018).

Despite their advantages, permissioned blockchains like Hyperledger Fabric have limitations, notably the centralization of control. Since only trusted entities validate transactions, there is a risk of tampering if these entities act maliciously or fail to reach consensus. As noted by Solat et al. (2021), the closed network of permissioned blockchains can allow block alterations if the trusted participants permit it, undermining the core tamper-resistance feature of blockchain. Unlike permissionless blockchain, where tampering with past transactions would require recalculating the entire blockchain, this centralization poses a potential vulnerability (Solat et al., 2021).

However, despite these challenges, Hyperledger Fabric remains a robust solution for credential verification due to its enhanced privacy features, scalability, and compliance with regulations like the Philippine Data Privacy Act of 2012 (DPA 2012) (Pulmano et al., 2023). Its ability to securely manage academic and professional credentials, while offering real-time verification, makes it a suitable solution for mitigating the risks associated with academic fraud and inefficiency. By using Chaincode (smart contracts), Hyperledger Fabric enables the automated issuance and verification of credentials, ensuring they are tamper-proof and easily accessible by authorized entities (Androulaki et al., 2018).

In conclusion, Hyperledger Fabric provides a secure, scalable, and compliant framework for academic credentialing that addresses the flaws of traditional verification systems. Its combination of privacy controls, real-time verification, and auditability makes it an ideal solution to ensure the authenticity and integrity of academic credentials in today's digital age.

1.2 Statement of the Problem

This section focuses exclusively on the data- and information-related root causes behind the inefficiencies, fraud, and limitations of traditional systems using the 5 Whys method. Each identified problem includes a justification of why it is a problem, a logically connected sequence of questions and answers, and real-life examples grounded in the review of related literature (RRL) and project appendix.

1.2.1 Specific Problem #1 - Reliance on Static Credential Formats Without Embedded Verifiability

Traditional systems issue and store credentials in static, non-machine-readable formats like PDFs or printed copies that lack embedded authenticity metadata such as cryptographic hashes, issuer signatures, credential IDs, and timestamps. These static formats cannot be independently validated, are highly vulnerable to tampering, and require human-driven processes to inspect.

1.2.1.1 5 Whys Analysis

1) Why do fake credentials go undetected?

Fake credentials often go undetected in traditional systems because the verification process used by HR departments and government agencies is largely manual and based on visual inspection of submitted documents, which typically involve PDFs or printed diplomas, which lack embedded validation data such as digital signatures, hashes, or cryptographic proofs. This lack of digital verifiability leads to trust being based on physical appearance or subjective judgment, which is easily flawed. Miss Tan, an HR recruiter at Avesco Marketing Corporation, mentioned that applicants are required to bring physical copies of their diplomas and transcripts, which HR recruiters inspect by either asking the applicant about the contents or visually checking for logos and seals; if further verification is needed, they would reach out to the issuing institution. This process takes five to ten working days and lacks standardized confirmation protocols balik (Tan, personal communication, June 28, 2025; Appendix A; Tan,

personal communication, July 21, 2025; Appendix C; Lagunilla, personal communication, July 21, 2025; Appendix D.2). Overall, because the documents contain no cryptographic or linked data, visually convincing forgeries easily bypass scrutiny. One high-profile example is Marilee Jones, the former MIT dean who submitted fabricated degrees from multiple institutions, which were never verified beyond face value and went undetected for 28 years (Hoover & Millman, 2007)

2) Why are institutions still relying on outdated methods?

Institutions continue to use printed documents and static PDFs because these are low-cost, familiar, and require no investment in infrastructure upgrades. Many schools and agencies are constrained by entrenched workflows and legacy systems that were never built to accommodate verifiable data. Additionally, there is limited awareness and understanding of technologies like verifiable credentials, blockchain anchoring, or credential APIs (Freeman et al., n.d; Appendix C; Appendix E). Unfortunately, even institutions that are aware of digital threats often underestimate the risk or believe that fraud detection can still be managed manually. For instance, MIT continued manual verification practices even after uncovering Marilee Jones' fraud (Hoover & Millman, 2007). In local cases, Bacoor LGU and Reliance Producers Cooperative admitted their systems are inefficient but remain unreformed due to budget constraints and lack of staff training (Appendix E).

3) Why are credentials not embedded with real-time validation mechanisms like hashes or blockchain references?

Credentials are not embedded with real-time validation mechanisms because most issuer systems still operate on legacy infrastructure designed for document storage rather than interoperable data. These systems do not support features like cryptographic hashes, issuer metadata, or blockchain anchors. Most static credential formats do not integrate with digital verification ecosystems, limiting trust and increasing dependency on human judgment. According to Freeman et al. (n.d.) observed that current databases lack schema fields necessary for verifiable attributes, making it difficult to modernize without overhauling the

entire structure. In practice, this means that credentials like diplomas or certificates exist only as flat files, without any built-in proof of issuance or links to authoritative registries, unless contacted directly. This was evident in the case of Scott Thompson, the former Yahoo CEO, who claimed to have a Computer Science degree that did not exist. The lack of embedded metadata or any automated reference to verify that claim allowed the fraud to persist until it was exposed by an external shareholder investigation (Lückerath-Rovers, 2024).

4) Why haven't institutions upgraded to secure, verifiable digital credential systems?

Most institutions have not adopted modern credentialing systems because of a combination of cost barriers, lack of technical capacity, and organizational resistance to change. Jimu (2018) highlights how institutional entrenchment leads to the continued use of outdated systems despite known risks. This is reinforced by local examples like Bacoor LGU and Reliance Producers Cooperative, where officials acknowledged inefficiencies but cited budget limitations and the steep learning curve as obstacles to upgrading (Appendix E; Appendix D.1). Moreover, there is a fear of complexity and uncertainty about regulatory compliance when adopting blockchain or digitally verifiable credentials, especially in government contexts where any systemic change involves long bureaucratic processes. As a result, institutions prioritize continuity over reform, even after encountering cases of fraud (Hoover & Millman, 2007)

5) Why is this reliance on unverifiable documents dangerous?

Relying on unverifiable, static documents creates systemic vulnerabilities that allow forged or altered credentials to enter legitimate pipelines such as employment, licensing, or certification. Without built-in validation logic or tamper-evidence, fraudulent documents can persist for years without detection. This poses legal, operational, and ethical risks, particularly in sectors like education and healthcare. The most alarming example is the 2023 U.S. case where over 7,600 fake nursing diplomas were sold and at least 2,800 individuals with fraudulent records were able to pass the licensure exam and enter the healthcare system (Putri, 2024).

1.2.1.2 Root Cause Analysis

The core issue in this problem lies in the traditional treatment of credentials as static records, either physical or digital documents, rather than dynamic, verifiable data objects. Traditional issuance and verification systems were never designed to capture or process authenticity metadata, resulting in a lack of cryptographic proofs, issuer signatures, and verifiable audit trails. Because of this, institutions are forced to rely on manual processes such as visual inspection, phone calls, or emails, which are slow, error-prone, and inconsistent. The lack of built-in integrity checks makes these credentials vulnerable to forgery, while institutional resistance to adopting modern solutions like blockchain is fueled by perceived cost, complexity, and unfamiliarity. As long as credential formats remain static and unverifiable, the education and employment ecosystem will remain exposed to fraud, inefficiency, and compromised trust.

1.2.1.3 Opportunity

Our blockchain-based system issues verifiable credentials with embedded cryptographic proofs such as hashes and issuer metadata. Each credential is recorded on a permissioned ledger, enabling real-time validation without relying on human inspection. This eliminates the forgery risk and significantly reduces turnaround time for verification. With built-in auditability, our system strengthens public trust, enforces accountability, and replaces static PDFs with dynamic, verifiable digital credentials.

1.2.2 Specific Problem #2 - Lack of Credential Lifecycle Management (Issuance, Expiry, Revocation, and Updates)

Traditional credentialing systems do not support dynamic metadata or credential states that reflect real-world changes, such as revocation, expiration, or version updates. Credentials are issued once and treated as immutable, static documents, with no mechanism to reflect whether they have been invalidated or updated. This creates an information asymmetry between the issuer and verifier, who has no way of knowing whether a credential is still valid at the time of review.

1.2.2.1 5 Whys Analysis

1) Why do traditional systems fail to indicate if a credential has been revoked or expired?

This is because most credentials exist only as static documents like PDFs or printed certificates, which carry no metadata that indicates validity period or revocation status. They are typically stored in folders, personal email inboxes, or printed files and lack links to real-time status systems. Traditional verification methods rely on manual cross-checking with registries, which is time-consuming and often skipped. Thus, if a diploma or certificate is revoked due to dishonesty or policy validations, there is no downstream notification to employers or regulators on the real-time status of the submitted credential.

2) Why are expiration and revocation statuses not integrated in the credential itself?

Static formats like PDFs do not support programmable metadata that can update over time. Credential files lack linked data or API endpoints that can reflect changes in status. As observed by Jimu (2018), traditional systems lack the infrastructure to automate credential lifecycles, making them blind to conditions such as renewal deadlines, certification suspensions, or expirations.

3) Why is it difficult to track updated versions of a credential or maintain version control?

There is no versioning framework in static credentialing systems. If someone reissues a diploma with updated names, corrected honors, or a change in affiliation, there is no way to track which version is the most recent or authoritative unless the user explicitly discloses the change or the verifier contacts the associated issuer institution. (Pandya, 2022). According to Freeman et al. (n.d), they argue that traditional systems do not include credential IDs or version hashes that can differentiate legitimate updates from duplicates or forgeries. This leads to confusion among verifiers, especially when discrepancies exist between multiple copies.

4) Why is this a risk for employers, institutions, or regulators?

Without lifecycle visibility, institutions risk accepting outdated, falsified, or revoked credentials. This is particularly dangerous in time-sensitive roles like medicine, aviation, or finance, where certifications expire or require revalidation. The 2023 fake nursing diploma scandal exemplifies how expired or fraudulent

credentials can remain undetected due to lack of real-time revocation awareness. (Putri , 2024)

5) Why haven't institutions implemented systems that support credential lifecycle management?

Many issuers operate legacy databases or rely on Excel spreadsheets, physical files, or email logs for credential issuance. These systems lack support for API-based revocation, audit trails, or expiration rules. Integrating lifecycle tracking requires redesigning credential schemas and building interoperable verification systems, efforts that institutions perceive as costly and complex. Moreover, there is no regulatory pressure to require revocation protocols, so many issuers do not prioritize it (Rustemi et al., 2023; Freeman et al., n.d; Jimu, 2018).

1.2.2.2 Root Cause Analysis

Traditional credentialing systems lack the capacity to manage dynamic states because they were designed around paper-based processes and later digitized into flat files. This results in credentials that are permanently static and disconnected from the real-world conditions that affect their validity. The lack of metadata standards, infrastructure, and legal mandates perpetuates this flaw, exposing employers and regulators to expired, revoked, or inaccurate records.

1.2.2.3 Opportunity

Our solution introduces lifecycle management for all issued credentials. In simple terms, all transactions are logged, where every transaction or credential has a timestamp, expiration date, and revocation status field embedded and tracked on-chain. Issuers can revoke or update credentials at any time, and verifiers will always see the most recent status during validation. This ensures that credentials used in high-stakes environments are always up-to-date, improving trust, safety, and compliance.

1.2.3 Specific Problem #3 - Absence of Standardized Credential Metadata and Format Structures

This is a data standardization issue because traditional issuers provide credentials in inconsistent layouts and unstructured formats. An example of which is PDFs with no unified fields for issuer, recipient, type, status, or dates. Without standard metadata schemas, systems and verifiers cannot reliably extract or validate data, leading to interpretation errors, inefficiencies, and the inability to automate cross-institutional checks.

1.2.3.1 5 Whys Analysis

1) Why is credential formatting inconsistent?

   Credentials are inconsistent because institutions independently design and issue credentials with no mandatory structure or metadata format. Each issuer uses its own layout, which leads to differences in how fields like name, issue date, and program title appear or are labeled. This inconsistency creates problems for downstream verifiers such as HR or government offices, who must manually identify information across documents with varying formats (Pandya, 2022). For example, Avesco HR has to manually interpret credentials submitted from different institutions; some credentials contain names in uppercase, while others hide key data inside paragraphs or images, resulting in longer processing time and increased error risks (Appendix C).

2) Why isn't there a single standard for formatting or storing credential data?

   There is no enforced credential data standard due to the decentralized nature of issuing bodies. Schools, training centers, and certification providers each adopt their own templates without a unifying guideline. As noted by Freeman et al. (n.d.), this leads to fragmented data and technical barriers for automation. Even when credentials contain similar data points, the labeling, position, and structure vary significantly. In local interviews, registrars mentioned that even within the same school, departments use different formats for certificate issuance, which complicates digital record keeping (Appendix D.1).

3) Why is standardization important for automation or interoperability?

Standardized metadata and credential schemas allow systems to parse, read, and verify credentials automatically. Without it, every verifier must create custom logic to interpret incoming documents. This is not scalable or sustainable. Systems like Credly and ESbi use JSON-LD or W3C VC formats to enable plug-and-play integration across issuers and verifiers (Stefaniak & Carey, 2019).

4) Why haven't institutions adopted these standards?

The adoption of credentialing standards is limited by lack of awareness, technical capacity, and guidance from regulatory bodies. Institutions are hesitant to switch to structured formats like JSON-LD or XML because of unfamiliarity and lack of local infrastructure. In the Philippines, no national framework exists for digital credential metadata, unlike in the EU, where projects like Europass and EBSI have begun implementing such standards (Pulmano et al., 2023).

5) Why is lack of standardization risky?

The absence of standards leads to human errors, delays in verification, and misinterpretation of critical details. Inconsistent formats allow fraudulent actors to modify or misinterpret data more easily. For instance, a fake diploma in image format can bypass checks simply because HR systems cannot process credential types, which means even legitimate documents are hard to cross-check efficiently.

1.2.3.2 Root Cause Evaluation

Credential documents are treated as static artifacts without embedded, structured metadata. The absence of a shared data model among institutions leads to fragmented, human-readable-only formats. Without schema enforcement, the potential for automation, integration, and security is severely limited.

1.2.3.3 Opportunity

Our system enforces the use of standardized credential metadata following the W3C Verifiable Credentials Data Model, ensuring that all credentials include consistent,

machine-readable fields. This allows instant parsing, efficient filtering, and automated verification across institutions, regardless of the issuing body.

1.2.4 Specific Problem #4 - Exclusion of Microcredentials and Informal Learning Proofs

Traditional credential systems are focused on degree-based or terminal certifications and often exclude short-term or modular achievements such as online courses, training workshops, or industry certifications. These microcredentials are harder to verify and rarely included in official registries or transcripts, leading to a fragmented view of lifelong learning.

1.2.4.1 5 Whys Analysis

1) Why are microcredentials often unrecorded?

   Traditional systems were not designed to handle high-frequency, small-scope certifications. They focus on end-stage outcomes like degrees or licenses. There's no infrastructure to accommodate frequent, low-stakes issuances. (Varadarajan, et al., 2023).

2) Why do systems struggle to verify informal learning?

   Systems lack the ability to ingest and recognize diverse formats and issuers. Microcredentials may come from online platforms such as Coursera, LinkedIn Learning, or community-based training. These are not centrally recognized or formatted consistently, making them hard to validate or include in employment workflows. (Varadarajan, et al., 2023).

3) Why are microcredentials important for employability?

   Employers increasingly value skill-based credentials to supplement or replace traditional degrees, especially in tech and vocational fields. According to microcredentials, they provide a flexible path for upskilling and meet labor market demands more rapidly than formal degrees. However, lack of integration into credential systems makes them invisible to most hiring systems. (Keystone Education Group, 2024).

4) Why haven't traditional systems adapted to include these?

There is resistance from educational institutions to recognize credentials they don't issue. Additionally, older database systems do not support frequent issuances or flexible credential categories. Many schools still rely on manual records or outdated ERPs, making integration of microcredentials a challenge (Appendix D.1)

5) Why is this exclusion problematic?

It creates a blind spot in evaluating a person's full skillset. Candidates who participate in short-term learning programs cannot effectively present these credentials during application or licensing processes. This leads to underemployment, inefficiencies in hiring, and wasted learning opportunities.

1.2.4.2 Root Cause Evaluation

It creates a blind spot in evaluating a person's full skillset. Candidates who participate in short-term learning programs cannot effectively present these credentials during application or licensing processes. This leads to underemployment, inefficiencies in hiring, and wasted learning opportunities

1.2.4.2 Opportunity

Our system supports microcredentials by enabling flexible issuance from multiple types of verified issuers. It allows recipients to store and share all learning records, formal and informal, in a unified wallet, enriching their credential portfolio and improving employability outcomes.

1.2.5 Specific Problem #5 - Decentralized and Paper-Based Storage Results in Credential Loss and Inaccessibility

Traditional credential systems rely on institution-specific storage and paper-based issuance with no digital backup, centralized registry, or redundancy. When credentials are issued only as physical documents or stored in local databases without synchronization or long-term preservation strategies, they are highly vulnerable to loss from natural disasters, poor archiving practices, and human error. Overall, individuals who lose access to credentials are unable to

recover or verify them, severely impacting their ability to pursue employment, licenses, or further education.

1.2.5.1 5 Whys Analysis

1) Why are credentials frequently lost or inaccessible?

   Credentials are lost or inaccessible because most are issued in printed versions or stored in local systems without cloud backup or integration into centralized registries. These formats are prone to loss from physical damage or institutional errors.

2) Why are paper-based systems still widely used despite these risks?

   Institutions continue to use paper-based systems because they appear simple and cost-effective, especially in settings with limited digital infrastructure. However, they carry hidden costs: physical degradation, data loss, and human misfiling. Jimu (2018) highlights that small schools or LGUs often do not have digitization initiatives or disaster recovery policies. Reliance Producers Cooperative confirmed that most credential-related documents are still stored manually, and unless scanned during onboarding, these are lost if misplaced or damaged (Appendix E). The absence of enforced digital transition policies allows these risky practices to persist.

3) Why don't institutions implement centralized or cloud-based archives?

   Many institutions do not implement cloud-based or redundant storage due to budget constraints, cybersecurity concerns, and lack of technical staff. Freeman et al. (n.d.) argue that decentralized, siloed credentials databases are common because schools operate independently without national mandates for credential backup or interconnectivity.

4) Why can't individuals recover lost credentials easily from institutions?

   Most institutions lack long-term digitized records or automated reissuance procedures. If a school's archives are damaged, disorganized, or only

kept for a few years, reissuance becomes impossible or heavily delayed. Freeman et al. (n.d.) note that credentialing authorities often fail to maintain permanent records.

5) Why are data loss and inaccessibility serious issues in credentialing?

Credential loss undermines the trust, continuity, and accessibility of education and employment systems. Without tamper-proof backups or decentralized registries, individuals risk permanent disqualification from job applications, licensing boards, or graduate programs. Rustemi et al. (2023) warn that siloed and analog credential practices disproportionately harm vulnerable populations who cannot afford multiple reissuance requests or who study in institutions with poor archival capacity. In the Philippines, disasters, institutional closures, and outdated storage habits continue to erase people's educational history, harming their long-term socioeconomic mobility.

1.2.5.2 Root Cause Evaluation

The root issue lies in the reliance on localized, paper-based credential storage and the absence of interoperable, secure digital systems for backups and access. Without national digital registries, cloud-based archives, or blockchain-anchored credential logs, educational records remain fragile, fragmented, and highly susceptible to permanent loss. This makes re-verification inefficient or impossible, denying individuals their earned qualifications and reducing institutional credibility in the process.

1.2.5.3 Opportunity

Our blockchain-based credentialing system addresses this problem by ensuring that every credential issued is recorded as a cryptographic hash on a decentralized ledger (Hyperledger Fabric), proving an immutable and tamper-evident proof of issuance. Additionally, credentials are stored in a user-owned digital wallet that is cloud-accessible, enabling individuals to retrieve them anytime, even if the issuing institution is no longer operational. Institutions are onboarded into a shared network, ensuring long-term access, redundancy, and disaster resilience. This approach eliminates single points of failure and builds permanent verifiability into the credential lifecycle.

1.2.6 Specific Problem #6 - Manual Verification Delays Due to Lack of Credential Interoperability

Most traditional credential verification processes involve manual procedures such as email exchanges, phone calls, or physical document checks. These delays are not just due to the static format of documents but also stem from the absence of interoperable systems and shared digital registries where credentials can be queried in real time. This hampers efficiency, especially in high-volume or time-sensitive settings.

1.2.6.1 5 Whys Analysis

1) Why are verification requests still manual and slow?

   Verification remains manual because there is no universal or federated system where issuers, employers, and verifiers can instantly check the authenticity of a credential. Most institutions do not expose APIs or searchable registries for external verification. Instead, HR departments must rely on time-consuming practices like emailing the issuing school or scanning printed documents for visual review. For instance, in the HR process at Avesco Marketing Corporation, recruiters spend 5-10 working days awaiting registrar confirmation. (Appendix D.2).

2) Why haven't institutions integrated verification APIs or interoperable platforms?

   Many issuers and verifiers lack both the technical capacity and the incentives to build or adopt interoperable systems. There is no consistent requirement or government mandate for this in many countries, including the Philippines. Freeman et al. (n.d.) point out that the credentialing ecosystem remains highly soiled, where each institution operates in isolation. Even when institutions have digital records, they are often inaccessible to external parties due to data privacy concerns or lack of authentication mechanisms.

3) Why can't verifiers access credential data directly from schools or systems?

   Even when credential data exists digitally within a school's internal system, it is rarely published or shared externally due to concerns over data protection, consent, and system design. There is no standard method for securely

querying records. In most cases, the verifier does not even know whether the credential exists unless the user provides a copy. As a result, verifiers must rely on manual outreach, creating a bottleneck.

4) Why is this problematic for recruitment and credential-based decision making?

   The lack of real-time access to credential data means that hiring decisions, scholarship awards, or license issuances are delayed. In high-stakes settings like healthcare, law enforcement, or government contracts, this slows down onboarding and creates risk. For example, the fake nursing diploma scandal in the U.S. (Putri, 2024) could have been detected earlier if real-time verification were available.

5) Why hasn't automation been prioritized in credential verification?

   There is still limited awareness among institutions about the possibility of secure automation. Many are also unsure how to implement verification without exposing sensitive data. In other cases, institutions assume that fraud is rare or manageable, so they tolerate delays.

1.2.6.2 Root Cause Evaluation

The core problem is that there is no shared, interoperable infrastructure for credential verification. Traditional credentialing systems are institution-specific and siloed, with no federated registry or verifiable credential wallet. This forces every verification request to be done from scratch, which is done by contacting the issuer or requesting a reprint, leading to delays, inconsistency, and higher operational costs.

1.2.6.3 Opportunity

Our system supports interoperable, API-accessible digital credential wallets where issuers can register credentials on a permissioned blockchain. Authorized verifiers can check credential validity instantly via accessing the hash sent by the user associated with the credential, eliminating the need for emails or manual confirmation. This not only reduces verification time from days to seconds but also supports better decision-making in time-sensitive contexts like employment, licensing, and scholarships.

1.2.7 Specific Problem #7 - No Tamper-Evident Logs or Version History

This is a data integrity problem because traditional credential systems do not maintain logs that track changes, issuance events, revocations, or reprints. Without version history or cryptographic trails, any modified or duplicated document appears indistinguishable from the original. The absence of tamper-evident mechanisms prevents detection of unauthorized changes, damaging trust and traceability.

1.2.7.1 5 Whys Analysis

1) Why can't tampered credentials be detected?

   Traditional systems treat credentials as single-version documents stored in folders or PDFs. There is no log of who issued it, when, or whether it was reissued or revoked.

2) Why aren't credential systems designed to record history?

   Legacy systems are focused on storage, not traceability. They lack append-only logs or change-tracking mechanisms, which are standard in blockchain or digital signature systems. This makes it impossible to determine if a credential has changed over time.

3) Why is this lack of version control dangerous?

   Without a history of issuance and updates, fraud cannot be easily detected. Employers might trust outdated, forged, or revoked credentials. This creates legal and reputational risks.

4) Why is fraud not detected without version control?

   Fraud is difficult to detect because there is no way to confirm if a credential was altered after it was issued or if it's the latest version. Without any metadata attached to the document, there's no chain of custody to follow.

5) Why aren't these systems designed with traceability?

   Legacy systems were built with a focus on simply storing data, not on its authenticity or traceability. There was little to no thought about protecting credentials from alteration or forgery.

1.2.7.2 Root Cause Evaluation

Credentials issued using traditional systems are static and versionless. The systems do not store metadata about issuance events, edits, or revocations, making all documents equally unverifiable once printed or shared. This opens pathways to fraud.

1.2.7.3 Opportunity

Our system uses blockchain's immutable audit trails to log every event in a credential's lifecycle—issuance, edits, and revocations. Employers and verifiers can view timestamps and change history, preventing forgery and reinforcing trust.

1.2.8 Specific Problem #8 - No Unified Registry or Searchable Credential Index

This is a discoverability and access problem. Credentials are stored in isolated silos like individual school databases or printed archives, which prevents global search or cross-institutional validation. Without a unified index or registry, credentials are hard to locate or verify.

1.2.8.1 5 Whys Analysis

1) Why can't credentials be found quickly?

   Most employers or institutions don't have access to school databases. If a school closes or changes staff, the credential may become unverifiable.

2) Why isn't there a centralized directory?

   Credential systems are built independently with no interoperability or national registry.

3) Why is there no interoperability between these systems?

   The lack of standardization and regulations means there's no push for institutions to adopt a common data format, further preventing any meaningful integration.

4) Why hasn't a national registry been implemented?

   There has been no strong policy mandate or initiative to establish a central registry. Each institution or agency operates in isolation, and they may not see the immediate need for such a system.

5) Why can't schools, employers, or verifiers easily access these credentials?

Since the data is decentralized and siloed, even those with legitimate access cannot quickly retrieve or validate credentials without relying on manual processes or contacting the issuer directly.

1.2.8.2 Root Cause Analysis

Credential data exists in isolated formats across institutions with no common lookup infrastructure, making verification difficult or impossible if the issuer is unresponsive.

1.2.8.3 Opportunity

Our system enables a searchable, permissioned registry of issued credentials. Verifiers with user consent can look up credentials in seconds using a unique ID or QR code—eliminating the need for manual chasing or guesswork.

1.2.9 Specific Problem #9 - Credentials Are Not Interoperable Across Systems

This is a data portability issue. Traditional credentials are often PDFs or scanned images that cannot be read by machines, imported to HR systems, or connected to digital wallets. This creates friction, especially in job platforms or institutional systems.

1.2.9.1 5 Whys Analysis

1) Why can't credentials be reused in different systems?

Credential formats lack structure and standards (e.g., JSON-LD, Open Badges). Avesco HR manually encodes details from certificates because PDFs can't be parsed (Appendix D.2).

2) Why weren't credentials built to be machine-readable?

Traditional focus was on visual presentation, not machine interpretation. As Freeman et al. (n.d.) note, current formats limit automation and scaling.

3) Why don't these systems use standardized formats?

The lack of awareness and understanding of the importance of standardization, along with a reliance on outdated technologies, prevented adoption of interoperable formats like JSON-LD or Open Badges.

4) Why does this lead to inefficiencies in credential verification?

Because the data isn't structured, it requires manual extraction and data entry into various systems, increasing the risk of error and reducing overall efficiency. This also slows down processes such as job applications or admissions, where fast verification is key.

5) Why does this lead to slow verification and increased errors?

Because traditional credential formats lack automated validation mechanisms, verifiers must manually check the authenticity of credentials by contacting issuers or performing additional steps, which delays the verification process and increases the chances of human error.

1.2.8.2 Root Cause Evaluation

Credentials are not structured using interoperable data formats. As a result, they can't be integrated with systems that require structured input (e.g., job portals, academic apps).

1.2.8.3 Opportunity

Our platform supports verifiable credentials in machine-readable, standards-based formats (W3C VC), ensuring seamless import/export with third-party systems.

1.3 Solutions Framework

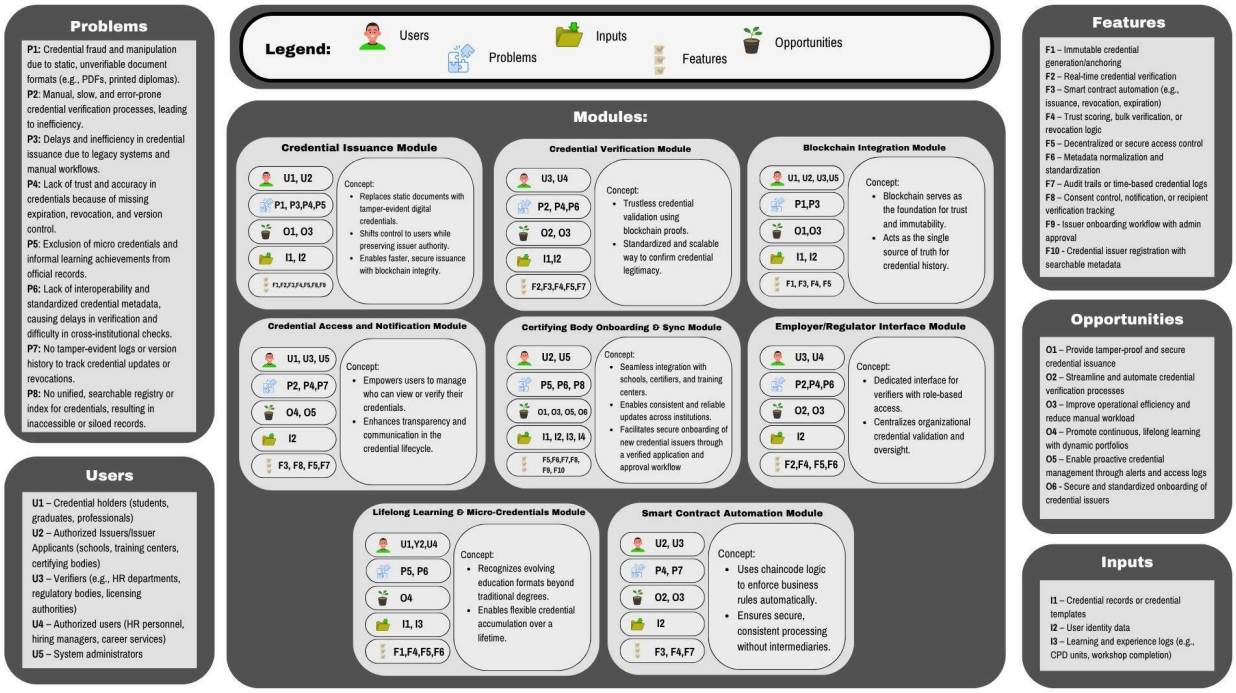


Diagram 1.3: Solutions Framework Diagram for ChainCert System Modules

This section presents the proposed Solutions Framework for ChainCert, a blockchain-based credential verification and management platform designed to solve critical issues such as credential fraud, inefficiency, and lack of interoperability. Built on Hyperledger Fabric, a permissioned blockchain, ChainCert provides a secure, tamper-evident, and real-time system for the issuance, upload, storage, access, and verification of credentials.

ChainCert addresses challenges like static document formats, manual verification, data fragmentation, and lack of credential lifecycle management by enabling verifiable, decentralized credentials accessible to employers, institutions, regulators, and learners. It empowers both authorized issuers and credential holders to participate in the credentialing ecosystem.

The framework is structured around five key elements: Users, Problems Addressed, Inputs, Features, and Opportunities. The system is composed of eight interconnected modules, each designed to support secure, trustworthy, and interoperable credential management across the entire credential lifecycle.

1.3.1 Target Users

- **U1** – Credential holders (students, graduates, professionals)
- **U2** – Authorized issuers/Issuer applicants (schools, training centers, certifying bodies)
- **U3** – Verifiers (e.g., HR departments, regulatory bodies, licensing authorities)
- **U4** – Authorized users (HR personnel, hiring managers, career services)
- **U5** – System administrators

1.3.2 Problems

- **P1:** Credential fraud and manipulation due to static, unverifiable document formats (e.g., PDFs, printed diplomas).
- **P2:** Manual, slow, and error-prone credential verification and onboarding processes, leading to inefficiency.
- **P3:** Delays and inefficiency in credential issuance due to legacy systems and manual workflows.
- **P4:** Lack of trust and accuracy in credentials because of missing expiration, revocation, and version control.
- **P5:** Exclusion of microcredentials and informal learning achievements from official records.
- **P6:** Lack of interoperability and standardized credential metadata, causing delays in verification and difficulty in cross-institutional checks.
- **P7:** No tamper-evident logs or version history to track credential updates or revocations.
- **P8:** No unified, searchable registry or index for credentials and credential issuers, resulting in inaccessible or siloed records.

1.3.3 Inputs:

- **I1** – Credential records or credential templates
- **I2** – User identity data
- **I3** – Learning and experience logs (e.g., CPD units, workshop completion)
- **I4** - Issuer application and organization identity data

1.3.4 Opportunities

- **O1** – Provide tamper-proof and secure credential issuance
- **O2** – Streamline and automate credential verification processes
- **O3** – Improve operational efficiency and reduce manual workload
- **O4** – Promote continuous, lifelong learning with dynamic portfolios
- **O5** – Enable proactive credential management through alerts and access logs
- **O6 -** Secure and standardized onboarding of credential issuers

1.3.5 Features

- **F1** – Immutable credential generation / anchoring
- **F2** – Real-time credential verification
- **F3** – Smart contract automation (e.g., issuance, revocation, expiration)
- **F4** – Trust scoring, bulk verification, or revocation logic
- **F5** – Decentralized or secure access control
- **F6** – Metadata normalization and standardization
- **F7** – Audit trails or time-based credential logs
- **F8** – Consent control, notification, or recipient verification tracking
- **F9 -** Issuer onboarding workflow with admin approval
- **F10 -** Credential issuer registration with searchable metadata

1.3.6 Modules and Functions

1.3.6.1 Credential Issuance Module

Module Description:

Allows authorized institutions to issue blockchain-based credentials with embedded cryptographic verification, replacing static documents with dynamic, tamper-evident records. Supports both formal and microcredentials with smart contract automation. Additionally, institutions can apply and register as authorized issuers, which is subject to approval by system administrators.

Users:

- U1: Credential holders (students, graduates, professionals)
- U2: Authorized issuers (schools, training centers, certifying bodies)

Problems Addressed:

- P1: Credential fraud and manipulation
- P3: Delay and inefficiency in credential issuance and verification
- P4: No expiration/version control
- P5: Exclusion of microcredentials and modular learning

Inputs:

- I1: Credential records or credential templates
- I2: User identity data

Features:

- F1: Immutable credential generation
- F2: Blockchain-anchored credential issuance
- F3: Smart contract-based creation and issuance
- F4: Real-time issuance notifications to recipients
- F5: Decentralized credential creation
- F8: Consent control for recipients
- F9: Issuer onboarding workflow with admin approval (new feature for registration process)

Opportunities:

- O1: Provide a tamper-proof solution for issuing credentials
- O3: Reduce processing time for issuing credentials

1.3.6.2 Credential Verification Module

Module Description:
Enables real-time, secure, and independent verification of credentials via the Hyperledger Fabric blockchain. QR-code-based verification and trust scoring increase transparency and security.

Users:

- U3: Verifiers (HR departments, licensing bodies, regulatory organizations)
- U4: Authorized users (e.g., hiring managers)

Problems Addressed:

- P2: Inefficiency and fraud due to manual verification
- P4: Lack of trust in credentials due to tampering risks
- P6: Delays due to lack of credential interoperability

Inputs:

- I1: Credential records or issued certificates
- I2: User identity data

Features:

- F2: Real-time access to credential verification
- F3: QR code-based lookup
- F4: Trust scoring system
- F5: Decentralized credential validation
- F7: Verification audit trail

Opportunities:

- O2: Enable streamlined, paperless credential verification
- O3: Enhance the verification process with blockchain transparency

1.3.6.3 Blockchain Integration Module

Module Description:

Manages the anchoring of credential data on Hyperledger Fabric, enforces access control, and handles all blockchain-related interactions to ensure tamper-evident and traceable credentials.

Users:

- U1: Credential holders
- U2: Authorized issuers
- U3: Verifiers
- U5: System admins

Problems Addressed:

- P1: Lack of transparency in credential verification
- P3: Risk of credential tampering and fraud

Inputs:

- I1: Credential records or templates
- I2: User identity data

Features:

- F1: Blockchain anchoring of credentials
- F3: Smart contract triggers for automatic updates
- F4: Blockchain access control for authorized participants
- F5: Immutable ledger recording

Opportunities:

- O1: Increase blockchain adoption in credentialing
- O3: Improve security and privacy by decentralizing data

1.3.6.4 Credential Access and Notification Module

Module Description:

Tracks credential activities and provides real-time notifications about access, expiration, revocation, and status changes. Keeps users and verifiers informed throughout the credential lifecycle.

Users:

- U1: Credential holders
- U3: Verifiers
- U5: System admins

Problems Addressed:

- P2: Difficulty in tracking credential status
- P4: Lack of awareness about expiration or revocation
- P7: No tamper-evident logs or version history

Inputs:

- I2: User identity and credential data

Features:

- F3: Real-time notifications
- F8: Verification tracking and consent control
- F5: Expiration/revocation alerts
- F7: Activity logs for credential interactions

Opportunities:

- O4: Enable proactive credential management
- O5: Simplify verification and renewal tracking

1.3.6.5 Certifying Body Onboarding and Sync Module

Module Description:
Handles both the onboarding and integration of certifying bodies into ChainCert. Institutions and organizations may apply for authorization as credential issuers through a secure application process. Once approved by system administrators, the module enables real-time syncing of

credential data, standardized metadata, and seamless upload of learning records between ChainCert and the authorized issuer's internal systems.

Users:

- U2: Certifying bodies (e.g., universities, training institutions)
- U5: System administrators (for issuer onboarding and vetting)

Problems Addressed:

- P5: Incomplete recognition of learning achievements
- P6: Lack of standardized and automated credential submissions
- P8: Absence of a unified registry of authorized issuers

Inputs:

- I1: Credential records and templates
- I2: Institution/organization identity data and supporting documents
- I3: Learning and experience logs
- I4: Issuer application forms and onboarding metadata

Features:

- F5: API integration for automated uploads
- F6: Metadata normalization
- F7: Real-time data syncing
- F8: Automated validation upon submission
- F9: Issuer onboarding workflow with admin approval
- F10: Credential issuer registry with searchable metadata

Opportunities:

- O1: Streamline credential uploading
- O3: Improve data accuracy and consistency
- O5: Proactive credential and issuer management

- O6: Secure and standardized onboarding of credential issuers

1.3.6.6 Employer/Regulator Interface Module

Module Description:
Provides a centralized dashboard for employers and regulators to verify single or bulk credentials securely. Enables trust-based filtering and secure record access with user consent.

Users:

- U3: Employers, regulatory bodies, licensing authorities
- U4: HR departments, hiring managers

Problems Addressed:

- P2: Slow and manual verification
- P4: Lack of a scalable verification system
- P6: No federated infrastructure for fast lookup

Inputs:

- I2: User identity data

Features:

- F2: Credential dashboard
- F4: Bulk or individual verification
- F5: Trust-based filtering
- F6: Secure record sharing options

Opportunities:

- O2: Improve hiring efficiency with real-time verification
- O3: Enhance accuracy using trust-scoring

1.3.6.7 Lifelong Learning & Micro-Credentials Module

Module Description:

Allows users to collect, track, and share non-degree achievements like CPD units, badges, and short courses across their careers. Promotes lifelong learning visibility and employment relevance.

Users:

- U1: Credential holders
- U2: Issuers of short courses/microcredentials
- U4: Career services departments

Problems Addressed:

- P5: Difficulty tracking non-traditional learning
- P6: Microcredentials excluded from formal records

Inputs:

- I1: Credential records, CPD units
- I3: Learning and experience logs

Features:

- F1: Tracking microcredentials and CPD
- F4: Career-long portfolio development
- F5: Employer integration
- F6: Real-time updates for milestones

Opportunities:

- O4: Promote continuous learning through a validated portfolio.

1.3.6.8 Smart Contract Automation Module

Module Description:

Automates expiration checks, renewal workflows, and revocation events using smart contracts. Keeps credential status updated in real time without manual intervention.

Users:

- U2: Authorized issuers
- U3: Verifiers

Problems Addressed:

- P4: Inefficient manual credential updates
- P7: No version history or update traceability

Inputs:

- I2: User identity and credential data

Features:

- F3: Automatic expiration handling
  F4: Revocation and renewal logic
- F7: Time-based updates and notifications

Opportunities:

- O2: Automate issuer and verifier workflows
- O3: Improve operational efficiency

1.3.7 Technologies to be Integrated

- Blockchain (Hyperledger Fabric): A permissioned blockchain for secure, decentralized credential issuance and verification. It ensures privacy, scalability, and tamper-proof records.
- Hashing Algorithms: Convert credentials into cryptographic hashes stored on the blockchain, ensuring data integrity and preventing tampering.

- W3C Verifiable Credentials Model: A standardized framework for issuing and validating credentials in a machine-readable format, integrated with Hyperledger Fabric for secure verification.
- Smart Contracts (Chaincode): Automates credential transactions like issuance, updates, and verification within Hyperledger Fabric, ensuring tamper-proof processes.
- React, Node.js & Express, CouchDB:
    - React: For building responsive user interfaces.
    - Node.js & Express: Handles backend logic and blockchain interactions.
    - CouchDB: base64/document itself
- QR Code Technology: Provides fast, secure credential verification by linking to the blockchain verification portal via QR codes.
- **PDF.js:**

  A browser-based PDF rendering engine used for securely displaying uploaded or issued credential documents within the system (e.g., diplomas, certificates).
- **PDF-lib:**

  A JavaScript library for programmatically modifying PDF files (e.g., inserting blockchain hashes, user data, and timestamps) before issuing or storing them.

1.3.8 Tools and Platforms to Be Used for Development

- Version Control: GitHub (for managing source code, version control, and team collaboration)
- Development Environment: Visual Studio Code (for writing, debugging, and testing code across frontend, backend, and smart contracts)
- Frontend: React.js, HTML, CSS, JavaScript (for building responsive and interactive user interfaces)
- Backend: Node.js with Express.js (for server-side logic, API handling, and blockchain integration)
- Blockchain SDK: Hyperledger Fabric SDK for Node.js (for managing credentials and invoking smart contracts)
- Containerization & Deployment: Docker and Docker Compose (for deploying Hyperledger Fabric components such as peers, orderers, CA, and CouchDB)
- Testing Tools: Postman (for testing and validating API endpoints); Hyperledger Fabric Test Network (for simulating blockchain operations and smart contracts)
- Database: CouchDB (for storing the actual credential (base64))

1.3.9 Integration with Existing Systems

- **Authorized Institutions' Information Systems:**
  ChainCert will integrate with existing records and credential management systems used by schools, universities, training centers, and certifying bodies. This enables authorized issuers to upload official credentials and templates directly into the blockchain-based system without manual re-entry. Data mapping, PDF credential import (via PDF-lib), and identity verification mechanisms will ensure secure onboarding of institutional records.
- **Employer and HR Recruitment Systems:**
  ChainCert provides a verification API that employers, HR departments, and regulators (e.g., PRC, POEA) can integrate into their recruitment or licensing systems. This allows them to verify credentials in real-time by scanning QR codes or accessing verifiable links sent by applicants, ensuring faster, fraud-resistant hiring and compliance processes.
- **Credential Holder Wallets and Cloud Storage Integration:**
  Credential holders can optionally upload credentials from personal cloud storage or external sources. ChainCert enables users to maintain a self-owned credential wallet,

supporting the upload of both issued and legacy credentials, with verification and hash generation handled automatically during onboarding.

- **Government and Regulatory Portals (Future Scope):**
  ChainCert is designed with modular APIs that could be extended to integrate with national ID systems (e.g., PhilSys), education databases (e.g., CHED or TESDA systems), or regulatory portals, supporting nationwide interoperability for credential access and audit.

1.3.10 Justification of Solution

The proposed blockchain solution, utilizing Hyperledger Fabric, overcomes the limitations of traditional PKI-based systems. Public Key Infrastructure (PKI) is susceptible to single points of failure, relying on trusted Certificate Authorities (CAs) that, if compromised, can invalidate the entire trust system. In contrast, blockchain offers a decentralized, immutable ledger where credentials are not only stored but also verified and auditable by multiple stakeholders in real-time. This eliminates the need for third-party intermediaries and ensures the integrity of credentials without relying on centralized authorities.

ChainCert ensures transparency, security, and scalability while aligning with data privacy regulations like the Philippine Data Privacy Act of 2012 (DPA). By using cryptographic hashes, it guarantees that credentials remain secure and verifiable without exposing Personally Identifiable Information (PII). Unlike traditional methods, which store only the digital signatures or hashes of credentials, blockchain provides a dynamic ledger capable of tracking the full lifecycle of credentials, from issuance to updates and revocations. As highlighted in Pulmano et al. (2023), blockchain's decentralized nature prevents tampering and enhances trust, especially when compared to centralized systems that are vulnerable to data breaches or manipulation (Pulmano et al., 2023).

Moreover, permissioned blockchains such as Hyperledger Fabric provide enhanced privacy, making them more suitable for credential management. Unlike public blockchains, where every transaction is visible to all participants, permissioned blockchains restrict access to trusted entities, ensuring that sensitive data remains secure. This selective access also helps maintain compliance with regulatory requirements, such as the Philippine Data Privacy Act of 2012 (DPA

2012), ensuring that only authorized participants can access or update academic credentials (Mitani & Otsuka, 2020).

The benefits of blockchain in credentialing have been demonstrated in various studies. For example, Pulmano et al. (2023) showed how permissioned blockchains can securely store and verify academic credentials while maintaining data privacy and compliance with regulations. Additionally, studies by Ding, Gao, Zhu, and Yuan (2020) and Shen et al. (2024) highlighted the effectiveness of blockchain in ensuring privacy, scalability, and compliance in complex, multi-party systems. These studies underscore the practicality of adopting Hyperledger Fabric for credential management, making it a robust solution for combating academic fraud, ensuring transparency, and enhancing the efficiency of credential verification processes.

In conclusion, Hyperledger Fabric's use in the ChainCert system offers a decentralized, tamper-proof solution to credential verification, addressing the flaws of traditional methods. The system's privacy features, real-time verification capabilities, and regulatory compliance make it an ideal choice for securely managing academic credentials and mitigating the risks associated with fraudulent academic qualifications.

1.4 Design Architecture

The ChainCert verification system leverages Hyperledger Fabric, a permissioned blockchain framework, to combat the growing issue of fraudulent academic credentials. The platform provides a secure, transparent, and efficient method for issuing, managing, and verifying academic and professional credentials. Authorized institutions and organizations can issue credentials via a secure dashboard, students can manage them through a digital wallet, and third parties can verify them in real time via a public portal, all while ensuring data integrity and user control. ChainCert implements an independent and fully customized system using Hyperledger Fabric, optimized for educational institutions and organizations within a trusted network. This section will discuss the architecture of the proposal, which include the system components, process workflows, and tools and technologies to be utilized

1.4.1 System Components:

a. Authorized Issuers (Schools, Companies, Organizations)

Authorized institutions serve as credential issuers in the system. Using a dedicated Issuer Dashboard, they can upload new credentials, update existing records (e.g., for corrections), or revoke erroneous or invalid credentials. Each institution is uniquely identified on the blockchain and operates under permissioned access to ensure authenticity.

b. Students (Credential Holders)

Students receive and manage their credentials through a secure, digital wallet interface. Each wallet is tied to the student's unique blockchain identity. From the wallet, users can view all their verified credentials, download them, share them via QR codes or secure links, and track their current status, whether active, revoked, or expired. This self-service functionality not only reduces institutional workload but also gives students full transparency and control over their records across their academic and professional careers.

c. Permissioned Blockchain Network

The backbone of the ChainCert platform is a Hyperledger Fabric permissioned blockchain network. Unlike public chains like Ethereum, Fabric allows fine-grained control over who can participate in the network, ensuring a secure and trusted environment. Each transaction—credential issuance, revocation, or update—is processed by the network's smart contracts (chaincode) and logged immutably to the distributed ledger. Only hashed data (credential fingerprint), issuer signatures, and related metadata are stored on-chain, while full documents remain off-chain. This reduces blockchain bloat while maintaining verification integrity.

d. Smart Contracts (Chaincode)

ChainCert uses custom-developed chaincodes to enforce the logic of credential operations. Chaincodes define how credentials are issued, updated, revoked, and verified. They also ensure that only authorized issuers can perform these operations and that records remain tamper-proof once written. This automation layer plays a critical role in

maintaining the platform's reliability, traceability, and adherence to defined policies without human intervention.

e. Data Privacy and Security

Privacy is a core principle in ChainCert's design. Only non-PII data such as credential hashes, timestamps, and issuer information are stored on-chain. Full credential documents are stored off-chain in secure repositories, accessible only by the student and authorized verifiers. This architecture balances the need for verifiability with strict privacy standards. Each credential is digitally signed, and any change—even a single character—would result in a different hash, instantly flagging tampering attempts during verification.

1.4.2 User Workflow and Process Flow

This section outlines the process flow, which includes credential issuance, credential verification, and credential lifecycle management processes, tackling the user roles interactions with the system.

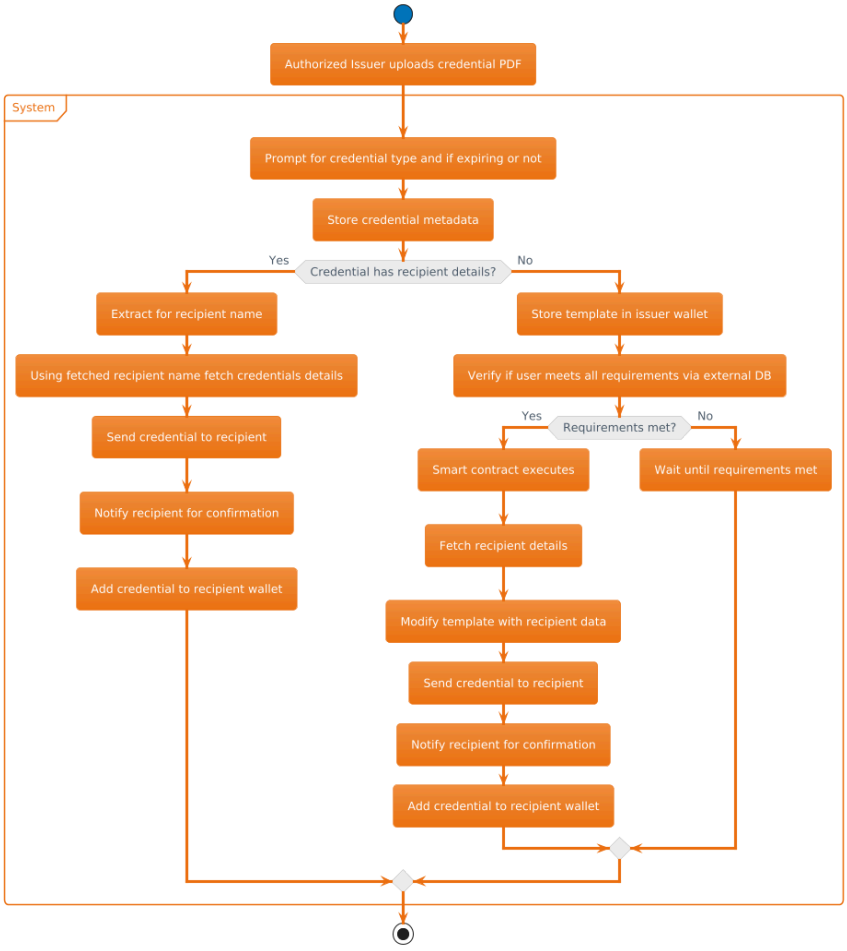
1.4.2.1 Credential Issuance Workflow

The Credential Issuance process begins when an Authorized Issuer, uploads a credential file (in PDF format). The file may either be a pre-designed template or a credential with recipient details already filled in. When a file (template or with recipient details) is uploaded, the system prompts the Issuer to select the type of credential (academic, professional, etc.) and determine if the credential has an expiration date. In the case of a credential that already includes recipient details, the system will ask the Issuer to input the recipient's name, and the platform will automatically fetch the recipient's data from the authorized Issuer's internal database. This process ensures that the recipient's information matches the details held in the database, ensuring accuracy.

After categorizing the uploaded file and verifying the recipient's details, the system automatically sends the credential to the recipient, provided that the recipient has a valid account on the system. The recipient is notified of the credential issuance and

must confirm receipt before the credential is added to their digital wallet. In the case where the uploaded credential was just a template, the system stores it in the Issuer's wallet for future use. The Issuers can issue it to a User once that user has satisfied the required criteria, such as completing a course or meeting professional requirements.

The system can integrate with external databases, such as the PRC system for license renewals, to verify the recipient's eligibility for receiving credentials. For example, once a recipient has completed the required Continuing Professional Development units for a PRC license renewal, the Smart Contract is triggered, fetching the recipient's details from the PRC system to validate eligibility, and then the credential is issued automatically. The recipient is notified once the credential is issued to their wallet.



1.4.2.1 Credential Issuance Workflow Diagram

1.4.2.2 Credential Management by Credential Holders

Students receive their credentials in a secure digital wallet interface, where each is associated with the user's blockchain identity, allowing them to view, download, and share credentials with customized access control by the user. Sharing is enabled through secure links or QR codes. Students can monitor their credentials and their status, whether still valid or revoked.

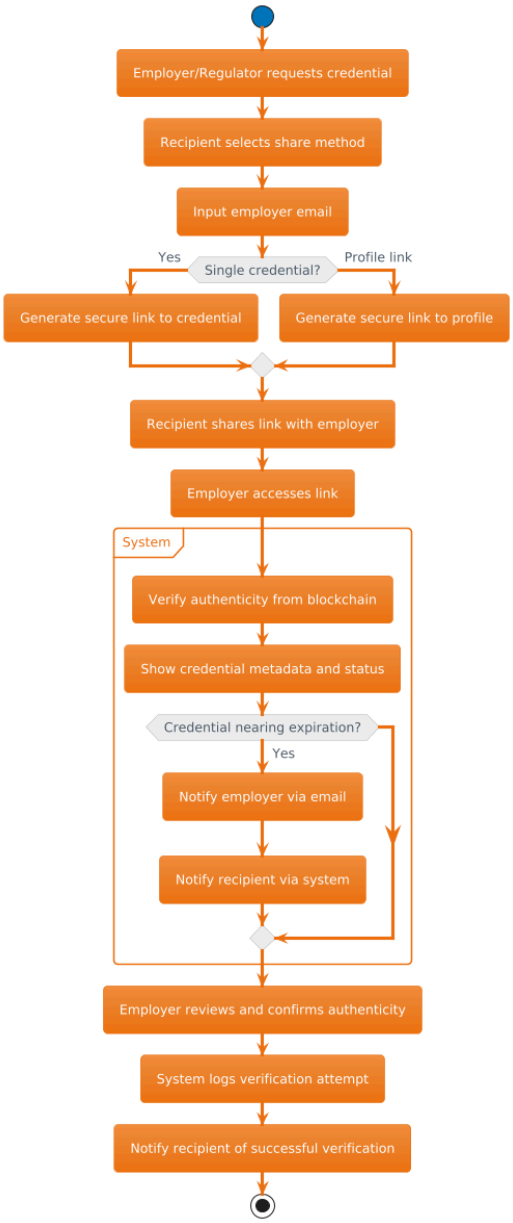
1.4.2.3 Credential Verification Workflow

For Credential Verification, the Employee/Regulatory Body who needs to verify the Recipient's credential can initiate the process by requesting the User to share a verification link. The Recipient has two options: they can either share a single credential link or a link to their entire profile, where all their credentials are listed, both active and expired. If the Employer chooses to verify a single credential, the Recipient simply shares the direct link to that specific credential, and the Employer can instantly verify its authenticity. Moreover, in compliance with data protection and privacy, links to be shared are customized such that specific emails are authorized to view and access the credential associated with the link sent.

Once the Employer/Regulatory Body receives the verification link, they can access the credential details and view important metadata such as the credential type, issuer, expiration date, and any historical updates or renewals. The system fetches real-time data from the blockchain, ensuring that the credential hasn’t been tampered with and that all details are authentic. If the Employer/Regulatory Body views the User’s profile link, they will also have access to the full credential history of allowed credentials to be seen by the user, allowing them to see how many times a certificate has been renewed and whether it is still valid or nearing expiration.

Moreover, the system tracks credential updates and expiration dates. If the credential is expiring soon, both the Employer/Regulatory Body and the Recipient are notified, ensuring that neither party misses important updates. The Employer/Regulatory Body are notified via email of the updates regarding the credentials shared with them. On the other hand, the recipient is notified of upcoming expirations or any updates via the

system itself. This process is streamlined, reducing the time it takes to verify credentials and ensuring the authenticity of every credential that is shared. Moreover, the system logs each verification attempt, allowing the Recipient to review who has accessed their credential, and adding a layer of transparency and accountability to the verification process. Once the Employer completes the verification, the process ends with confirmation of authenticity, and the Recipient is notified that their credentials have been verified.



1.4.2.3 Credential Verification Workflow Diagram

Technological Integration

The ChainCert system is powered by a permissioned blockchain network using Hyperledger Fabric, which serves as the foundation for secure, decentralized, and transparent academic credential management. Hyperledger Fabric was chosen for its enterprise-grade features such as modular architecture, fine-grained access control, and high throughput, making it ideal for inter-institutional credential issuance and verification.

ChainCode (Smart Contract logic)

Custom-built chaincodes (Hyperledger's smart contracts) define the business logic for credential issuance, revocation, and validation. These contracts ensure that only authorized issuers can write to the ledger and all credential actions are tracked in an immutable audit log. While automation is already embedded via chaincodes, future iterations may integrate more advanced smart contract logic to support conditional access, expiration policies, and multi-institutional workflows.

1.5 Study Objectives

1.5.1 General Objective

To develop a permissioned blockchain-based credential management system using Hyperledger Fabric that empowers students to securely store and manage their credentials, while enabling trusted institutions and organizations to issue and verifiers to validate these credentials in a scalable, tamper-proof, and privacy-conscious environment.

1.5.2 Specific Objectives

1.5.2.1 Development of a Hyperledger Fabric-Based Credential System

- *Explanation*: Build a permissioned blockchain network using Hyperledger Fabric that allows authorized institutions and organizations to issue credentials to users. The actual credential files (e.g., diploma PDFs) are encoded (e.g., base64) and

stored in CouchDB, Fabric's world state database, ensuring tamper-resistance and traceability.

- *Measure of Success*: Successful deployment of a functioning blockchain network (with multiple organizations, peers, and smart contracts) supporting credential issuance, on-chain referencing, and CouchDB-based storage and retrieval of actual documents.

1.5.2.2 Integration of Issuer Web Dashboard and Verifier Portal

- *Explanation*: Develop a web-based interface that allows authorized institutions and organizations to issue, update, and revoke actual credential documents; users to access and manage their credentials through a user-friendly digital wallet interface; and employers and verifiers to view and validate credentials by querying the blockchain and retrieving documents from CouchDB.
  Measure of Success: Fully operational web dashboard connected to the Fabric network and CouchDB, supporting secure document issuance and verification workflows end-to-end.

1.5.2.3 Implementation of Credential Lifecycle Features

- *Explanation*: Ensure that the system supports the full credential lifecycle, including issuance, revocation, renewal, and reissuance. Credential state changes (e.g., revoked, expired) are updated in the chaincode logic and reflected in CouchDB.
- *Measure of Success*: State transitions for credentials (e.g., valid → revoked) are accurately recorded and verifiable; previous versions remain auditable within the system.

1.5.2.4 Privacy and Access Control Management

- *Explanation*: Protect sensitive data by encrypting credential documents before storing them in CouchDB. Implement Attribute-Based Access Control (ABAC) via smart contract logic to ensure only authorized institutions can issue

credentials, and only designated users can access them. Personal data exposure is minimized in line with the Data Privacy Act of 2012 (DPA).

- *Measure of Success*: Credentials are securely stored in encrypted form; document access is controlled through smart contracts; compliance with DPA 2012 is demonstrated by limited and permissioned data visibility.

1.5.2.5 Wallet Creation and Management for Credential Ownership

- *Explanation*: Develop a digital wallet interface for users to view, download, and share their credentials. The wallet retrieves documents directly from CouchDB and reflects any changes to credential state in real time.
- *Measure of Success*: Users can securely access and manage their credentials; the wallet interface successfully reflects document status (e.g., revoked, reissued).

1.5.2.6 System Testing and Performance Optimization

- *Explanation*: Test the system for performance and reliability when storing, retrieving, and verifying credential documents in CouchDB through smart contract interactions. Simulate real-world usage scenarios to assess speed, stability, and correctness.
- *Measure of Success*: System handles credential workflows under simulated load with minimal latency and errors; document retrieval and verification are consistently successful.

1.5.2.7 User Feedback Collection and Continuous Improvement

- *Explanation*: Collect feedback from users, institutions, and employers to ensure the platform addresses real-world credentialing needs and usability concerns. Use insights to refine features and improve system relevance.
- *Measure of Success*: Feedback loop is established and maintained; refinements are implemented based on user suggestions; high stakeholder satisfaction is achieved.

1.6 Significance of the Study

This study proposes a blockchain-based credential management system using Hyperledger Fabric, a permissioned blockchain framework that ensures secure, efficient, and privacy-conscious handling of

actual academic documents. Unlike traditional systems that store only cryptographic hashes or rely on external storage, this system stores the actual credential files directly in CouchDB, the state database of Hyperledger Fabric, encoded and securely linked through smart contracts. Authorized academic institutions can issue or revoke credentials, while users retain full control and ownership via a digital wallet interface. Employers and verifiers can access real-time credential validation directly through the system, eliminating delays and reducing fraud. With full support for credential lifecycle events (issuance, update, and revocation), encrypted storage, and real-time verification, the system offers practical, scalable, and tamper-resistant benefits to users, schools, employers, and government agencies.

1.6.1 Theoretical Contributions

- Exploring Capabilities of Blockchain Technology

  This study contributes to blockchain research by demonstrating how Hyperledger Fabric can be used not only for tracking hashes or references but also for storing actual digital documents directly in its world state. It highlights the potential of permissioned distributed ledger systems to handle sensitive data with fine-grained access control, rich data querying (via CouchDB), and robust chaincode logic for managing document state transitions. The study also enhances understanding of how enterprise-grade blockchain platforms can support real-world applications, such as credential lifecycle management, while remaining compliant with privacy laws like the Data Privacy Act of 2012.

1.6.2 Practical Applications

- Combating Fake Credentials

  By storing the actual document data directly in CouchDB (e.g., a base64-encoded diploma), the system makes it virtually impossible to forge or tamper with academic credentials. Only verified institutions can create or update credentials, and all changes are logged securely through smart contracts. This ensures that every credential in the system is trustworthy and traceable, helping combat widespread issues with fake diplomas, certificates, and unverified online course completions.

- Improving Verification Efficiency

  The system streamlines the traditionally slow and manual verification process by allowing verifiers (e.g., employers, government bodies) to directly access credential data stored in CouchDB through a blockchain-backed web portal. Real-time access means

there's no need for back-and-forth emails or institutional delays. Unlike systems that rely on external file storage or blockchain hashes, this platform allows immediate access to the full credential document while ensuring security and authenticity.

- Supporting Credential Portability and Global Accessibility

  With credentials stored directly in a permissioned blockchain system, users can access and present their verified records anytime, anywhere—via a secure digital wallet. Whether applying for jobs abroad, continuing studies, or pursuing certifications, users have direct access to their records without needing to contact their alma mater. The system supports credential reissuance, revocation, and updates, ensuring that all records remain current and portable.

1.6.3 Beneficiaries

- Authorized institutions

  Authorized institutions benefit by issuing tamper-proof digital credentials, reducing the administrative burden of manual verification, and maintaining control over the full lifecycle of each document. They can confidently revoke, update, or reissue credentials without fear of misuse.

- Online Learning Platforms and Certifications Providers

  MOOC platforms and skills-based certification providers can issue verifiable, blockchain-backed documents that are recognized in the same ecosystem as traditional academic records. This increases trust in non-traditional learning methods and supports learners in building a unified credential portfolio.

- Employers and Recruitment Agencies

  Organizations can instantly verify academic credentials without delays or intermediaries. Since documents are stored and verified within a trusted system, HR departments reduce the risk of hiring candidates with fake qualifications and save time and money on background checks.

- Government and Regulatory Agencies

Agencies like CHED, DICT, and TESDA can use the system as a model for national digital credential registries, improving policy enforcement and standardizing digital qualifications across institutions and industries.

- Future Researchers and Developers

  This system provides a practical foundation for students, developers, and researchers exploring blockchain applications in fields like education, legal records, healthcare, identity management, and more. It serves as a blueprint for secure, real-world use cases where full document integrity and lifecycle tracking are essential.

1.7 Scope and Limitations

This section outlines the scope and limitations of the ChainCert project to ensure that the system's scope is deliberately limited to essential features.

1.7.1 Users

Authorized Institutions and Credentialing Bodies

These include schools, universities, and other authorized certifying organizations. They are responsible for issuing and uploading either the credential template or actual credential documents to the blockchain system through the Admin Dashboard.

Users

Credential holders receive issued documents through a blockchain-linked digital wallet. They have full ownership of their credentials and can control access, sharing, and retrieval securely through the system interface. They can also add credentials from non-authorized issuers but will be categorized in their wallet as "Unverified."

Verifiers

Entities such as HR recruiters, government agencies, and other institutions can verify credentials by accessing the link sent by the users.

System Administrators and Anonymous Verifiers

In addition to the main user groups, the system includes a designated system administrator role. Administrators are responsible for managing user access levels, reviewing and approving new issuer applications, and ensuring system integrity. Moreover, anonymous verifiers, such as potential employers without registered accounts, may still perform credential verification via public QR code scanning. Their access is limited to the document hash, issuer information, and credential status (e.g., verified, unverified, or revoked).

1.7.2 Features and Functions

Credential Issuance

Authorized issuers upload either the credential template or the credential itself (with recipient details) through the Admin Dashboard. If a credential template was uploaded, the system encodes the **actual document** and submits it to the blockchain via smart contract. The document is stored in **CouchDB (world state)** and anchored via the blockchain ledger, ensuring it cannot be tampered with or forged.

For users without issuer privileges (e.g., individuals attempting to upload their own credentials), the system will still accept the credential but will tag it as "Unverified." The system does not validate or authenticate self-issued or unlinked credentials. Only documents from registered issuers are recognized as "Verified" within the platform.

Wallet Management

Each user has a secure digital wallet that links to their credentials stored on the blockchain. These wallets are accessible upon login and enable users to view, download, and share their verified records while retaining full control. Each credential access or sharing activity is logged within the system to maintain a verifiable audit trail and support traceability.

Credential Verification

Employers and verifiers use the Verifier Portal to validate credentials by uploading the actual file or scanning a QR code. The system retrieves the original document from CouchDB and performs:

- Document Integrity Check

- Issuer Verification
- Credential Status Check (valid, and revoked)

Lifecycle Support (Updates and Revocations)

The system supports full credential lifecycle events, such as name changes, updates, or revocations due to fraud or clerical errors. These actions are handled via smart contracts and stored immutably on the blockchain, with updated records replacing the previous version in CouchDB.

Interoperability and Export

Credentials can be exported in standardized formats (e.g., PDF or JSON) and validated even outside the platform by referencing blockchain records. This ensures portability and long-term usability of credentials in global or offline settings.

1.7.3 Technology

Blockchain Infrastructure: Hyperledger Fabric

Hyperledger Fabric is used as the core permissioned blockchain framework. It supports fine-grained access control, chaincode (smart contract) logic, and a modular architecture that ensures scalability, security, and trust among participating institutions without the gas fees or anonymity of public chains.

CouchDB: Credential Storage

Actual credential files are stored directly in **CouchDB**, Fabric's world state database, as base64-encoded data. This allows for fast retrieval and stateful credential management while ensuring that each change is validated and committed via the blockchain ledger.

As of now, the system only supports PDF files as the accepted format for credential uploads, whether a full document or a template. Other formats such as Word documents, image files (e.g., PNG, JPEG), or structured formats (e.g., JSON-LD) are not supported at this stage. This constraint ensures consistent encoding and compatibility with CouchDB but may limit adoption for institutions using diverse credential formats

1.7.4 Credentials

1.7.4.1 Non-Expiring Credentials

1) Academic Credentials
    - Ex: Bachelor's Degree, Master's Degree, Doctorate (PhD), Diploma
2) Award and Recognition Credentials
    - Employee of the month/year, Industry Awards, etc.
3) Career Milestones and Experience Credentials
    - Years of experience in a field or role
4) Micro-credentials
    - Ex: Online Course Completion Badges (Coursera, Cisco, etc.)

1.7.4.2 Expiring Credentials

1) Professional Certifications
    - Ex: PRC Licenses, AWS Certification, PMP, CISSP, etc.
2) Micro-credentials
    - Ex: Online Course Completion Badges (Coursera, Cisco, etc.)
3) Industry and Trade Credentials
    - Ex: Trade Certifications (Electricians, Plumbers, etc.), Safety Certifications (Forklift Operators, First Aid), etc.
4) Continuing Education and Lifelong Learning Credentials
    - Ex: Continuing Medical Education, Professional Development Courses

Chapter 2

2.1 Review of Related Concepts

2.1.1 Prevalence, Causes, and Impact of Fake Academic Credentials

In today's digital world, academic credential verification has become increasingly crucial for educational institutions and employers, as the rise of digital technologies and globalization has made it easier to create convincing fake diplomas. According to Attewell and Domina (2010), most individuals who claim to have utilized fake degrees have actually attended the institution in question and completed most of the credits but failed to finish all degree requirements. These

individuals are further encouraged to fake their credentials, as diploma mills or fraudulent institutions are now producing certificates that are nearly indistinguishable from legitimate ones, complicating verification efforts. This phenomenon has escalated, especially with the increased ease of creating fake institutions on the internet, where fraudulent schools can mimic legitimate ones with professional-looking websites and certifications that resemble real diplomas (Hallak & Poisson, 2007). Moreover, the global demand for higher education has created increased competition among students and institutions and is further exacerbated by corruption in accreditation and quality assurance mechanisms. Many higher education systems, especially in developing regions, face challenges with inconsistent accreditation processes, where bribes and favoritism can influence the legitimacy of institutions and their credentials. This lack of effective accreditation procedures creates a breeding ground for fraud, leading to widespread issuance of fake diplomas and undermining trust in educational systems (Johnson, 2006). These reasons, unfortunately, have given rise to more opportunities for academic fraud. Overall, these fraudulent documents are often created with the intent to deceive employers, leading to delays in hiring, wasted resources, and a loss of trust between educational institutions and employers.

A notable example of fraudulent practice is the case of Marilee Jones, a former dean of admissions at the Massachusetts Institute of Technology (MIT). Despite being a respected figure in higher education, advocating for a less stressful and more honest college admissions process, it was revealed that she was falsifying her academic credentials for nearly three decades, which is alarming, as this revelation was made possible because she admitted to it. These falsified credentials were degrees from Albany Medical College, Union College, and Rensselaer Polytechnic Institute. But investigations revealed that neither Albany Medical College nor Union College had any record of her attendance, and her enrollment at Rensselaer Polytechnic Institute was limited to part-time studies (Hoover & Millman, 2007). In addition, in January 2012, Scott Thompson became CEO of Yahoo, replacing Tim Morse. However, in May 2012, Daniel Loeb, an activist shareholder, revealed that Thompson falsely claimed to have a degree in Computer Science from Stonehill College, in addition to his degree in accounting. This revelation was confirmed by the Third Point's investigation, as well as Stonehill College, leading to Thompson's resignation (Lückerath-Rovers, 2024). This high-profile case highlights the significant flaws in traditional academic credential methods, as it took a significant period of time for the false claims to be revealed.

Moreover, according to reports, many individuals, particularly in developing regions like Nigeria, resort to using fake academic credentials to secure job opportunities, bypassing qualifications and overall hiring criteria. This has resulted in substantial losses for companies and a misalignment between education and workforce readiness (Jimu, 2018). This issue is not isolated to one region, as demonstrated by a 2025 operation by the Manila Police District (MPD), which led to the arrest of Mario Acantilado for producing fake diplomas. During the entrapment operation, counterfeit diplomas and receipts were recovered. (Calucin, 2025) In addition to these local cases, studies have shown the global scale of academic fraud. For instance, in Brazil, a large-scale investigation uncovered thousands of fake adult high school diplomas sold to individuals who had never completed their education. These fake diplomas resulted in wage gains of 5% to 8% for individuals compared to those without high school diplomas. This suggests that employers may perceive diplomas as a basis for productivity, even when they do not reflect actual skills or education, further complicating the recruitment process (Firmo, 2021). These examples of fake diplomas or academic fraud incidents highlight the need for more robust verification.

The recent case involving two Filipina nationals sheds light on vulnerabilities in Singapore's employment pass system. Both the women submitted forged university certificates, one from De La Salle University and the other from Ateneo de Manila University in order to secure positions at CL Enterprise, a retail company operating Mr. Fix outlets. Their deception remained undetected for over a year, until a random inspection by the Ministry of Manpower (MOM) uncovered the falsified documents. The case highlights the significance of thorough document verification, as false educational qualifications and inflated salary declarations were used to meet the employment pass criteria.

The involvement of an unlicensed agent who instructed the women to falsify their applications and charged one of them $4,500 further emphasizes the risk of underground networks exploiting job seekers. MOM's prosecuting officer, Teo Sijing, stressed that these fraudulent actions deprived other qualified individuals of job opportunities and undermined the credibility of the system. This incident underscores the importance of upholding the integrity of employment systems to prevent illegal employment and protect fair access to job opportunities (Tan, 2025)

2.1.2 Legal Implications and Responsibilities in Credential Verification

In the context of credential verification, particularly within the Philippines, we are governed by a complex framework of laws that emphasize the responsibility of employers to verify the authenticity of the qualifications of their prospective employees. These laws ensure the integrity of the hiring process and protect against the growing concern of credential fraud. Specifically, employers are mandated to exercise due diligence when hiring employees, especially in verifying the authenticity of credentials. A failure to do so can result in significant legal, financial, and reputational risks. These are brought about by the following laws:

2.1.2.1 Labor Code of the Philippines | Republic Act No. 6715

Employers are required by law to ensure that all credentials presented by employees are authentic. The labor code mandates employers to exercise due diligence during the hiring process, which includes the verification of non-educational and educational qualifications. Due diligence is technical requirements, which implies that employees must take reasonable steps to verify the accuracy of the credentials presented to them. This includes checking in with educational institutions, licensing boards, or using third-party verification services. Negligence in this regard can result in "negligent hiring" claims, which is particularly pertinent if the employee causes harm or fraud within the organization. (R.A. 6715, n.d.)

2.1.2.2 PRC Modernization Act | Republic Act No. 8981

The professions regulated by the Professional Regulation Commission (PRC), employers must and are included to verify the authenticity of professional licenses before hiring, as the PRC Modernization Act emphasizes on ensuring that only qualified individuals who have passed the standardized licensure exams, are employed in regulated professions such as healthcare, engineering, law, and accountancy. Employers who fail to do so may face sanctions from regulatory bodies, such as fines, revocation of accreditation, and even potential for criminal charges, should the unverified professional be involved in fraudulent activities. (R.A. 8981, n.d.)

2.1.2.3 Revised Penal Code | Republic Act No. 3815

The Revised Penal Code plays a critical role in the regulation of fraudulent credentials, as through provisions on the "Falsification of Documents", employers can be held criminally liable for hiring individuals with falsified educational qualifications or

professional licenses. Employers who are found guilty of this offense could face severe penalties including imprisonment, depending on the gravity of the fraud. This includes a penalty for the person who produced and used the forged documents for the purpose of employment or any other fraudulent activities. (Bautista, n.d.)

2.1.3 Existing Anti-Fraud Measures of Educational Institutions and its Challenges

With respect to these flaws of employer's existing verification methods, educational institutions' use of anti-fraud or anti-forgery methods are no different, despite their intention to safeguard integrity of academic credentials, they still have notable limitations. Anti-forgery techniques in the physical aspect involve the use of watermarks and holograms, the chemical aspect utilizes special substances, and the optical aspect incorporates visual elements like special printing or optical effects, all of which are not foolproof and can be counterfeited by advanced forgery techniques. An example of which are digital signatures, which verify the authenticity of a diploma. However, this method also comes with its own challenges, including the need for long signature codes that are stored in chips, which can be expensive and have a limited lifespan of around 10 years. Lastly, unique identification systems, which assign distinct ID numbers to each student, help in verification, but in the event that databases are compromised, insertion of false qualifications will be difficult to detect. Overall, these gaps in existing anti-forgery methods prove to show that there is a need for a more secure means to combat fraudulent academic records.

As proof of such, in a case in the United States, over 7,600 fake nursing degrees were issued by South Florida nursing schools, with approximately 2,800 recipients passing the national exams. This not only exposes the vulnerabilities in the credential verification system ,but also poses risks to public safety, particularly in healthcare where qualified professionals are critical. This scandal led to legal actions and raised alarms about the dangers of fraudulent degrees, most especially in high-stakes professions like healthcare. Similarly, a fake degree syndicate in Malaysia exploited social media platforms to sell counterfeit degrees from reputable universities, utilizing advanced printing technologies to produce convincing replicas. This fraudulent operation harmed the reputation of legitimate institutions, creating challenges for employers trying to differentiate between genuine and fake academic credentials. (Putri, 2024)

These high-profile cases underscore the urgent need for educational institutions and employers to implement more secure and efficient anti-fraud measures. The increasing operations of fake degrees highlights the current methods which are watermarks, holograms, and digital signatures, which are all insufficient in preventing fraud. Educational institutions must adapt to these challenges by integrating digital verification technologies, such as blockchain, to ensure the authenticity of academic credentials. Overall, relying on outdated verification methods not only jeopardizes the credibility of institutions, but also compromises public safety and trust.

2.1.4 Blockchain Technology and Its Role in Academic Verification

Blockchain technology serves as the foundational architecture for secure digital credentialing systems, offering a decentralized, cryptographically enforced, and tamper-resistant ledger for storing and verifying academic records. Unlike a traditional centralized database, where a single point of failure can compromise data integrity and availability, blockchain operates on a distributed ledger system maintained across a network of nodes. Each credential issuance is treated as a discrete transaction, recorded in a block that includes a timestamp, the issuing institution’s digital signature, and a unique cryptographic hash representing the credentials' content. These blocks are changed chronologically, with each new block referencing the hash of the previous one, creating an immutable audit trail that cannot be retroactively altered without recalculating and modifying all subsequent blocks, making it computationally infeasible on public blockchains due to consensus constraints.

Consequently, the use of consensus mechanisms, such as Proof of Work (PoW), Proof of Stake (PoS), or Byzantine Fault Tolerance (BFT), ensures that only valid transactions, in this case, verifiable credentials issuance authorized by legitimate institutions, are appended to the chain. This decentralized validation process eliminates the need for a central verification authority, significantly reducing the risk of data manipulation, unauthorized access, and administrative errors. Moreover, blockchain's reliance on public-key cryptography allows for secure issuance and verification as institutions sign credentials using their private keys, and verifiers, such as employers, can authenticate the legitimacy of these credentials using the corresponding public keys without the need to contact the educational institution directly.

To showcase blockchain capabilities, existing blockchain-based academic verification systems, such as OpenCerts, have demonstrated the power of blockchain in solving the challenges of academic credential management. OpenCerts, for instance, allows educational institutions to issue certificates that are cryptographically secured and easily verifiable. Each certificate is associated with a unique cryptographic proof, stored on the blockchain, that guarantees its authenticity. (Pandya, 2022) These systems not only enhance security and efficiency but also empower students and employers by providing a simple, reliable way to confirm academic credentials. These capabilities of blockchain prove to show its compatibility with combating academic fraud, as it is in alignment with legal frameworks and professional standards outlined in judicial precedents, such as the Philippine Supreme Court's emphasis on the authenticity and accuracy of credentials in formal employment (LAWPHIL, 2010). Moreover, the use of blockchain in these systems showcases its inherent characteristics, such as immutability, decentralization, and transparency, which allow for the creation of tamper-proof academic records. Once a document, such as a diploma or certificate, is stored on the blockchain, it cannot be altered or forged, ensuring the authenticity of the academic credentials.

2.1.5 Barriers to Adopting Modern Technology for Academic Credential Verification and Anti-Fraud Measures

The integration of modern technologies such as blockchain offers significant potential in the verification of academic credentials for employers and anti-fraud measures for educational institutions. However, institutions face substantial barriers in adopting such technologies, one of which is institutional resistance to change. Many institutions are reluctant to shift from traditional methods, as the adoption of modern technology like blockchain requires a fundamental change in organizational culture and workflows. In addition, there are technical barriers associated with implementing modern technology in educational institutions, as it is complex and requires specialized skills. Moreover, they also do not have resources to develop or maintain these systems, as blockchain technology requires significant technical expertise and infrastructure. Another barrier worth noting is the financial burden that comes with the implementation, as educational institutions struggle to justify the high upfront costs of developing and deploying modern technology systems, particularly when the benefits of such systems are not deemed necessary, most often due to lack of knowledge and intimidation of its complexities. Institutions may find it challenging to allocate resources for such solutions, especially for those whose resources are already stretched to thin. (Rustemi et al., 2023)

Along with these financial constraints, concerns about security and privacy also play a crucial role in hindering modern technologies such as blockchain. Despite its reputation for secure data handling, blockchain's nature of immutability raises concerns regarding data privacy laws, especially in jurisdictions with stringent data protection regulations. Even though the demand for blockchain professionals is increasing, educational institutions often lack the expertise to implement and maintain such systems. The shortage of skilled blockchain professionals hampers the ability of institutions to adapt to blockchain. In alignment with this, adoption challenges are also due to lack of awareness and trust in modern technology as most administrators are not familiar with the complexities of blockchain, and this understanding leads to skepticism about its potential application. (Rustemi et al., 2023)

Despite these barriers, real-world examples of blockchain implementation in education suggests that overcoming these challenges is possible. An example of which, Nicosia University and Sony Global Education have successfully experimented with blockchain technology issuing and verifying digital certificates. With this, blockchain continues to evolve and prove its potential in other sectors, its adoption in education, particularly in issuing and verification of academic credentials. (Rustemi et al., 2023)

2.1.6 Blockchain Technology vs Other Modern Technology Solutions

In recent years, numerous modern technologies have been explored to enhance the security, efficiency, and transparency of various processes, including academic credential verification. However, blockchain technology stands out due to its unique characteristics that address the weaknesses of traditional solutions.

2.1.6.1 Blockchain vs. Centralized Databases

Centralized databases have traditionally been used for academic credential management, where a central authority, such as a university or credentialing agency, manages and verifies records. While these systems are widely adopted, they come with notable limitations. One major issue is their vulnerability to hacking and data breaches. Centralized databases are attractive targets for cybercriminals, and a single breach can compromise all the data stored within the system, exposing sensitive academic records. High-profile data breaches, such as the Target breach or the Equifax data leak,

demonstrate the risks associated with centralized systems (Zhao et al., 2020). Another limitation is the concept of a single point of failure. If the central authority is compromised, all records within the system are at risk. Additionally, there is potential for administrative errors, which could lead to incorrect credentialing or delays in verification.

In contrast, blockchain technology offers several advantages. One of the key benefits is decentralization. Blockchain eliminates the single point of failure by distributing data across a network of nodes, ensuring that no single entity can alter or compromise the integrity of the data without consensus from the network (Crosby et al., 2016). Furthermore, blockchain's immutability ensures that once data is entered, it becomes permanent and cannot be modified, making fraudulent changes impossible. This is a key feature that traditional technologies like digital signatures, QR codes, or encrypted databases cannot match (Narayanan et al., 2016).

In summary, blockchain offers superior security, transparency, and decentralization relative to centralized databases, rendering it a more dependable solution for the management and verification of academic credentials.

2.1.6.2 Blockchain vs. Digital Signatures

Digital signatures are a prevalent tool for authenticating digital documents, particularly academic credentials. They act as a safeguard to make sure that a signed document or record hasn't been altered. Digital signatures, however, are mostly used for document verification and do not provide a transparent or impenetrable system for confirming the full credentialing process. Additionally, there is a chance of fabrication or poor management because digital signatures depend on a centralized certificate authority to sign the papers. Digital signatures can verify that a document came from a certain source, but they are unable to stop counterfeit papers from being created or ensure the integrity of the issuing institution's entire procedure.

Blockchain technology offers a more complete solution, which overcomes these constraints. In addition to using digital signatures to confirm a document's legitimacy, blockchain also keeps track of the academic credential's complete history on a decentralized, unchangeable ledger (Narayanan et al., 2016). The system is extremely

resistant to fraud because of this tamper-proof ledger, which makes sure that no data can be changed once it has been recorded. Additionally, the open nature of blockchain ensures that the entire verification process is decentralized and trustless by enabling all pertinent stakeholders—employers, universities, and students—to independently verify credentials in real time.

In summary, blockchain offers a more reliable, transparent, and impenetrable system that guarantees the security and integrity of the complete credentialing process, even though digital signatures are helpful for verifying individual papers.

2.1.6.3 Blockchain vs. Public Key Infrastructure (PKI)

Public Key Infrastructure (PKI) is a cryptographic framework that enables entities to sign and verify digital documents using asymmetric key pairs. It is effective in proving the authenticity and integrity of documents through digital signatures. However, PKI systems are typically centralized, relying on Certificate Authorities (CAs) to issue and manage keys. This introduces a single point of failure: if a CA is compromised or revoked, all signed credentials may lose their validity (Ristenpart et al., 2017) Furthermore, PKI provides no native mechanism for public audit trails or decentralized consensus. It is not well-suited for multi-institutional or longitudinal academic credentialing, where credentials may be issued, revoked, and re-issued over time as part of modular or lifelong learning frameworks.

In contrast, blockchain technology provides a distributed trust model, eliminating dependence on any one authority. All credential transactions are recorded on a public ledger, creating an immutable and auditable trail. Blockchain supports real-time, trustless verification, meaning employers of other verifiers can confirm authenticity without contacting the issuing institution. This improves verification efficiency and removes latency caused by slow institutional response times. While both technologies use cryptographic signatures, blockchain systems like Ethereum also support smart contracts, enabling automated issuance, revocation, or conditional disclosure of credentials, which are features not natively supported in PKI.

Therefore, while PKI is a strong tool for authenticating static documents, blockchain is better suited for complex, multi-party, evolving ecosystems like academic credentialing.

2.1.6.4 Blockchain vs. Traditional Paper-Based Certificates

Academic credentials have always been issued using traditional paper-based certificates, however these have a number of serious disadvantages. First, paper certificates are highly susceptible to forgery and alteration, especially when institutions do not implement sophisticated anti-forgery measures like holograms or watermarks. Because of this, it is comparatively simple for dishonest people to copy or alter credentials, which could put businesses and educational institutions at risk (Ristenpart et al., 2017). Furthermore, the paper certificate verification process is laborious and ineffective. Each document must be manually reviewed by the institution, which can lead to inefficiencies and delays in the hiring process. This is particularly true when verification necessitates getting in touch with the issuing institution, which further prolongs the schedule (Zhao et al., 2020). Paper certificates are also expensive and logistically difficult to distribute and store physically. Educational institutions are often required to maintain physical storage systems for large volumes of certificates, which becomes particularly problematic when managing alumni requests or document retrieval. The process also incurs significant printing, shipping, and handling costs associated with distributing physical certificates (Pandya, 2022).

Blockchain technology offers distinct advantages over traditional paper-based certificates. With blockchain, academic credentials are converted into digital credentials that are stored in a tamper-proof decentralized ledger. This ensures that once a credential is recorded, it cannot be altered or forged, addressing the vulnerabilities of paper certificates. Blockchain's immutability and decentralized nature ensure that credentials can be verified in real-time, streamlining the process and saving time and resources compared to manual paper-based verification (Crosby et al., 2016). Blockchain's security features also prevent fraud in ways that paper certificates cannot, as the data recorded on the blockchain is protected by cryptographic algorithms that guarantee its authenticity and integrity (Narayanan et al., 2016). Additionally, blockchain eliminates the need for physical storage, printing, and mailing, resulting in significant cost savings for educational institutions and employers. The digital nature of blockchain credentials

allows for easy access and verification, further reducing the administrative burden associated with paper certificates (Pandya, 2022).

In conclusion, when compared to conventional paper-based certifications, blockchain technology provides a more secure, effective, and economical option. It is a better choice for academic credential verification since it removes the dangers of counterfeit, delays in verification, and expenses associated with physical storage. Employers and educational institutions alike ultimately gain from blockchain's tamper-proof and transparent nature, which offers a degree of security and efficiency that paper-based credentials cannot equal.

2.1.7 Digital Signatures and Public Key Infrastructure

Digital signatures are a core component of blockchain-based credentials, used to verify the authenticity and integrity of each credential issued. When a university or certifying body issues a credential, it signs the data using its private key—a cryptographic tool unique to the institution. Anyone who wants to verify the credential can use the public key to confirm that the credential was indeed issued by the legitimate organization and has not been tampered with. This is made possible through Public Key Infrastructure (PKI), a system that manages the creation, distribution, and revocation of digital keys. PKI ensures that credential holders and verifiers can trust the signatures attached to credentials without needing to contact the issuer directly. This secure method of authentication dramatically reduces fraud and simplifies verification, making it particularly valuable in large-scale academic and professional settings. Overall, digital signatures and PKI ensure that digital credentials are not only verifiable but also secure from forgery and unauthorized alterations.

2.1.8 Verifiable Credentials

Verifiable credentials are a standardized way to present and verify digital qualifications and achievements. Developed under the W3C (World Wide Web Consortium) guidelines, VCs include structured metadata that makes credentials machine-readable and interoperable across platforms. Each verifiable credential contains detailed information such as the type of credential, who issued it, the recipient, the date of issuance, and the criteria for earning it. The structure allows for privacy-preserving features such as selective disclosure, where users can choose what part of the credential to share. This is especially useful in professional environments where candidates may want to verify a specific skill without revealing unrelated personal details.

Verifiable Credentials are also foundational to the concept of Self-Sovereign Identity (SSI), empowering individuals to own and control their digital identities. With growing adoption in education, healthcare, and industry, VCs are becoming the global standard for secure, portable, and trustworthy digital credentialing.

2.1.9 Zero-Knowledge Proofs (ZKPs)

Zero-Knowledge Proofs (ZKPs) are cryptographic techniques that allow one party to prove to another that a certain statement is true without revealing the actual information. In the context of blockchain credentials, ZKPs enhance privacy by enabling credential holders to prove the validity of their qualifications without disclosing sensitive details. For instance, a job candidate can prove they have a university degree without showing the exact graduation year or the full document. This balance between privacy and verification is crucial in today's digital ecosystem, where data protection regulations like GDPR place strict requirements on how personal data is shared and stored. ZKPs prevent data overexposure, reduce identity theft risk, and foster trust between parties who may not know each other. As credentialing systems scale and integrate with global platforms, ZKPs are becoming a key technology for secure, privacy-respecting digital interactions. Their use in blockchain credentialing is a significant step toward secure, decentralized identity management.

2.1.10 Digital Wallets and Self-Sovereign Identity

Digital wallets are secure platforms that store a user's digital credentials and allow them to manage, share, and access their achievements with ease. Unlike traditional systems where credentials are stored by the issuing institution, digital wallets give users full control over their data, aligning with the principles of Self-Sovereign Identity (SSI). SSI is a model of digital identity where individuals own their personal information and credentials, rather than relying on centralized authorities to manage it. These wallets often support integration with professional networking sites and job portals, making credential sharing fast and seamless. Users can store multiple credentials from various institutions in one place and present them selectively depending on the context. Moreover, because the credentials are tied to cryptographic proofs and stored securely, they remain safe from tampering or unauthorized access. Digital wallets are central to the future of credentialing, putting power into the hands of the individuals and enhancing data privacy, portability, and control.

2.1.11 Smart Contracts

Smart contracts and decentralized applications (Dapps) are key blockchain innovations that support trustless and automated processes, making them ideal for credential verification systems. Smart contracts, first conceptualized by Nick Szabo, are digital protocols stored on a blockchain that automatically execute actions when predetermined conditions are met. These contracts are immutable and transparent, ensuring that once academic credentials are verified, they cannot be altered or tampered with. Dapps are decentralized applications that run on blockchain networks and rely on smart contracts to function, enabling secure and transparent data management. In a credential verification system, Dapps can provide user interfaces for institutions and employers to access verified academic records without intermediaries. Blockchain technology ensures that data is encrypted, trackable, and resistant to fraud, making it ideal for combating fake academic credentials. Its decentralized nature removes the risk of single points of failure, enhancing both security and reliability. As seen in other sectors like healthcare and finance, blockchain improves efficiency by eliminating lengthy manual verification processes. The use of smart contracts can automate credential issuance and validation, ensuring real-time, tamper-proof academic records.

2.1.12 Decentralized Identifiers (DIDs)

The rapid evolution of the digital ecosystem has exposed the limitations of traditional identity management systems. Centralized identifiers such as emails and phone numbers, which depend on third-party issuers, are increasingly vulnerable to control, revocation, and security breaches. In this context, Decentralized Identifiers (DIDs) emerge as a transformative solution, offering a secure, user-centric, and tamper-proof identity model built on blockchain and decentralized ledger technologies. DIDs are globally unique identifiers that enable individuals and entities to manage their digital identities without relying on a centralized authority. Unlike traditional identifiers, DIDs are cryptographically secured and decentralized, enhancing privacy, security, and control. As digital identity verification checks continue to rise, projected to exceed 71 billion in 2024 from 61 billion in 2023, the adoption of DIDs becomes increasingly essential to streamline authentication processes and mitigate the risks of identity theft and fraudulent credentials.

2.1.12.1 Architecture

DIDs function as a foundation for decentralized identity systems, designed with four core properties: Permanence, Resolvability, Cryptographic Verification, and Decentralization. They are persistent over time, allowing users to retrieve associated information through DID documents, verify identities using cryptographic keys, and maintain control independent of centralized entities. These characteristics make DIDs an ideal candidate for digital identity applications across platforms. A DID is linked to a DID Document, which holds essential data about the identity subject, such as public keys, service endpoints, and metadata required for verification. This document ensures consistency, integrity, and public validation. A DID Controller manages updates to the document, determining how and by whom the identity is managed. Notably, DIDs can have multiple controllers, adding flexibility and adaptability in real-world use cases. DID Resolution refers to the process of accessing the DID document using a DID resolver. This operation is method-specific and relies on a structure like did:method:uniqueidentifier, allowing DIDs to function across different ledgers such as Ethereum, Bitcoin, and other distributed technologies. As of July 2022, there were over 100 experimental DID method specifications, reflecting active innovation and implementation diversity.

2.1.12.2 Cryptography and Security Principles

Cryptographic methods are the backbone of DIDs, ensuring confidentiality, integrity, authentication, and non-repudiation. Both symmetric and asymmetric encryption are employed, with the latter enabling secure identity verification through public-private key pairs. This allows users to prove identity ownership without exposing sensitive data, thereby enhancing privacy and reducing exposure to breaches. The cryptographic robustness of DIDs directly contributes to the elimination of single points of failure, which are common in centralized systems. DIDs reduce the likelihood of identity theft by distributing control and embedding verification capabilities directly into the digital identity framework.

2.1.12.3 Types and Implementation

There are several types of DIDs, each catering to different operational and security requirements:

- Ledger-Based DIDs: Anchored on blockchains, offering resilience and decentralization.
- Ledger Middleware DIDs: Use an intermediate layer for scalable DID management.
- Peer DIDs: Support secure exchanges within private networks.
- Static DIDs: Simplified identifiers suitable for basic, immutable identity references.

Each DID type supports the overarching goal of decentralized, secure, and scalable digital identity management.

2.1.13 Digital Credentials

Digital credentials are modern, digital alternatives to traditional paper-based certifications that represent a person's achievements, skills, or qualifications. As outlined in the Accredible article, these credentials include digital certificates, badges, micro-credentials, stacked credentials, and blockchain-based records. Each type serves a specific purpose, from recognizing participation in short courses to validating formal educational accomplishments. What sets digital credentials apart is their embedded metadata, which allows for instant verification, increased security, and detailed context behind the achievement.

Digital badges, for example, are commonly used to represent smaller milestones such as course modules, skill acquisition, or event participation. These badges follow the Open Badge standard and can be easily displayed on social media profiles, websites, and resumes. Digital certificates, on the other hand, provide a more formal recognition of accomplishments, often supported by transcripts or references. Micro-credentials and stacked credentials offer a modular approach to learning, allowing individuals to build up qualifications over time. Blockchain credentials further enhance security by making records tamper-proof and verifiable through decentralized technology.

One of the primary advantages of digital credentials is their convenience and shareability. Recipients can access and display their credentials online, add them to digital wallets, and share

them with employers or institutions with just a few clicks. For issuers, the benefits include reduced administrative time, improved fraud prevention, and increased brand visibility. Accredible highlights real-world examples, such as the Digital Marketing Institute reducing its credentialing workload from hours to minutes by switching to a digital system. These advantages make digital credentials an effective tool for modern education and workforce development.

Despite their benefits, digital credentials also face challenges in adoption. Some institutions may be hesitant due to costs, integration difficulties, or uncertainty about user engagement. Accredible recommends careful planning, clear communication of value, and evaluation of platform features to address these concerns. As digital transformation accelerates across industries, digital credentials are becoming a vital part of how individuals validate and showcase their skills, offering a scalable, secure, and future-ready solution for lifelong learning and professional recognition.

In addition to traditional digital certifications and badges, micro-credentials are becoming essential in modern education systems. They provide flexible pathways for learners to acquire job-relevant skills and adapt to evolving industries. Blockchain enables these credentials to be securely issued as individual entries, supporting stackability, transparency, and cross-institution verification. By anchoring each credential in a ledger, schools and employers can view not just the credentials themselves, but the verified sequence in which they were earned, reinforcing academic trust and granularity.

2.1.14 Permissioned Blockchain, its Advantages, and Compatibility with Credential Management

Blockchain technology has gained significant attention for its ability to provide secure, transparent, and immutable records. However, the limitations of public blockchains like Bitcoin and Ethereum, particularly in terms of privacy, scalability, and transaction costs, have led to the emergence of permissioned blockchains as a more practical solution for enterprise applications. Unlike public blockchains, where anyone can join the network and validate transactions, permissioned blockchains, such as Hyperledger Fabric, restrict the access to trusted participants who are granted specific permissions to interact with the network, ensuring that sensitive information and transactions are kept confidential. In the context of credential management, a

permissioned blockchain restricts access to a set of authorized entities, such as universities, certification bodies, and other trusted institutions, ensuring that only designated participants can access or update sensitive credentials such as academic records. (Mitani & Otsuka, 2020)

A key benefit of permissioned blockchains like Hyperledger Fabric is enhanced privacy, which is essential for handling sensitive data. Public blockchains require that every transaction be visible to all participants in the network, which can expose sensitive information. This openness is a major concern in industries like academic credentialing or medical records management, where confidentiality is critical. In contrast, permissioned blockchains allow only authorized participants to access and validate transactions, ensuring that data remains private and secure as it restricts large-scale public visibility of personal information by requiring minimal necessary data. According to Pulmano et al. (2023), he demonstrated the use of Hyperledger Fabric for securely storing and verifying academic credentials and national IDs, showcasing the framework's ability to meet stringent data protection laws, such as the Philippine Data Privacy Act of 2012 (DPA 2012). Overall, this selective access prevents unauthorized entities from viewing or manipulating the data, providing a higher level of security compared to public blockchains.

Furthermore, permissioned blockchains are cost-effective, thus more scalable than public blockchains. In traditional public blockchains like Ethereum, every node must validate and store every transaction, which involves transaction/gas fees to incentivize miners who process the transaction. This implies that as the transactions and the network grow, so do the transaction fees needed to incentivize the miners. Overall, this full replication of data across all nodes leads to slower transaction speeds and higher costs. Permissioned blockchains, however, use modular consensus mechanisms, where only a subset of trusted nodes validate transactions, eliminating the need for mining, thus reducing the computational load and transaction fees, enabling the network to process a higher volume of transactions more efficiently. This scalability is particularly important for systems that handle large volumes of data and transactions, such as credential management systems.

Moreover, permissioned blockchains ensure regulatory compliance. Since access to the network can be strictly controlled, only authorized parties can participate in the validation and storage of data. This controlled access makes it easier to comply with data privacy regulations like the Philippine Data Privacy Act of 2012 (DPA 2012) and the General Data Protection Regulation

(GDPR). Additionally, the ability to create audit trails for all transactions enhances transparency and accountability, which is crucial for industries that must adhere to regulatory standards.

As proof of permissioned blockchain capabilities, the Ateneo de Manila University study demonstrated that permissioned blockchain systems like Hyperledger Fabric are not only capable of securely managing academic credentials but are also well-suited for integration into decentralized e-governance systems. The study highlighted the development of a blockchain-based digital credential system using Hyperledger Fabric for national identification and academic credentials, proving that such systems can effectively handle and validate sensitive personal information while maintaining privacy, scalability, and compliance with data protection regulations. (Pulmano, Estuar, De Leon, Tan, Co, & Tamayo, 2022) In addition, another effective use case of permissioned blockchain is the the double-layer framework developed by Ding, Gao, Zhu, and Yuan (2020), which is designed to address the limitations of traditional blockchain-based product traceability systems, specifically focusing on privacy, scalability, and regulatory participation. By leveraging permissioned blockchain technology, the framework introduces a main layer consisting of a consortium blockchain that facilitates data sharing and access for government regulators and other enterprises. On the other hand, the sub-layer consists of enterprise-specific private blockchains that store sensitive product data securely. This structure ensures that transparency is maintained, while also protecting enterprise data privacy by means of smart contracts. (Ding, Gao, Zhu, & Yuan, 2020). Moreover, in the study by Shen et al. (2024), a model-driven framework for multi-party collaboration was developed by making use of Hyperledger Fabric, a permissioned blockchain frame, which integrates BPMN (Business PRocess Model and Notation) and DMN (Decision Model and Notation) to execute business processes and decisions within a secure blockchain environment Overall, the successful application of Hyperledger Fabric for these use cases supports the feasibility of adopting similar blockchain frameworks for managing credentials, making it an ideal solution for systems like ChainCert.

.

In conclusion, these capabilities mentioned prove to show that permissioned blockchains, such as Hyperledger Fabric, are ideal for ChainCert that involves academic and professional credentialing, as they provide a controlled environment for trusted participants, ensuring privacy, compliance, and efficiency, which are key requirements for credentialing systems in various industries, including education, professional certification, and beyond. By leveraging these

capabilities, both academic credentials and professional certifications can be securely stored, easily verified, and resistant to tampering or fraud.

2.2 Review of Related Systems

2.2.1 Non-Blockchain Systems

2.2.1.1 Professional Regulation Commission (PRC)

The Professional Regulation Commission (PRC) was established by Presidential Decree No. 223 in 1973, with the primary function of licensing and regulating the practice of over 40 professions in the country, which include but are not limited to engineers, doctors, nurses, teachers, etc. The commissions ensure that specific qualifications adhere to ethical standards and contribute to the protection of public interest and safety (History | Professional Regulation Commissions, n.d.) These are done by administering licensure examinations, granting professional licenses, monitoring the practice of regulated professions, license renewals, and ensuring compliance with ethical standards. Moreover, it also enforces the Code of Ethics for professionals, handles disciplinary actions for violations, and mandates Continuing Professional Development (CPD) for professionals to stay updated and relevant in their fields. Overall, PRC is crucial for maintaining credibility and competence of professionals in the Philippines, fostering trust within the industries they serve (Romano 2025).

To become a licensed professional in the Philippines, individuals must first pass the licensure examination administered by the Professional Regulation Commission (PRC). The process begins with the examination application, where candidates are required to apply through Licensure Examination and Registration Information Systems (LERIS), an online platform provided by the PRC (PRC, n.d.). In order to do so, applicants must create an account, complete an application form, and upload essential documents such as educational qualifications, identification, and clearance. Once the application is submitted and the exam fee is paid, candidates receive notifications regarding the exam schedule and location, which are posted through the LERIS portal. The licensure exams are typically held annually and are specific to each profession and may involve both written and practical components. After passing the examination, candidates proceed to the initial registration steps, where they must complete necessary documentation and pay

registration fees to obtain their profession identification card (PIC). The PIC serves as proof of licensure and is required to practice legally in their respective field. The PRC's online system allows for real-time verification of professional credentials, enabling employers and institutions to confirm the authenticity of the professionals qualifications (Nama-Admin, 2023; PRC, n.d.).

In addition, PRC licenses are valid for three years from the date of issuance, after which they must be renewed. The PRC has made the renewal process more accessible through the use of online systems, allowing professionals to manage their licenses more efficiently. In renewing, professionals must log into the PRC online portal (LERIS) to start the renewal process, wherein the portal allows users to check their CPD, unit compliance, update their personal information, and initiate the renewal application. Professionals can start the renewal process up to a year in advance if their license is nearing expiration (FilipiKnow, 2025). As part of the renewal requirements, professionals must accumulate CPD units, which vary by profession. These units can be earned through accredited seminars, training programs, or self-directed learning (Romano, 2025). As of 2025, professionals can still renew their licenses without completing the CPD units but must sign an undertaking to comply by December 2025 (FilipiKnow, 2025). The other requirements for renewal include documents such as an updated photo (2x2), a complete application form, etc. Payment for the renewal fee is done online and once the application and payment are processed, professionals must schedule an appointment at their preferred PRC office to claim their renewed license. However, PRC offers home delivery of the license, and special provisions exist for Overseas Filipino Workers (OFWs) and dual citizens, where the PRC has implemented a courtesy lane to expedite the process (FilipiKnow, 2025).

To conclude, PRC provides a reliable verification platform that enhances trust between employers, clients, and regulatory agencies. Using the LERIS system, employers can easily verify whether a professional holds a valid license, helping prevent fraudulent claims of professional credentials (Nama-Admin, 2023). These online systems support employers in confirming the authenticity of licenses, thereby safeguarding both professionals and clients from potential risks. Furthermore, the PRC's regulatory role is crucial in ensuring the quality and integrity of services in professions that directly impact

public safety, such as healthcare, engineering, etc. Lastly, by making license renewal mandatory, this ensures that the workforce remains well-trained, knowledgeable, and up-to-date, benefiting the professional landscape at large. (Romano, 2025).

2.2.1.2 DMW, POEA, POLO, and OWWA: Interconnected Agencies in OFW Deployment and Welfare

The Department of Migrant Workers (DMW) is the central government agency in the Philippines mandated to oversee and streamline all aspects of overseas employment, including deployment regulation, welfare provision, and credential verification. It now absorbs the functions of previously independent bodies such as the Philippine Overseas Employment Administration (POEA) and the Philippine Overseas Labor Office (POLO) and works in close coordination with the Overseas Workers Welfare Administration (OWWA). POEA historically handled recruitment agency licensing, job order approvals, and issuance of Overseas Employment Certificates (OEC), while POLO, operating through Philippine embassies and consulates abroad, managed contract verification and on-site labor concerns. OWWA, as the welfare arm, provides benefits such as scholarships, social assistance, and reintegration programs to documented Overseas Filipino Workers (OFWs). Together, these agencies create an interconnected system that governs the life cycles of OFWs, from recruitment and contract validation to deployment and welfare.

The verification and deployment process begins when employers or workers submit documentation, such as employment contracts, work permits, and company credentials, to POLO (now referred to as Migrant Workers Offices, or MWOs). POLO verifies the authenticity and compliance of these documents with Philippine and host country labor standards. Once verified, the documents are endorsed to DMW (formerly POEA), which approves the deployment and issues the OEC, a required document for legal overseas employment. OWWA uses the verified status of workers to determine eligibility for membership and access to welfare benefits. For agency-hired workers, DMW also processes job orders and agency compliance. Throughout this process, each agency plays a specific role in ensuring that only legitimate, properly documented OFWs are deployed and supported.

Despite the structural integration under DMW, the existing system remains largely manual, paper-based, and prone to inefficiencies. Credential verification by POLO involves physical review and communication via email or phone, with no real-time access to issuing institutions like CHED, TESDA, or PRC. DMW continues to rely on scanned or hardcopy documents for deployment processing, while OWWA depends on previously verified records without independently validating credentials. The lack of centralized databases, automated validation tools, and secure digital infrastructure contributes to delays, redundant verification steps, and potential fraud. These limitations indicate the need for a tamper-proof, interoperable credentialing solution, such as a blockchain-based system, that can improve transparency, speed, and trust across all agencies involved in OFW deployment and welfare.

2.2.1.4 Corporate Sector

Credential verification remains a vital process in recruitment, ensuring that candidates possess the necessary academic and identity credentials before employment. Interviews with HR officers from Reliance Producers Cooperative, Avesco Marketing Corporation, and Sunlife Philippines revealed that verification typically begins with the submission of original documents, including diplomas, transcripts of records (TORs), birth certificates, and government-issued IDs such as SSS, PhilHealth, Pag-IBIG, and TIN. For roles involving safety and compliance, police clearances and medical certificates are also required. HR personnel manually inspect these documents for authenticity, evaluating the layout, seals, logos, and consistency of information. Preliminary interviews or reference checks are sometimes conducted before formal interviews to serve as an early screening layer. However, this process is often fraught with vulnerabilities. Several HR professionals noted that verification depends largely on the initiative and available time of the employer, which means that not all applicants undergo thorough background checks. If suspicions arise, employers may attempt to verify records by contacting schools or previous employers, but the effectiveness of this step varies depending on the responsiveness and systems of the institutions involved (Appendix C). Many schools, particularly smaller ones, lack digital registries or online verification tools, forcing HR to rely solely on physical documents and manual communication, often resulting in delays. In such cases, university rankings are occasionally used as a proxy for legitimacy, though this approach is flawed. As Kayyali

(n.d.) points out, rankings are subjective and can be skewed by methodological bias or data manipulation.

The limitations of current practices open the door to diploma mill fraud. According to Attewell and Domina (2010), individuals can exploit verification gaps by presenting forged credentials or purchasing degrees from unaccredited institutions, knowing that verification may be skipped altogether. This exposes businesses to negligent hiring risks, with potential consequences including financial loss, legal liability, and reputational damage. Findings from a credential verification feasibility survey conducted among HR officers further support these insights. Most HR departments in the private sector still rely on paper-based methods, collecting documents such as TORs, certificates, and IDs, which are stored in personnel files or localized digital repositories (Appendix F). While some respondents reported confidence in their data handling systems, others expressed concerns about long-term data integrity and security (Appendix F).

Despite these operational limitations, there is growing openness to adopting more advanced solutions. The majority of surveyed HR officers acknowledged the legal and reputational risks posed by fraudulent credentials (Appendix F). Blockchain technology, in particular, was viewed as a promising innovation that could mitigate these risks by offering tamper-proof, real-time verification—especially by eliminating the need to contact issuing institutions directly (Appendix F). However, concerns were also raised regarding legal compliance, particularly on how blockchain systems would adhere to national data privacy regulations like the Philippine Data Privacy Act of 2012 (Appendix F). Implementation challenges were also cited. Respondents emphasized the need for technical resources, IT training, and system integration support (Appendix F). While some HR professionals expressed a willingness to invest in blockchain, others prioritized familiar tools such as applicant tracking systems or payroll platforms (Appendix F). Integration was seen as moderately difficult due to the need for field mapping and alignment with existing HR workflows.

In conclusion, while traditional methods still dominate credential verification in the private sector, both qualitative interviews and survey data reflect a strong interest in modernizing these systems. The sector appears to be moving toward more secure,

standardized, and scalable technologies—such as blockchain—provided that issues related to compliance, integration, and resource availability can be addressed. This shift is essential to mitigate hiring risks and ensure the integrity of academic and identity verification in an increasingly competitive labor market.

2.2.1.5 Government Sector (Public)

In contrast, the public sector operates under more standardized hiring protocols, as mandated by civil service regulations. According to an interview with the Department Head of the Human Resources Development and Management Department (HRDMD) of the City Government of Bacoor, Cavite, the verification process starts with the submission of a notarized personal Data Sheet (PDS) using the Civil Service Commission (CSC) Revised 2017 form. This legal document is treated as a sworn statement and includes a section subject to penalties for falsification. Applicants must also provide certified true copies of their educational records, such as diplomas and TORs, in accordance with the Omnibus Rules on Appointments and Other Human Resource Actions (ORA-OHRA). For roles that require eligibility or licensure, HR staff verify these using government portals such as those of the Professional Regulation Commission (PRC) or the CSC itself. Background checks are routinely conducted by contacting former employers and educational institutions. Additionally, psychometric and intelligence tests are required, while neuropsychiatric evaluations are mandatory for sensitive roles, particularly those related to peace and order.

While this process appears comprehensive, it is not without its limitations. The system heavily relies on physical documentation, which is susceptible to forgery if security features are outdated or inconsistent across institutions. Although online government portals have streamlined parts of the verification, HR officers noted that delays still occur when waiting for responses from schools, particularly those without digitized systems. Moreover, suspicious applicants are rarely dealt with face-to-face due to pre-screening procedures that limit direct contact. Applicants are typically submitted via email or through designated government windows, and security protocols, including CCTV surveillance and restricted office access, are used to mitigate potential threats. While this reduces risk, it also means that HR staff may miss cues that would be apparent in face-to-face interactions. Furthermore, without a unified national database for academic

records, discrepancies may go unnoticed unless the HR personnel are exceptionally diligent. As with the private sector, the lack of real-time verification tools leaves the door open for fraudulent credentials to slip through the cracks.

2.2.1.6 Credential Issuance Processes in Philippine Government Agencies

Credential issuance by government institutions in the Philippines plays a crucial role in employment authorization, worker protection, and verification. The processes implemented by agencies such as the Department of Migrant Workers (DMW), Overseas Workers Welfare Administration (OWWA), and the Technical Education and Skills Development Authority (TESDA) highlight both the traditional and digital methods used to manage and verify credentials.

2.2.1.6.1 Department of Migrant Workers (DMW) / POEA: Overseas Employment Certificate (OEC)

The Overseas Employment Certificate (OEC) is a mandatory credential issued by the DMW (formerly POEA) to authorize the overseas deployment of Filipino workers. The OEC serves as proof that the worker's employment contract has been vetted and approved by the Philippine government. It also ensures compliance with host-country labor standards. The issuance process includes verification of documents such as valid passports, work visas, employer ID, and a contract authenticated by a Philippine labor attaché or POLO (Department of Migrant Workers, 2024). The OEC is issued through both domestic DMW offices and Migrant Workers Offices abroad. Before the issuance, employment contracts must undergo validation to confirm legitimacy and adherence to labor regulations. These safeguards are critical in protecting OFWs from illegal recruitment, contract substitution, and workplace abuse (Honolulu PCG, 2024).

2.2.1.6.2 Overseas Workers Welfare Administration (OWWA): Membership and OFW eCard

OWWA issues membership credentials to overseas Filipino workers (OFWs), which entitle them to benefits such as health coverage, training, repatriation aid, and social programs. Membership is valid for two years and

requires a completed information sheet, proof of overseas employment, and a $25 membership fee (Kabayan Remit, 2024). In addition, OWWA introduced the OFW eCard, a digital identification and access credential. The application process includes online form submission, privacy policy acceptance, and regional office pickup. Applicants can track their card status using a reference number. The OFW eCard serves not only as proof of membership but also streamlines access to government services and repatriation support (OWWA Member, 2024). The OWWA's digitization efforts reflect a shift toward more convenient and secure credential management. However, applicants still need to submit requirements in person or via email depending on location, showing that a hybrid model of verification is still in place.

2.2.1.6.3 Technical Education and Skills Development Authority (TESDA): National Certificates (NC) and Digital Credentials

TESDA is responsible for certifying technical skills and competencies through the issuance of National Certificates (NC) and Certificates of Competency (CoC). These are essential credentials for workers in skilled trades, both domestically and internationally. To address fraud and improve verification, TESDA has begun issuing digital certificates embedded with QR codes. These allow employers, licensing bodies, and government agencies to quickly verify the authenticity of credentials online. This initiative enhances the security, portability, and efficiency of TESDA certifications, particularly for OFWs or applicants undergoing cross-border verification (TESDA, 2024). TESDA's adoption of QR-based e-certificates is part of its broader strategy to reduce credential fraud and align with digital transformation goals in education and labor sectors.

The processes used by DMW, OWWA, and TESDA for credential issuance illustrate the evolving landscape of verification in the Philippine public sector. While manual verification and in-person applications remain in use, all three agencies are gradually shifting toward digitized and tamper-resistant systems. These include QR-coded e-certificates, online portals, and centralized credential databases. Moving forward, integration and standardization across these agencies—potentially through blockchain or interoperable systems—could greatly improve efficiency, reduce fraud, and ensure that

Filipinos are equipped with secure, verifiable credentials in both local and global contexts.

2.2.2 Blockchain Systems

2.2.2.1 Exonum: A Blockchain Framework for Secure and Transparent Systems

Exonum is a permissioned blockchain framework designed to build secure and scalable blockchain applications, making it an ideal candidate for applications requiring high levels of data integrity, transparency, and security. Unlike permissionless blockchains like Bitcoin, Exonum provides a controlled environment where only authorized parties can commit transactions, offering greater flexibility and control over the system. The framework utilizes *Byzantine fault-tolerant* consensus to ensure data synchronization, even in the presence of compromised or faulty nodes, making it highly resilient to attacks. Exonum's use of cryptographic methods for transactions ensures the integrity and authenticity of data, which is valuable for applications like academic credential verification. Additionally, Exonum supports advanced storage capabilities, enabling efficient handling of large data collections with immutable and verifiable audit trails. With high throughput capabilities, supporting up to 7,000 transactions per second, it is well suited for decentralized applications requiring transparent and reliable transaction logs. Its ability to integrate multiple programming languages, such as Rust and Java, provides a flexible environment for developers, further expanding its potential use cases in secure and transparent systems like credential verification. For example, Synergy University in Russia utilized Exonum to create a blockchain-based platform for recording and storing student education information. The solution enabled the university to securely store diplomas and other academic records, providing a transparent, tamper-proof, and instantly verifiable system that improved trust and reliability for both students and employers. This blockchain solution addressed issues of academic certificate fraud, significantly reducing labor costs for archiving and certification while offering real-time verification capabilities for employers (Exonum, n.d.).

2.2.2.2 Akowe's Blockchain Solution for Academic Credential Verification in Africa

Akowe, a Nigerian startup, is revolutionizing academic credential verification in Sub-Saharan Africa with its blockchain-based solution. Traditional methods of verifying

academic credentials in the region are plagued by inefficiencies, errors, and fraud due to manual checks and reliance on physical documents . Akowe addresses these challenges by offering a secure, transparent platform that stores academic credentials on an immutable blockchain, ensuring they are tamper-proof and auditable. This digital solution simplifies the verification process, allowing employers to confirm qualifications in real time while reducing the risk of fraud and delays.

In addition to improving the verification process, Akowe's platform helps educational institutions digitally transform their operations, enabling them to issue and verify academic certificates efficiently. The use of blockchain technology also opens up new revenue opportunities for schools through smart verification services. By offering a more reliable and streamlined system for academic record management, Akowe is set to enhance the transparency and efficiency of academic credential verification across Africa (Web3Africa, 2023).

2.2.2.3 BCdiploma: A Blockchain-Enabled Solution for Secure Digital Credentialing

BCdiploma is a blockchain-backed digital credentialing solution that provides secure, tamper-proof certificates for educational institutions. Unlike traditional paper-based or PDF certificates, BCdiploma ensures that academic credentials are permanently verifiable and easily accessible through a unique, shareable link. By leveraging blockchain technology, BCdiploma eliminates the reliance on third-party platforms, offering institutions full access over their credentialing process while maintaining full compliance with data protection regulations. The platform guarantees enhanced security, preventing fraud and ensuring the integrity of academic records. With its user-friendly interface and seamless API integrations, BCdiploma supports the efficient issuance and real-time verification of credentials. Notably, BCdiploma has been adopted by over 250 institutions worldwide, including renowned universities like Stanford and the University of Toronto, demonstrating its effectiveness in integrating the credential verification process and reducing administrative costs (BCdiploma, n.d.).

2.2.2.4 European Blockchain Services Infrastructure (EBSI) for Academic Credential Verification

The European Blockchain Services Infrastructure (EBSI) is an initiative developed by the European Commission and the European Blockchain Partnership (EBP), consisting of the 27 EU countries, Norway, Liechtenstein, and the European Commission. EBSI leverages blockchain technology to build a secure, transparent, and decentralized infrastructure for public sector applications, with a roadmap for expanding to the private sector. One of its key use cases is the verification and management of academic credentials, allowing educational institutions to issue and store diplomas and certificates on a blockchain platform. This system ensures tamper-proof, easily verifiable records that can be accessed instantly by employers, institutions, and other authorized parties without relying on centralized authorities. By digitizing and decentralizing academic credentials, EBSI aims to reduce verification costs, increase trust in the authenticity of academic records, and streamline the process for both institutions and graduates. Furthermore, EBSI's blockchain infrastructure provides a secure method for notarization, trusted data sharing, and the creation of European digital identities, further enhancing transparency and compliance across borders. As EBSI evolves, additional use cases, such as financing for small and medium-sized enterprises (SMEs) and social security services, will expand its utility in the public and private sectors (European Commission, n.d.).

2.2.2.5 NUS Green Bond Reporting Tokenisation Initiative

The National University of Singapore (NUS) has partnered with Northern Trust and UOB to launch a pioneering green bond reporting tokenization initiative, significantly advancing transparency and integrity in environmental, social, and governance (ESG) reporting. This initiative uses blockchain technology to create a secure, immutable token that holds the environmental impact reporting data from NUS' third green bond, issued in 2023. The tokenization process ensures that this data remains unchanged and verifiable, providing investors with reliable information for their sustainability reporting obligations, even as the bond changes hands. By using blockchain for reporting, NUS eliminates the need for manual updates, ensuring real-time, tamper-proof tracking of green bond data.

The groundbreaking effort is part of Project Guardian, a global initiative led by the Monetary Authority of Singapore, aimed at exploring the benefits of asset

tokenization in enhancing market efficiency and liquidity. As the first university in Singapore to implement blockchain for ESG reporting, NUS is setting a new standard in green finance innovation. The tokenization of green bond credentials enhances investor confidence by offering full transparency and accountability, furthering NUS' commitment to its Climate Action Plan in 2030. This collaboration is expected to improve sustainability reporting and contribute to advancing sustainable financial practices, making NUS a leader in the integration of blockchain into green finance (Northern Trust, 2024).

2.2.2.6 OpenCerts: Singapore's Blockchain-Powered Credential Verification System

OpenCerts is a blockchain-based digital credential verification system developed by Singapore's Government Technology Agency (GovTech) in partnership with Ngee Ann Polytechnic. It was designed to address the growing problem of academic fraud by enabling educational institutions to issue tamper-proof digital certificates anchored on the Ethereum blockchain. The system uses a cryptographic hash of the certificate, stored immutably on the blockchain, which allows third parties such as employers to instantly verify the authenticity of academic credentials without contacting the issuing institution (GovInsider, 2023).

According to the OECD's Observatory of Public Sector Innovation, OpenCerts operates using an open-source framework called OpenAttestation, which is designed for scalable adoption across different sectors, including education, healthcare, and trade. The initiative aligns with Singapore's Smart Nation strategy, focusing on transparency, data sovereignty, and user empowerment. The Smart Nation initiative explains that each certificate is represented as a signed JSON file containing metadata and a cryptographic proof. The authenticity of these certificates can be verified by recalculating the hash and comparing it to the one recorded on the blockchain, ensuring data integrity and eliminating forgery (OECD OPSI, n.d.).

OpenCerts was piloted in 2018 and formally adopted in 2019 by 18 educational institutions across Singapore. As highlighted by GovInsider, the implementation came after the Ministry of Manpower reported several incidents involving fake foreign

certificates (GovInsider, 2023). The system allows graduates to securely receive and share their academic credentials via a unique link or QR code, which streamlines processes like job applications, admissions, and licensing.

The platform promotes decentralization, transparency, and long-term accessibility, offering a cost-efficient and fraud-resistant alternative to traditional paper or PDF diplomas. Moreover, it is fully interoperable with Singapore's national digital identity system (SingPass), further enhancing security and convenience (GovInsider, 2023).

2.2.2.7 TruScholar: Blockchain-Based Digital Credentialing Platform in India

TruScholar is a blockchain-powered platform that aims to address the growing concerns related to the verification of academic credentials, particularly the rise in fraudulent diplomas and certificates. By leveraging blockchain technology, TruScholar allows educational institutions to issue, store, and verify tamper-proof digital credentials. This approach ensures that once a credential is issued, it cannot be altered or forged, offering a higher level of security than traditional systems. The platform's decentralized nature guarantees that the data recorded on the blockchain is immutable, making it a reliable and secure solution for credential verification (TruScholar, n.d.).

A key feature of TruScholar is its use of blockchain to enable real-time verification. Traditional verification methods often involve lengthy manual checks and direct communication with the issuing institution, leading to delays. TruScholar eliminates this inefficiency by allowing employers and other stakeholders to instantly verify the authenticity of academic credentials directly through the blockchain (TruScholar, n.d.). This is particularly important in reducing the time spent on recruitment processes and ensuring that fraudulent credentials do not go undetected.

In addition to its real-time verification capabilities, TruScholar provides users with ownership and control of their digital credentials. This is achieved through self-sovereign identity (SSI), which allows students and professionals to securely manage, share, and store their credentials without relying on third-party intermediaries. By giving users full

control over their data, TruScholar enhances privacy and ensures that the credentialing process is more secure and user-centric (TruScholar, n.d.).

The platform also supports global interoperability, which enables educational institutions and employers worldwide to verify digital credentials without relying on centralized systems. This integration across multiple platforms and institutions enhances the global recognition and acceptance of digital credentials, promoting trust and reliability in academic records (TruScholar, n.d.). Furthermore, by digitizing the credentialing process, TruScholar reduces administrative costs associated with paper-based certificates, printing, and postage (LinkedIn, 2024), making the system more cost-effective for educational institutions.

Transparency and auditability are also crucial aspects of TruScholar's blockchain-based system. The blockchain creates a permanent, verifiable audit trail for each credential issued, allowing for easy tracking and ensuring the credibility of academic records (TruScholar, n.d.). This transparency is crucial for preventing fraudulent activities and fostering trust among educational institutions, employers, and students.

TruScholar's system is highly relevant to the development of secure academic verification systems, aligning with the goals of this research, which seeks to create a transparent, efficient, and tamper-proof method for verifying academic credentials. By utilizing blockchain, TruScholar addresses key issues such as fraud prevention, verification delays, and the lack of standardized systems in academic credentialing. The platform's emphasis on decentralization, cryptographic security, and user control presents a compelling model for blockchain-based verification systems in education (TruScholar, n.d.; LinkedIn, 2024).

2.2.2.8 DiplomaSafe: A Digital Credentialing Solution for the Future

DiplomaSafe is a blockchain-backed platform that provides a secure and transparent solution for issuing and verifying digital credentials. It focuses on enabling educational institutions and employers to issue tamper-proof certificates and academic credentials, ensuring the authenticity and security of digital diplomas and certificates. The

platform addresses the growing need for efficient, secure, and reliable verification systems, especially in response to the increasing prevalence of counterfeit academic credentials (DiplomaSafe, n.d.).

DiplomaSafe utilizes blockchain to eliminate the reliance on third-party intermediaries for credential verification, making the process faster, more secure, and cost-effective. Unlike traditional paper-based methods, which are prone to errors and delays, DiplomaSafe allows institutions to issue credentials digitally, with each credential encrypted and stored securely on the blockchain. Once a credential is issued, it is immutable and tamper-proof, meaning that no unauthorized changes can be made, which significantly reduces the risk of fraud (DiplomaSafe, n.d.).

One of the key benefits of Diploma Safe's blockchain integration is real-time verification. Employers, recruiters, or other stakeholders can quickly validate the authenticity of a certificate without the need to contact the issuing institution directly. This feature streamlines the hiring process and reduces the administrative burden on both educational institutions and employers. Additionally, DiplomaSafe offers user-centric control, allowing students and professionals to manage and share their credentials independently, ensuring their privacy and reducing reliance on third-party verification services (DiplomaSafe, n.d.).

The platform also emphasizes digital transformation in education by providing a user-friendly interface for both institutions and students. According to research on student perceptions of digital credentials, students have increasingly shown interest in digital diplomas due to their ease of use, security, and portability. This shift towards digital credentials is seen as a positive evolution, allowing students to share their achievements with potential employers in a more efficient and verified manner (DiplomaSafe, 2023).

In the workforce, its solution is designed to address the need for secure workforce credentials. As the job market becomes more competitive, employers are increasingly relying on verified, digital certifications to ensure that candidates possess the required skills and qualifications. The platform's ability to verify credentials in real time reduces

the risk of hiring individuals with fraudulent diplomas, ensuring the quality and reliability of the workforce (DiplomaSafe, 2023).

DiplomaSafe presents a significant development in the field of academic credential verification. Its focus on blockchain technology, real-time verification, and user empowerment makes it a valuable model for the system being developed in this study. By ensuring that academic credentials are both secure and easily verifiable, DiplomaSafe's approach addresses the core challenges of fraud prevention, verification delays, and user control in the credentialing process (DiplomaSafe, n.d.). Additionally, its support for digital transformation in education aligns with the goals of creating a streamlined, efficient, and tamper-proof verification system for academic diplomas.

2.2.2.9 Verifi-Chain: A Credentials Verifier using Blockchain and IPFS

Verifi-Chain is a blockchain-based credential verification system developed to address the widespread issue of fraudulent academic certificates, particularly in regions like Southeast Asia where document forgery is prevalent. This prototype utilizes a hybrid architecture combining the Ethereum blockchain and the InterPlanetary File System (IPFS) to ensure secure, tamper-proof, and transparent academic record verification. In this system, applicants first upload their credentials, which are then verified by an administrator. Once verified, the documents are uploaded to IPFS, a decentralized storage platform that generates a unique content identifier (CID) or hash for each file.

This CID acts as a digital fingerprint and is stored immutably on the Ethereum blockchain through a smart contract. Because blockchain transactions are irreversible and publicly verifiable, the certificate's authenticity can be validated without relying on a centralized authority. Employers and other verifiers can retrieve the certificate by entering the CID hash and requesting access through the system. The applicant has full control over which companies can view their records, maintaining a degree of privacy while ensuring verification transparency.

The system follows a role-based access model, assigning distinct functions to four types of users: applicants (who upload and share credentials), administrators (who verify documents), companies (who search and verify applicants), and general users (who can explore the system's functionalities). By separating responsibilities and securing each

step of the process, Verifi-Chain minimizes the risk of fraud, unauthorized access, and data loss. The use of IPFS reduces the cost of storing large files directly on the blockchain, addressing one of the common scalability challenges in decentralized systems. Meanwhile, Ethereum's smart contracts handle logic execution securely without human intervention. This approach provides significant improvements over manual and paper-based verification methods in terms of efficiency, traceability, and cost-effectiveness.

Verifi-Chain demonstrates a promising approach for modernizing academic credential verification by combining decentralized storage, cryptographic assurance, and user-centric access control. It paves the way for institutions and employers to adopt more robust and scalable verification workflows, especially in environments where traditional methods are inefficient or susceptible to manipulation (Rahman et al., 2023).

2.2.2.10 Cerberus: A Blockchain-Based Accreditation and Degree Verification System

Cerberus: A Blockchain-Based Accreditation and Degree Verification System is a robust solution designed to combat academic credential fraud using a permissioned blockchain network managed by accreditation bodies, universities, and observers. Its core features include on-chain issuance and revocation of credentials via smart contracts, QR code-based verification that does not require cryptographic key management, and selective data disclosure to protect student privacy. By incorporating Merkle tree structures, Cerberus ensures scalable and efficient data handling, all while maintaining a user-friendly experience suitable for non-technical users.

Compared to other blockchain-based systems like Blockcerts, Hypercerts, EchoLink, and EduCTX, Cerberus offers a more integrated and practical approach. While these systems have introduced innovations such as decentralized credential storage and Ethereum-based smart contracts, they often require users to manage digital wallets or lack essential features like QR code verification and oversight from accreditation bodies. Cerberus addresses these gaps by providing real-time, third-party verifiable credentials without compromising usability or data privacy.

Legacy verification systems, which rely on institutional databases and manual processing, are inefficient and vulnerable to various forms of fraud, including diploma mills and insider manipulation. These systems often fail to meet the needs of modern, fast-paced, and global academic and professional environments. Cerberus surpasses these outdated methods by leveraging blockchain's immutability and transparency, enabling secure, efficient, and accessible verification of academic credentials for all stakeholders (Yuan et al., 2019).

2.2.2.11 CredSec: A Blockchain-based Secure Credential Management System for University Adoption

In the effort to combat the growing issue of fake academic credentials, the study titled "CredSec: A Blockchain-based Secure Credential Management System for University Adoption" introduces a robust and decentralized solution using blockchain technology. Traditional university credential management systems rely on local dedicated servers (LDS), making them vulnerable to threats like SQL injections, credential tampering, and single-point failures. The proposed system—BCMS—replaces centralized authority with a blockchain-enabled infrastructure that ensures high integrity, transparency, and tamper-resistance (Ghosh et al., 2024). Central to its approach is the use of modified two-factor encryption (m2FE), combining RSA cryptography with DNA encoding, and a character-to-integer (C2I) mapping method that enhances both security and performance.

The literature reviewed in the study reveals critical shortcomings of existing credential management systems, especially those relying solely on LDS or permissioned blockchains without sufficient privacy measures. Systems like Blockcerts, Recordkeeper, and SmartCert offer blockchain-based verification but lack robust encryption mechanisms or allow too much control by central authorities, increasing the risk of credential manipulation. In contrast, CredSec's BCMS not only eliminates the need for third-party involvement but also introduces a multi-entity architecture involving a Certification Authority (CTA), instructors, students, and dual servers (LDS and blockchain server). This distributed model ensures secure storage, verification, and retrieval of academic records, making it significantly more resilient to internal or external attacks (Ghosh et al., 2024).

Furthermore, the proposed BCMS is validated through experimental and comparative analyses. Results show improved performance over other systems, including faster registration, encryption, uploading, and retrieval processes, even for credentials of varying sizes. The m2FE encryption not only maintains confidentiality but also reduces cipher size and processing time compared to earlier two-factor encryption schemes. Security models such as IND-MOD, UNF-MOD, and IMU-MOD demonstrate the system's resistance to unauthorized access, credential forgery, and tampering. Altogether, the CredSec system provides a comprehensive and scalable blockchain-enabled solution that directly addresses the core challenges of verifying and securing academic credentials in higher education (Ghosh et al., 2024).

2.2.2.12 Blockchain for Academic Integrity: Developing the Blockchain Academic Credential Interoperability Protocol (BACIP)

The Blockchain Academic Credential Interoperability Protocol (BACIP) emerges as a pioneering solution designed to combat the growing issue of academic fraud, particularly the circulation of fake degrees and credentials. BACIP leverages a dual blockchain architecture, smart contracts, and zero-knowledge proofs to build a secure, scalable, and privacy-preserving framework for verifying academic records. Its primary objective is to enhance the interoperability and trustworthiness of academic credential verification systems on a global scale. By decentralizing the issuance, storage, and verification of digital certificates, BACIP eliminates the reliance on traditional third-party authorities, thereby reducing the potential for forgery, tampering, and unauthorized access (Abubakar et al., 2024). The system aims to streamline academic and professional mobility while maintaining compliance with data privacy regulations such as GDPR.

The protocol is grounded in a robust theoretical framework and incorporates operational components adapted from successful existing models. These include smart contract-based credential management, decentralized storage solutions like IPFS, AES encryption for data security, and a global credit transfer system. Additionally, BACIP integrates features such as student digital identity management and a decentralized educational resource marketplace, drawing insights from prior implementations like BCERT and the European Blockchain Services Infrastructure (EBSI). These technologies not only support

automation and verification but also facilitate international accreditation, scalability, and adaptability. The use of zero-knowledge Ethereum Virtual Machine (zkEVM) and modular blockchain structures further ensures privacy and performance without sacrificing transparency or interoperability (Abubakar et al., 2024).

Despite its promise, BACIP acknowledges the critical need for real-world validation and stakeholder collaboration. Challenges such as system interoperability, onboarding governance, compliance with international legal frameworks, and user accessibility remain central to its implementation strategy. The methodology underpinning BACIP includes a four-phase development cycle: articulating core principles, integrating operational components, standardizing technical specifications, and evaluating ethical viability. These phases are supported by a mixed-methods research approach combining empirical studies and literature reviews. Ultimately, BACIP sets a benchmark for future developments in blockchain-enabled academic credential verification by addressing current inefficiencies and providing a universally applicable, secure solution to fake academic credentials (Abubakar et al., 2024).

2.2.2.13 TESDA's Blockchain-Based Credential Verification System Using Polygon

The Technical Education and Skills Development Authority (TESDA) in the Philippines has been exploring the potential of blockchain technology to improve the verification and security of academic credentials, particularly National Certificates (NCs). The primary goal of TESDA's blockchain-based system is to provide a secure, tamper-proof method for issuing, storing, and verifying vocational and technical credentials. Blockchain's decentralized nature ensures that credentials are not only immutable but also easily accessible for verification by employers, government agencies, and other educational institutions. This solution aims to address the issue of fake diplomas and credentials, which has been a growing concern in both local and international job markets.

TESDA's implementation leverages the core principles of blockchain technology, including smart contracts and distributed ledgers, to ensure that the issuance of National Certificates is transparent and verifiable. By storing the credential data on the blockchain, TESDA eliminates the need for traditional, centralized databases that are often vulnerable

to unauthorized access and fraud. Furthermore, the system integrates with existing platforms, allowing students to easily share their verified credentials with employers or other entities via secure QR codes or hash values, which can be quickly checked against the blockchain. This process not only streamlines the verification process but also increases the trustworthiness of the credentials being issued.

Despite its promising features, TESDA's blockchain system faces several challenges. These include ensuring system interoperability with other institutions and government bodies, managing data privacy in compliance with local regulations such as the Philippine Data Privacy Act, and ensuring scalability as the system expands to accommodate more credentials and users. Nevertheless, the use of blockchain represents a significant step toward digitizing and modernizing the credential verification process, making it more efficient and secure for both students and employers. TESDA's blockchain system serves as a model for how public institutions in the Philippines can harness emerging technologies to combat academic fraud and improve the reliability of academic credentialing (Bautista et al., 2023).

2.2.2.14 Blockchain in Higher Education—Partido State University & University of the Cordilleras

A study by Kennedy C. Cuya (Partido State University) and Thelma D. Palaoag (University of the Cordilleras) presents a blockchain-based framework for securing and verifying academic credentials in the Philippines. The system uses the Polygon blockchain to store hashed credentials, ensuring immutability and transparency. By decentralizing the verification process, the system allows quick, secure, and tamper-proof validation of academic records, such as diplomas and transcripts. This method eliminates the need for manual checks, preventing fraud while providing trustworthy verification that is accessible to employers and institutions without direct contact with the issuing schools.

The verification process is simple yet secure. When a student presents their credentials, they provide a QR code or hash linked to the document stored in the IPFS (InterPlanetary File System). The verifier scans the code and queries the blockchain to confirm the authenticity of the document, ensuring that it has not been tampered with and matches the

original data. IPFS allows for decentralized document storage, ensuring that large files, such as transcripts, are securely stored off-chain, with only the hash recorded on the blockchain. This method provides a scalable, cost-efficient solution, as it eliminates the need for centralized databases and enables secure, real-time verification.

In conclusion, the blockchain-based framework developed by Partido State University and the University of the Cordilleras demonstrates the feasibility of using blockchain technology in the Philippines' education sector. By using a blockchain platform for credential storage and verification, the system effectively reduces fraud, streamlines the verification process, and provides secure and easily accessible academic credentials for students, employers, and educational institutions. This model serves as an effective example for future blockchain-based academic credential systems aiming to improve efficiency, security, and trust in the education sector.

2.2.2.15 Hyperledger Fabric: Permissioned Blockchain Framework

Hyperledger Fabric is a leading open-source blockchain framework under the Hyperledger project, which is hosted by the Linux Foundation. Unlike public blockchains such as Ethereum and Bitcoin, Hyperledger Fabric is a permissioned blockchain designed specifically for enterprise use, making it an ideal solution for applications that require confidentiality, scalability, and robust access control. Its modular architecture allows organizations to tailor the blockchain network to meet specific business requirements, making it highly customizable for various use cases, which include management of credentials such as academic diplomas, professional licenses, and government-issued IDs. (Androulaki et al., 2018)

One of the key features of Hyperledger Fabric is its permissioned nature, which implies that only authorized entities such as universities, government agencies, and other trusted organizations can participate in the network. This promotes confidentiality as sensitive data remain private, as entities with proper permission are the only ones who can access and validate transactions within the network. Moreover, unlike traditional blockchain systems that require every node in the network to validate every transaction, which can lead to inefficiencies and slower processing times as the network grows, Hyperledger Fabric makes use of a module consensus mechanism, where validation is only necessary

for specific nodes (peers) involved in the transaction. This significantly reduces computational load and allows the network to process higher volumes of transactions, overall promoting scalability. Lastly, it provides strong privacy controls through features like data collections and channels. Private data collections enable the storage of sensitive information off-chain, ensuring that only authorized parties can view or access it, while channels allow for the isolation of data between different participants within the network, similar to the concept of subnets in networking. These features are particularly relevant for credentialing systems, where different types of data such as academic credentials, professional licenses, etc., may need to be kept private and only shared with specific entities. (Androulaki et al., 2018)

Furthermore, Hyperledger Fabric makes use of Chaincode, which is its version of smart contracts, to define the rules and logic for transactions. In the context of credentialing, Chaincode ensures that the processes of issuing, updating, verifying, and removing credentials are automated, secure, and transparent. This implies that when a credential is issued, it is recorded immutably on the blockchain, making it tamper-proof and easily verifiable. Moreover, the system makes it possible to allow tracking of changes to credentials such as name updates or license renewals, overall providing a full audit trail of all its transactions. (Androulaki et al., 2018)

To conclude, Hyperledger Fabric is a highly relevant and powerful framework for blockchain-based credentialing systems, given that its permission structure, scalability, privacy features, and support for smart contracts make it a robust solution for managing sensitive data securely and efficiently.

2.2.1.16 Credly by Pearson: Enterprise-Grade Digital Credentialing

Credly, a leading digital credentialing platform now owned by Pearson, enables organizations to issue, manage, and verify digital credentials at scale. This platform is trusted by over 3,500 issuers worldwide and has issued more than 110 million credentials to over 25 million individuals (Pearson, 2022). By using digital badges and certificates, Credly provides organizations with an efficient and secure way to validate skills and accomplishments. The credentials issued through Credly are enhanced with rich metadata, offering verifiable evidence of achievements, such as skill mastery and course

completion (Stefaniak & Carey, 2019). One of Credly's key strengths is its integration with various Learning Management Systems (LMS) and Human Resource Information Systems (HRIS), enabling seamless credentialing within an organization's existing workflow (Pearson, 2022). Additionally, Credly offers comprehensive workforce analytics, allowing employers and organizations to gain insights into their workforce's skills and competencies, which helps guide strategic planning and talent development (Pearson, 2022).

In Credly, the issuance and verification process starts when a member organization creates a template for a digital credential. This template can be a badge or certificate, which can be issued to users who will receive an email notifying them of the credential issuance. Upon acceptance, the badge or credential is added to the user's digital wallet within the Credly Platform. Users can then share their credentials in various ways, such as generating a link to the badge, downloading it, or sharing it on various social media platforms like LinkedIn, Facebook, or Twitter. Additionally, users can download a PDF version of their transcript that includes all their credentials (public credentials), or they can share a specific credential directly through a link. Moreover, they also have the option to share their public profile, which displays the badges and credentials they have chosen to make visible. Overall the process is designed to be seamless, secure, and flexible, offering a variety of ways for users to manage, verify, and showcase their digital credentials. (Credly Platform Walkthrough, n.d.)

Credly also ensures the security and privacy of its users through ISO certification and GDPR compliance, providing a robust framework for managing personal data (Stefaniak & Carey, 2019). This level of compliance is critical for educational institutions and employers looking to protect the integrity and privacy of their digital credentials. Pearson's acquisition of Credly in 2022 for $200 million strengthens its presence in the workforce skills sector, providing Pearson with a comprehensive suite of services ranging from learning and assessment to credentialing. This move addresses the growing demand for verified skills and competencies in the global labor market (Reuters, 2022; Corridor Business, 2022).

Credly has been effectively implemented by various organizations. For example, Nestlé Purina utilized Credly's platform to enhance their workforce's agility by providing data-driven insights into how new technologies impacted their employees' skills, driving strategic workforce planning (Credly, 2022). Additionally, Pearson itself integrated Credly's system within its own iSell initiative, upskilling its sales team and achieving a measurable return on investment through the acquisition of key competencies (LinkedIn, 2022).

2.3 Review of Related Methodologies

2.3.1 Agile Methodology

The Agile methodology, as outlined by Sarah Laoyan (2025), is a dynamic project management framework designed to promote flexibility, collaboration, and continuous improvement. Unlike traditional models such as the waterfall method, Agile breaks projects into iterative phases called sprints, allowing teams to regularly evaluate progress and adapt strategies in response to changing customer needs. Rooted in the Agile Manifesto, the framework is built on four core values—such as prioritizing individuals and interactions over processes and tools—and twelve guiding principles that emphasize early delivery, sustainable pacing, and face-to-face communication. Agile's adaptability makes it particularly effective in fast-paced environments like software development, where frequent updates, collaborative teamwork, and customer feedback are essential for success (Laoyan, 2025).

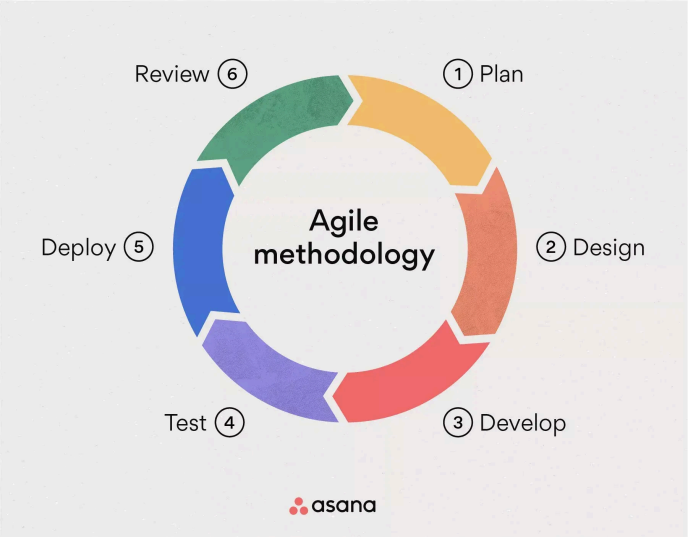


Diagram 2.1: Agile Methodology Diagram

2.3.1.1 Scrum

Scrum, as described by Atlassian, is a structured agile framework built around fixed-length iterations called sprints—typically two weeks long—that emphasize continuous delivery and improvement. Core roles include the Scrum Master, who ensures adherence to the methodology; the Product Owner, who prioritizes the product backlog; and the Development Team, which self-organizes to execute sprint work. Each sprint cycle involves four key ceremonies—Sprint Planning, Daily Stand-ups, Sprint Review, and Retrospective—which collectively support planning, progress tracking, stakeholder feedback, and process refinement. The use of visual tools like Scrum Boards enhances transparency by showing sprint backlog items moving through stages such as To Do, In Progress, and Done. Overall, Scrum offers a repeatable, collaborative, and incremental approach to product development, aligning agile philosophy with practical execution.

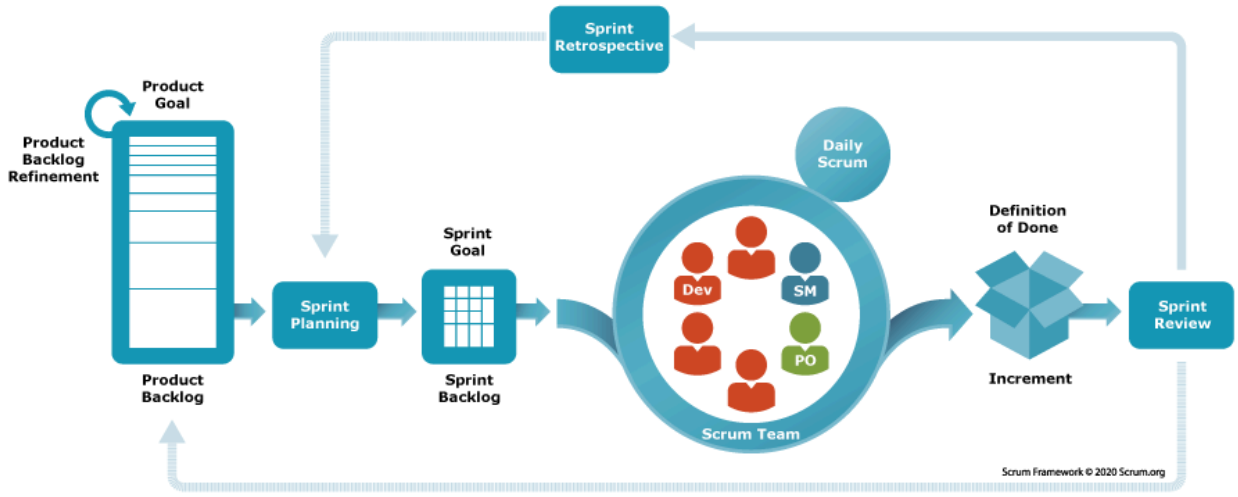


Diagram 2.2: Scrum Diagram

2.3.1.2 Kanban

Kanban is a visual workflow management methodology rooted in Agile and lean manufacturing principles, emphasizing continuous improvement and task flow efficiency. Its core components include Kanban boards, which use columns and cards to represent stages and tasks, allowing teams to visualize and manage work in real time. Key principles include starting with current processes, encouraging incremental change, respecting existing roles, and promoting leadership at all levels . Digital Kanban tools enhance this framework by offering automation, analytics, and integration features that support better visibility, efficiency, and decision-making (Martins, 2025).

2.3.2 Lean Software Development

Lean Software Development (LSD) is a methodology adapted from Lean manufacturing and popularized by Mary and Tom Poppendieck in 2003 to improve software development efficiency by minimizing waste and optimizing processes. It is grounded in seven key principles: eliminate waste, build in quality, amplify learning, delay commitment, deliver fast, respect people, and optimize the whole. LSD shares many similarities with Agile, emphasizing collaboration, continuous improvement, and flexibility, but it focuses more on improving

processes rather than just products. While it offers benefits like enhanced efficiency, quality, and customer satisfaction, LSD also presents challenges such as the need for team training, identifying the right metrics, managing evolving requirements, and avoiding suboptimization (Lutkevich, 2021).



Diagram 2.3: Lean Software Development Diagram

2.3.3 Waterfall Methodology

The Waterfall methodology, as outlined by Adobe for Business Team (2025), is a structured and linear project management approach that progresses through distinct phases—requirements, design, implementation, testing, and deployment—where each phase must be completed before the next begins. This model is especially effective in environments where project requirements are well-defined from the start and where minimal changes are expected during development. Its strength lies in thorough front-end planning and documentation, providing clear deliverables and predictable timelines. However, its rigidity can be a drawback in dynamic environments that require flexibility and frequent client feedback, making it less adaptable than iterative models like Agile (Adobe for Business Team, 2025).

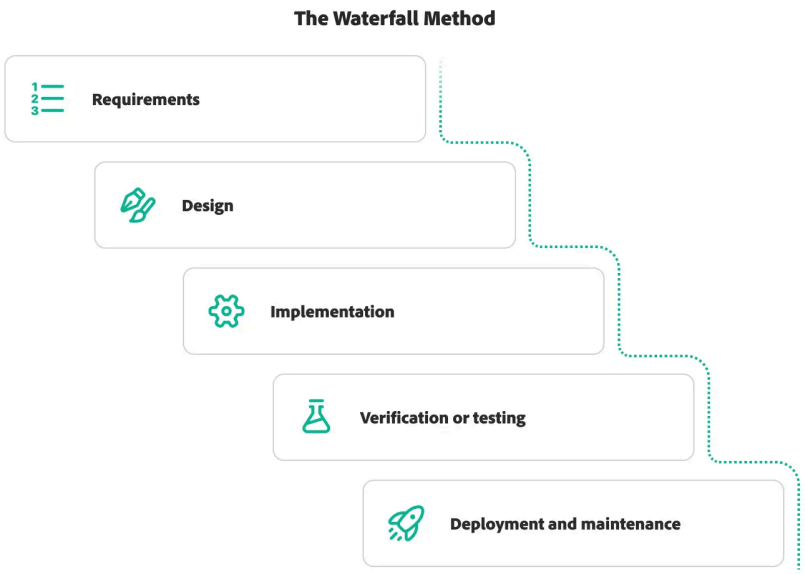


Diagram 2.4: Waterfall Diagram

2.3.4 DevOps Methodology

The DevOps methodology focuses on breaking down the traditional barriers between development and operations teams to improve collaboration, streamline workflows, and accelerate the software delivery process. As explained by Atlassian, DevOps combines a set of practices and tools—including continuous integration, continuous delivery, automated testing, and real-time monitoring—that help teams deploy code more frequently, with fewer errors and shorter lead times. This approach fosters a culture of transparency, shared responsibility, and continuous feedback, allowing teams to quickly detect and resolve issues, adapt to changes, and continuously improve system performance. Furthermore, DevOps supports better alignment between technical operations and business objectives, ensuring that software development efforts are closely tied to customer needs and organizational goals. By enabling faster, more reliable, and more secure software releases, DevOps helps businesses remain competitive in today's rapidly evolving digital landscape (Atlassian, n.d.).

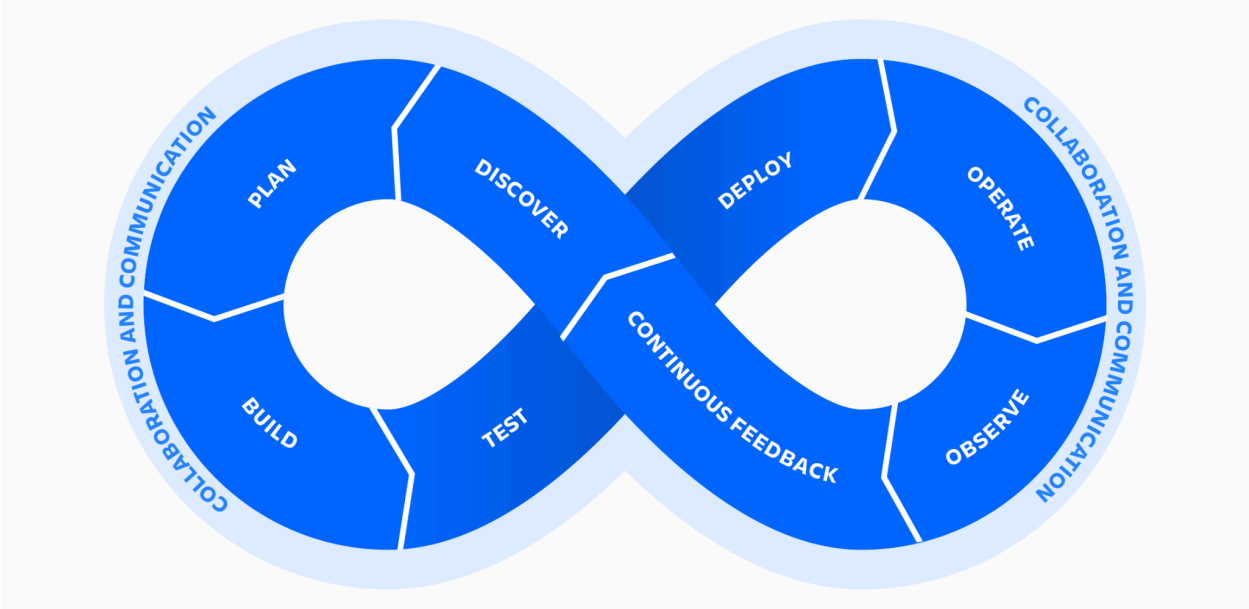


Diagram 2.5: DevOps Lifecycle

2.3.5 Whys Analysis

The 5 Whys Analysis is a root cause analysis methodology that involves drilling down the problem by repeatedly asking the question "Why?", specifically five times to explore the underlying cause of a problem. It was originally developed by Sakichi Toyoda and widely used with Toyota's manufacturing process, which is a technique that is simple yet effective for identifying the root cause of an issue rather than just addressing the symptoms. Overall, its strength lies in its clarity and accessibility; it does not require statistical tools or complex data, making it suitable for both technical and non-technical teams. (Wolniak & Grebski, 2023).

2.3.6 User Acceptance Testing (UAT)

User Acceptance Testing (UAT), also known as Application Testing or End-User Testing, is the final phase of the software development lifecycle where actual users validate whether a system meets business requirements and functions as intended in real-world scenarios. It serves as a last checkpoint before production release, ensuring that all critical features work correctly and intuitively from the end-user perspective. UAT involves planning test objectives, designing real-world scenarios, preparing a production-like environment, executing tests, logging issues, retesting, and securing user sign-off. Common types of UAT include Alpha Testing, Beta Testing,

Black Box Testing, Operational Acceptance Testing (OAT), Contract Acceptance Testing (CAT), and Regulation Acceptance Testing (RAT).

UAT is essential for catching defects that internal testing may overlook, especially those related to usability, clarity, or business alignment. It improves transparency, reduces the risk of costly post-release errors, and ensures the solution delivers value to its intended users. Successful UAT requires collaboration between product owners, business analysts, QA testers, and actual end users. Best practices include thorough documentation, proper tester selection, and use of test management and issue tracking tools like TestRail and Jira. Ultimately, UAT ensures that the software not only works, but also works for the people it's meant to serve (Gillis, 2022; TestRail, n.d.)

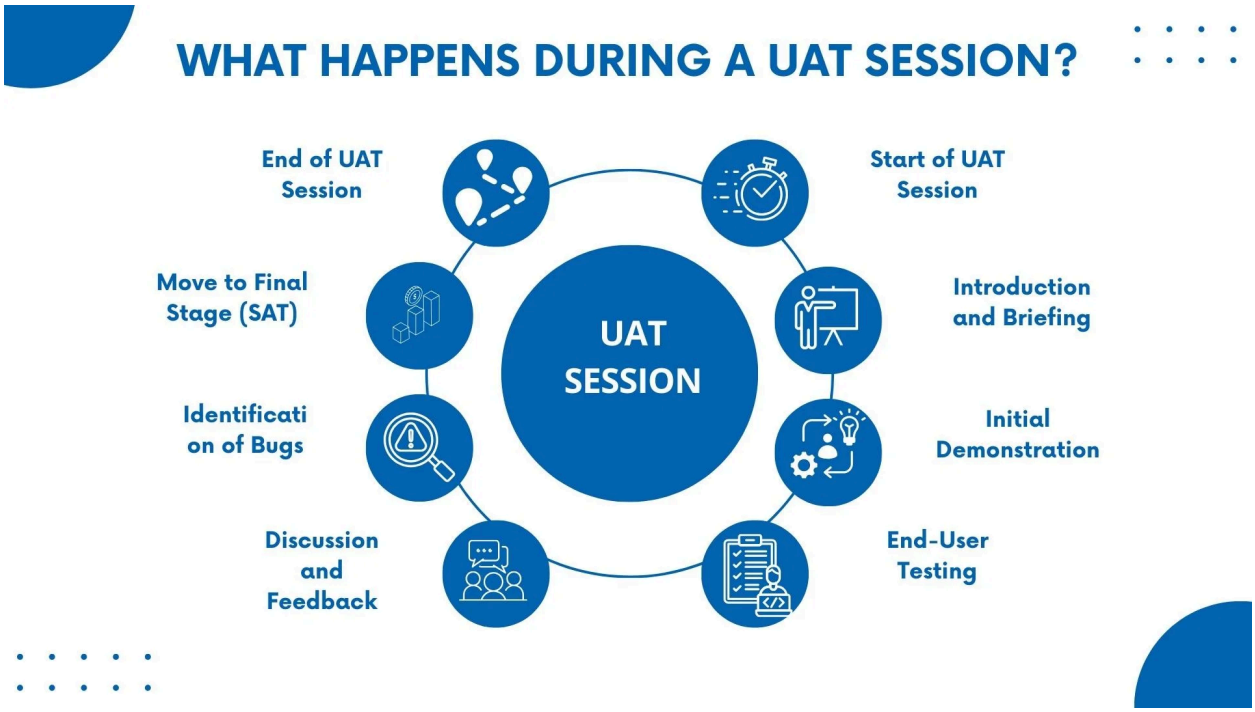


Diagram 2.6 User Acceptance Testing (UAT) Session Workflow

Chapter 3

3.1 Description

This study will utilize the Agile Scrum Framework as its primary methodology. Agile Scrum is a flexible and iterative software development framework that allows rapid and responsive progress through regular iterations known as sprints. The Scrum process is especially effective in projects requiring frequent adjustments and incremental improvements, such as software development for credential verification systems like ChainCert. The rationale for choosing Agile Scrum over traditional models like the Waterfall is its emphasis on adaptability, stakeholder collaboration, and early delivery of functional components. Unlike Waterfall, which follows a linear path, Scrum encourages the continuous delivery of workable outputs at the end of each sprint. This results in early detection of issues, quick feedback loops from stakeholders (including users, institutions, organizations, and employers), and rapid system enhancement. The iterative nature of Scrum makes it well-suited for ChainCert, a system integrating blockchain technology where requirements may evolve based on user testing and real-world challenges.

3.2 Scrum Methodology Phases

3.2.1 Initiation

The initiation phase established the framework for developing the Academic Credential Verification System. In this stage, the product vision was set to create a decentralized, blockchain-based platform to securely verify academic credentials and combat fraudulent claims. The product owner outlined the essential features, including academic credential issuance, real-time verification, and secure data storage. A project team was formed, with roles such as the Scrum Master, Product Owner, and Product Developers. Initial discussions were conducted with HR recruiters, registrars, and research advisers to gather requirements and ensure that the system's objectives are aligned with the needs of educational institutions and employers, focusing on creating a trustworthy and efficient verification process.

3.2.2 Planning

In the planning phase, the team conducted sprint planning sessions to break down the product backlog into manageable sprint backlogs. Prioritized features included credential issuance via smart contracts, document encryption, and user access management. Early sprints targeted the

implementation of core blockchain functionality and wallet integration, while later sprints focused on verification, role-based dashboards, and security enhancements. Tasks were clearly defined with time estimates, ensuring that each sprint delivered a functional, testable product increment that moved the system closer to its overall goal.

3.2.3 Implementation Phase

The initiation phase laid the foundation for the development of the ChainCert Academic Credential Verification System. During this stage, the product vision was clearly defined—to create a decentralized, permissioned blockchain platform for secure credential issuance and verification. Key roles such as the Product Owner, Scrum Master, and Development Team were assigned, and initial planning sessions were conducted. Stakeholder consultations with HR recruiters, university registrars, and the research adviser helped gather system requirements and align project goals with real-world needs. Core features, including tamper-proof credential issuance, blockchain-based validation, and encrypted document storage in CouchDB, were identified as development priorities.

3.2.4 Review and Retrospective Phase

At the end of each sprint, the Scrum team held review meetings to present completed features to stakeholders for feedback. Demonstrated components included issuer dashboards, verifier portals, and wallet functionalities. Feedback from registrars, employers, and users guided backlog refinement and helped prioritize future development. Retrospective meetings followed, where the team assessed what went well, identified pain points, and discussed actionable improvements. These continuous reflections promoted better collaboration, higher efficiency, and overall process improvement for upcoming sprints.

3.2.5 Release

The release phase marked the deployment of a stable, functional version of ChainCert. After several completed sprints and thorough validation, the system’s core modules—secure

credential issuance, blockchain verification, wallet management, and encrypted storage—were finalized and tested. A comprehensive quality assurance process ensured system reliability, security, and user-friendliness. Upon stakeholder approval, the platform was rolled out to target users, including registrars, students, and verifiers. Training materials and user guides were provided, and the team committed to ongoing monitoring, bug fixes, and iterative enhancements based on user feedback.

3.3 Phases

3.3.1 Product Backlog

The product backlog is a dynamic, prioritized list of all features, improvements, technical tasks, and bug fixes required to develop the ChainCert credential verification system. Maintained by the Product Owner, it evolves continuously based on stakeholder feedback, sprint reviews, and technical discoveries. Initial backlog items include modules for credential issuance via smart contracts, secure document storage in CouchDB, blockchain-based verification logic, wallet integration, admin and verifier portals, and QR code functionality. Each item is documented with clear acceptance criteria and refined during backlog grooming to ensure alignment with the Definition of Ready (DoR).

3.3.1.1 Tools

Google Workspace

The team will use Google Workspace for creating and organizing documentation, managing spreadsheets, and storing shared files. This ensures centralized access to all project-related materials.

Messenger

Messenger will be used for quick, real-time communication among team members to coordinate tasks and discuss updates efficiently.

Discord

The team will use Discord for voice meetings, collaborative discussions, and progress tracking, providing a more interactive communication platform during development.

3.3.1.2 Input

The inputs to the product backlog come from various stakeholders and project-related sources, including but not limited to

- **Stakeholder Requirements**: Initial project goals and features identified by clients, educational institutions, employers, and regulatory bodies.
- **User Stories**: Feedback from users such as students, employers, and university staff is gathered through interviews, surveys, or system testing.
- **Technical Constraints and Opportunities**: Recommendations by developers based on current technologies (e.g., blockchain limitations or opportunities with smart contracts).
- **Security and Compliance Needs**: Requirements driven by data privacy laws (e.g., GDPR, Philippine Data Privacy Act) and the need to secure academic records.
- **Past Sprint Reviews**: Items identified during retrospectives for improvement or newly discovered bugs and inefficiencies.
- **Research Insights**: Findings from literature reviews and benchmarking of similar systems like OpenCerts or BCdiploma.

3.3.1.3 Output

The outputs of the product backlog include a well-defined, prioritized, and constantly evolving list of features and tasks, specifically

- **Prioritized Backlog Items**: Organized by value, urgency, and dependencies, covering features such as
    - Secure issuance and revocation of credentials
    - Blockchain anchoring and state management via Hyperledger Fabric
    - QR code generation and scanning for credential verification
    - Encrypted digital wallets for secure credential storage access
    - Web-based employer and verifier portal for real-time credential verification
- **Sprint-Ready User Stories**: Refined entries that meet the Definition of Ready (DoR), complete with acceptance criteria.
- **Technical Tasks and Bugs**: Development-level breakdown of features, performance optimizations, and resolutions of system issues.

- **Documentation Tasks**: Needed documentation for usage, security protocols, and API integration guides.
- **Updated Project Roadmap**: Reflected in tools like Google Sheets or Jira, showing completed, in-progress, and upcoming features.

3.3.2 Sprint Planning

Sprint Planning is the initial step of every sprint cycle and serves to align the group's efforts toward specific, achievable goals. The group will conduct a Sprint Planning session before the start of each sprint to define what can be delivered and how the work will be completed. This session will involve the Product Owner, Scrum Master, and development members of the team. The group will use Sprint Planning to establish clear objectives for each sprint by selecting user stories or tasks from the product backlog and translating them into actionable items. This ensures that all members are synchronized and working efficiently toward shared goals.

3.3.2.1 Tools

Google Workspace

The group will use Google Workspace to create and maintain documents related to sprint planning, including the sprint backlog, task breakdowns, and sprint goals. Google Sheets will be used for organizing task assignments and tracking progress, while Google Docs will serve as a repository for planning notes and documentation. This ensures centralized and collaborative access to all sprint-related materials.

Messenger

The group will use Messenger for quick and informal communication during the planning phase. It will allow team members to clarify tasks, confirm availability, and provide timely updates regarding sprint preparations and task coordination.

Discord

The group will use Discord as the primary platform for conducting voice meetings and collaborative sprint discussions. It will serve as a space for real-time coordination, sprint goal confirmation, and resolution of questions

during the sprint planning session. Discord's channel system will also help keep conversations organized by topic.

3.3.2.2 Input

The group will use several key inputs during Sprint Planning. These include the refined product backlog, which contains well-defined and prioritized items; the team's historical velocity, which helps in determining how much work can be taken on; the Definition of Ready (DoR), which ensures backlog items are sprint-ready; and a proposed sprint goal that outlines the high-level objective for the sprint.

3.3.2.3 Output

The outputs of Sprint Planning will include a clearly defined sprint goal, a sprint backlog containing selected and broken-down tasks, assigned responsibilities for each member, and an initial schedule for completing the work. The group will also leave the session with a shared commitment to completing the sprint's objectives within the given timeframe.

3.3.3 System Deployment

System deployment marks the transition from the development phase to a live, functional environment where the ChainCert system becomes accessible to its end users—students, issuers, and verifiers. This process involves deploying the Hyperledger Fabric network using Docker containers, configuring peer nodes and ordering services, and integrating CouchDB as the world state database for storing base64-encoded credential documents.

A central part of the deployment process is the development and installation of custom chaincode (smart contracts), which the team will write specifically for the project. These smart contracts are responsible for managing credential issuance, revocation, document status changes, access control, and audit logging. Once developed and tested, the chaincode is packaged and installed onto Fabric peer nodes and instantiated on the channel.

Backend services developed in Node.js with the Hyperledger Fabric SDK handle the interaction between the blockchain and the system's web applications. These web interfaces—developed using React.js or Vue.js—serve as the Admin Dashboard, Verifier Portal,

and Student Digital Wallet. The system will undergo thorough integration and validation testing, which includes chaincode execution, API testing, role-based access simulation, encryption checks, and real-time document retrieval via CouchDB. Proper system configuration and environment variable management will also be completed to ensure secure and reliable operation in the deployed setting.

3.3.3.1 Tools

Github

The group will use GitHub to store, manage, and version the source code of the Academic Credential Verification System. It will serve as the central platform for collaboration, allowing developers to track changes, resolve issues, and coordinate updates during deployment.

Docker & Docker Compose

Docker and Docker Compose will be utilized to deploy and manage the Hyperledger Fabric network infrastructure. This includes setting up peers, orderers, CouchDB, and Certificate Authorities in a containerized and reproducible environment.

Node.js & Express

Node.js and Express will power the backend server, enabling it to handle API requests and communicate securely with the blockchain network through the Hyperledger Fabric SDK. This layer is responsible for executing smart contract functions and managing user operations.

React.js

React.js will be used to develop the front-end interfaces for students, credential issuers, and verifiers. It ensures a responsive and interactive user experience across the system's portals.

Hyperledger Fabric SDK

The Hyperledger Fabric SDK for Node.js will facilitate interaction between the backend server and the blockchain network. It allows the system to invoke and query smart contracts deployed on the Fabric peers.

Postman

Postman will be used for testing and validating API endpoints throughout development and deployment. It helps ensure that backend services correctly process requests and respond with expected data.

VS Code

Visual Studio Code (VS Code) will serve as the main development environment for writing and debugging the system's chaincode, backend logic, and frontend components. Its extensions and integrations enhance productivity during system implementation.

3.3.3.2 Input

The successful deployment of the ChainCert system will rely on a range of key inputs. These include the finalized and tested source code for the front-end, backend, and custom-developed chaincode (smart contracts) written by the team. Docker configuration files will be used to set up the Hyperledger Fabric network components, including peers, orderers, Certificate Authorities, and CouchDB. Environment variables and secure configuration files containing credentials, peer addresses, and API keys will be prepared to ensure proper connectivity between system components. Additional inputs include test credential data used for validating issuance and verification workflows, as well as deployment documentation that outlines infrastructure setup, system architecture, and integration procedures. Together, these inputs ensure that all elements of the system are properly configured, tested, and ready for deployment.

3.3.3.3 Output

The outputs of the system deployment phase will result in a fully operational and accessible version of ChainCert running in a live or staging environment. A complete Hyperledger Fabric network will be deployed with functional chaincode developed by the

team to manage credential logic. All major system components—such as credential issuance, wallet access, blockchain-based verification, and QR code validation—will be confirmed to operate correctly through comprehensive system testing. The backend and frontend will be fully integrated, with real-time interactions between the web interfaces, CouchDB, and the blockchain network. Logs, performance metrics, and runtime monitoring tools will provide insights into the system's behavior, while pilot users such as registrars and students will receive access credentials for initial testing. Feedback gathered during this stage will guide final refinements and inform future enhancements of the platform.

3.3.4 Unit Testing

User Acceptance Testing (UAT) will be conducted to validate that the ChainCert Credential Verification System meets both functional and non-functional requirements as defined by its stakeholders. This includes verifying that core features such as credential issuance, blockchain-based verification, QR code validation, and secure document access through the wallet interface are working as intended in real-world scenarios. The group will collaborate with representative end users—including students, employers, and university staff—to simulate actual workflows and gather qualitative and quantitative feedback. UAT will focus on identifying usability challenges, ensuring data accuracy, validating security features, and confirming the overall reliability of the system.

In line with usability testing best practices, the team will apply sample size recommendations to guide participant selection. For qualitative testing, a group of 5 to 8 participants will be recruited to observe user interactions, identify usability issues, and evaluate design clarity. For broader quantitative validation, especially to measure task success rates and support stakeholder confidence, the team may extend testing to 30 to 40 participants or more. Larger sample sizes (e.g., 15 for ~90% issue detection or 50 for ~98%) may be considered for high-risk areas such as credential revocation, blockchain anchoring, or digital wallet security. The team will also take into account factors such as testing goals, decision-maker expectations, confidence levels (typically 90–95% for industry-level UX validation), and user availability within the Philippine academic and employment sector (UserTesting, 2025). Feedback collected during UAT will directly inform final interface refinements, feature adjustments, and system stability improvements before official launch.

3.3.4.1 Tools

User Acceptance Testing

The group will conduct structured UAT sessions to gather direct feedback from users and stakeholders. This will validate whether the system meets real-world expectations and is ready for production deployment.

Test Cases

Detailed test cases will be prepared, outlining specific inputs, expected outputs, and acceptance criteria for each major feature. These serve as benchmarks during UAT to ensure that functionality, data integrity, and system responsiveness are thoroughly evaluated.

3.3.4.2 Input

Inputs for UAT will include the fully deployed ChainCert system in a controlled environment, access credentials for selected test participants, structured UAT feedback forms or digital questionnaires, and the original user requirement documents. These resources will guide test participants in validating if system behavior aligns with expectations for critical functionalities such as credential issuance, QR scanning and verification, wallet retrieval, and role-based access controls.

3.3.4.3 Output

The outputs of UAT will consist of completed feedback forms, usability issue logs, test result summaries, and recommended changes. These documents will help the team assess which aspects of the system require refinement—whether in terms of user interface design, technical performance, or blockchain integration. The feedback will also help finalize system readiness, guide bug fixes, and improve user satisfaction prior to the system's public rollout.

3.4 Work Plan/Gantt Chart

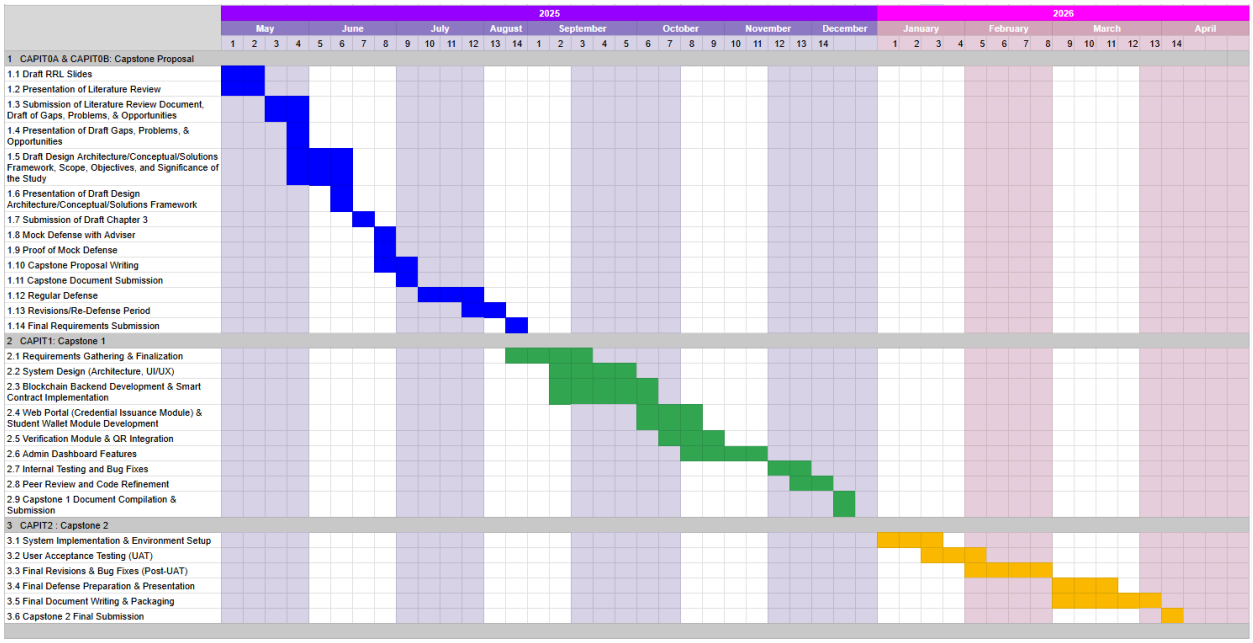


Figure 3.1: Capstone Project Phases and Timeline

The Capstone project is structured into three key phases: Capstone Proposal, Capstone 1, and Capstone 2. The Capstone Proposal phase (May to July 2025) focuses on the foundational groundwork, including the literature review, identification of research gaps, and development of the conceptual and solution frameworks. It concludes with proposal presentations, mock defenses, and final requirement submissions based on panel feedback. Capstone 1 (August to December 2025) centers on the initial development of the system, covering core modules and major technical components. Capstone 2 (January to April 2026) follows with the completion, testing, and refinement of the project. The updated timeline reflects a clear division of tasks and progress, with approximately 80% of the system completed during Capstone 1 and the remaining 20% finalized in Capstone 2. This phased and iterative approach supports continuous improvement, timely deliverables, and alignment with academic milestones.

3.5 Capstone 1 and 2 Targets

3.5.1 Capstone 1 (80% of the development)

3.5.1.1 System Architecture, Requirements, and User Roles

The team will begin by gathering and finalizing system requirements through consultations with stakeholders. This will include defining user stories and role-based access for four primary users: students (credential holders), registrars (credential issuers), employers (verifiers), and system administrators. The system architecture will be designed to ensure modularity, scalability, and secure role separation. UI/UX mockups will be created for all core interfaces, emphasizing intuitive navigation and accessibility across different devices.

3.5.1.2 Blockchain Backend and Smart Contract Implementation

The blockchain backend will be developed using Hyperledger Fabric to securely handle credential issuance, verification, and revocation. Smart contracts (chaincode) will be written and deployed to automate these processes and record credentials immutably on the ledger. Logic will be implemented to support lifecycle events such as issuance timestamps, revocation flags, and credential identifiers, ensuring data integrity and auditability.

3.5.1.3 Credential Verification System

Multiple functional modules will be implemented, starting with the Credential Issuance System, which will allow registrars to issue digital credentials and write them to the blockchain. The Student Wallet Module will be developed to allow students to receive, view, and store credentials, with QR generation and public share links for verification. The Verification System will be created to enable third-party verifiers to scan QR codes or enter credential IDs to validate credentials against blockchain records in real time.

3.5.1.4 Administrator Dashboard

An Administrator Dashboard will be developed to provide visibility into system activity, including real-time logs of issuance and verification events, user management, and platform status. Initial rounds of internal testing will be conducted for all core features, focusing on integration stability and logic correctness. Peer code reviews and security checks will also be performed to ensure quality before progressing to Capstone 2.

3.5.2 Capstone 2 (20% of the development)

3.5.2.1 System Implementation & Environment Setup

In this phase, the team will prepare the production-ready version of the system and deploy it to a secure, cloud-hosted environment. This includes configuring the application server, database connections, blockchain components, and front-end interfaces. Environment variables, APIs, and smart contract interactions will be finalized to ensure seamless communication between all modules. Monitoring and backup protocols will also be configured to support system stability and availability.

3.5.2.2 User Acceptance Testing (UAT)

After deployment to the testing environment, User Acceptance Testing (UAT) will be conducted with key stakeholders such as students, university registrars, and verifiers. Each user group will validate their assigned workflows, including credential issuance, wallet access, and real-time verification. Feedback will be gathered using structured test scripts and observation logs to identify issues related to usability, functionality, or performance.

3.5.2.3 System Testing and User Acceptance Testing (UAT)

Following UAT, the team will implement final revisions based on the feedback received. This may include fixing bugs, enhancing user flows, optimizing system performance, and refining interface elements. All fixes will undergo regression testing to ensure that existing features remain functional and no new issues are introduced.

3.5.2.4 Deployment and Final Documentation

Upon successful revision and validation, the system will be officially deployed

in its final environment. The team will also complete and submit all required documentation, including user manuals, technical documentation, deployment guides, architectural diagrams, and presentation materials. These documents will support end users and developers and serve as the official record for Capstone 2 evaluation and defense.

References:

A.M. No. P-10-2833. (n.d.). https://lawphil.net/judjuris/juri2010/dec2010/am_p-10-2833_2010.html

Androulaki, E., Barger, A., Bortnikov, V., Cachin, C., Christidis, K., De Caro, A., ... & Yellick, J. (2018, April). Hyperledger fabric: A distributed operating system for permissioned blockchains. In Proceedings of the thirteenth EuroSys conference (pp. 1-15). https://doi.org/10.1145/3190508.3190538

Atlassian. (n.d.-a). *What is DevOps? | Atlassian*. https://www.atlassian.com/devops

Attewell, P., & Domina, T. (2010). Educational imposters and fake degrees. Research in Social Stratification and Mobility, 29(1), 57–69. https://doi.org/10.1016/j.rssm.2010.12.004

Bapat, C. (2019). *Blockchain for Academic Credentials*. https://arxiv.org/pdf/2006.12665

Bautista, P. S. (n.d.). Act No. 3815. https://lawphil.net/statutes/acts/act1930/act_3815_1930.html

BCdiploma. (n.d.). *BCDiploma: Leading Digital Credential Software*. BCdiploma. https://www.bcdiploma.com/en

BCdiploma. (n.d.). *Diplomas certified on the blockchain: Trusted certification. Forever.* https://www.bcdiploma.com

Castro, R. Q., & Au-Yong-Oliveira, M. (2021). Blockchain and higher education diplomas. *European Journal of Investigation in Health, Psychology and Education, 11*(1), 154–167. https://doi.org/10.3390/ejihpe11010013

Castro, R. Q., & Au-Yong-Oliveira, M. (2021). Blockchain and Higher Education Diplomas. *European Journal of Investigation in Health, Psychology and Education*, *11*(1), 154–167. https://doi.org/10.3390/ejihpe11010013

Credly Platform Walkthrough. (n.d.). https://learn.credly.com/credly-platform-walkthrough?_gl=1*4um5fl*_gcl_au*MTExMjQyMjU0NS4xNzU0MDI1OTg4*_ga*NDY2ODA2NzMuMTc1Mzk2MjUwNQ..*_ga_Z763G7NB5L*czE3NTQwMzUyMzMkbzIkZzEkdDE3NTQwMzczNzUkajYwJGwwJGgw

Credly. (2020). *Why choose Credly?* Credly.com. https://info.credly.com/why-choose-credly

Credly. (2024, April 26). *Pearson's Standardized Credentialing Framework to Enable Business Success*. Credly.com; Credly, Inc. https://learn.credly.com/case-study/pearson

Crosby, M., Pattanayak, P., Verma, S., & Kalyanaraman, V. (2016). Blockchain technology: Beyond bitcoin. Applied Innovation Review, 2, 6-10.

Cuya, K. C., & Palaoag, T. D. (2024). Blockchain in Higher Education: Advancing Security, Verification, and Trust in Academic Credentials. Nanotechnology Perceptions, 20(23), 373–386. https://www.researchgate.net/publication/381887376_Blockchain_in_Higher_Education_Advancing_Security_Verification_and_Trust_in_Academic_Credentials

Ding, Q., Gao, S., Zhu, J., & Yuan, C. (2020). Permissioned blockchain-based double-layer framework for product traceability system. IEEE Access, 8, 6209-6225. https://doi.org/10.1109/ACCESS.2019.2962274

Dmw, P. B. I. I. B. (n.d.). *DMW | Department of Migrant Workers*. https://dmw.gov.ph/

Diplomasafe. (2025, April 29). Empower Success with Advanced Digital Credentials - Diplomasafe. Diplomasafe. https://diplomasafe.com/

Eliminating fraud in the verification of academic certificates. (2019). Exonum.com. https://exonum.com/story-russia

Embassy of the Republic of the Philippines. (2025, July 17). Other services - Embassy of the Republic of the Philippines. https://philippineembassy-dc.org/other-services/

Emma. (2024, February 3). Verification Employment Contract Guide for OFWs. Techy Kick. https://techykick.com/employment-contract-verification-guide/

Esguerra, M., Keno Piad, Tano, I., Victoriano, J., Espino, J., & Lagman, A. (2024). Smart Credentialing and Verification System for National Certificates using Blockchain Technology. 183–187. https://doi.org/10.1145/3696230.3696266

European Blockchain Services Infrastructure. (n.d.). Shaping Europe's Digital Future. https://digital-strategy.ec.europa.eu/en/policies/european-blockchain-services-infrastructure

FilipiKnow. (2025, February 27). How to renew PRC License 2025 online without CPD units. FilipiKnow. https://filipiknow.net/prc-license-renewal/

Firmo, M. G. (2021). Fake Diplomas and Signaling in Labor Markets: Evidence from Brazil.

Freeman, S., Jr, Karkouti, I. M., & Douglas, T. (n.d.). Avoiding fake degrees from diploma mills: Recommendations for educators and academic administrators. AUC Knowledge Fountain. https://fount.aucegypt.edu/faculty_journal_articles/1687/ cincon

Gillis, A. S. (2022, March 14). What is user acceptance testing (UAT)? Search Software Quality. https://www.techtarget.com/searchsoftwarequality/definition/user-acceptance-testing-UAT

Habib, M. A., Rahman, M. M., & Neom, N. H. (2024, June 4). *CreDSEC: a blockchain-based secure credential management system for university adoption*. arXiv.org. https://arxiv.org/abs/2406.05151

Hallak, J., & Poisson, M. (2007). Academic fraud, accreditation and quality assurance: learning from the past and challenges for the future. http://hdl.handle.net/2099/8108

History | Professional Regulation Commission. (n.d.). https://www.prc.gov.ph/history

Hoover, E., & Millman, S. (2007). Shocking Admission. The Chronicle of Higher Education, 53(36). https://link.gale.com/apps/doc/A163163095/AONE?u=anon~7d53bb4&sid=googleScholar&xid=dd1341e9

Hyperledger Foundation. (2025, April 23). Hyperledger Fabric. Hyperledger Fabric. https://www.lfdecentralizedtrust.org/projects/fabric

Jimu, I. M. (2018). Fake Qualifications and the Challenge of regulating Higher Education in Southern Africa. Modern Africa Politics History and Society, 6(1), 107–134. https://doi.org/10.26806/modafr.v6i1.236

Johnson, C. (2006). Degrees of deception: Are consumers and employers being duped by online universities and diploma mills? Journal of College and University Law, 32(1), 101–123. Ohio State Public Law Working Paper No. 78. Center for Interdisciplinary Law and Policy Studies Working Paper No. 51. https://ssrn.com/abstract=925237

Kayyali, M. (n.d.). University rankings as a tool to fight diploma mills. https://eric.ed.gov/?q=fake+diploma&id=ED626358

Keystone Education Group. (2024, May 2). The growing importance of micro-credentials in higher education. Keystone Education Group. https://www.keg.com/news/the-rising-significance-of-micro-credentials-in-higher-education

Laoyan, S. (2025, February 20). What is Agile Methodology? (A Beginner's Guide) [2025] • Asana. *Asana*. https://asana.com/resources/agile-methodology

Lückerath-Rovers, M. (2024). The Case of Yahoo and the CEO Who Lied on His Resume. In Moral Dilemmas in the Boardroom (pp. 147–154). https://doi.org/10.1007/978-3-031-65269-1_11

Lutkevich, B., & Silverthorne, V. (2021, December 6). *Lean software development*. Search Software Quality. https://www.techtarget.com/searchsoftwarequality/definition/lean-programming

Lyoshin, V., & Lyoshin, V. (2025, April 12). *What is Lean Software Development (LSD)*. Vit Lyoshin. https://vitlyoshin.com/blog/lean-software-development/

Mallory, M. (2014). *A Research Proposal: Negligent Hiring and Fraudulent Resumes*. https://scholarsarchive.jwu.edu/cgi/viewcontent.cgi?article=1031&context=mba_student

Manila Bulletin. (2025, February 27). Man arrested for falsification of diplomas in Manila. Manila Bulletin. https://mb.com.ph/27/2/2025/man-arrested-for-falsification-of-public-documents-in-manila

Marketing. (2024, November 19). The impact of certificate fraud and how education institutions can prevent it? | Advanced Secure Technologies. *Advanced Secure Technologies*. https://www.advancedsecure.co.uk/what-is-the-impact-of-certificate-fraud/

Martins, J. (2025, January 19). What is Kanban? A Beginner's Guide for Agile Teams [2025] • Asana. *Asana*. https://asana.com/resources/what-is-kanban

Mitani, T., & Otsuka, A. (2020). Traceability in permissioned blockchain. IEEE Access, 8, 21573-21588. https://doi.org/10.1109/ACCESS.2020.2969454

Mortensen, K. S. (2023, November 22). *The Effect of Digital Credentials on the Modern Workforce | Diplomasafe*. Diplomasafe; Diploma Safe. https://diplomasafe.com/digital-credentials-in-the-workforce/

Moya, J. a. B. (2024, June 17). *Blockchain for Academic integrity: Developing the Blockchain Academic Credential Interoperability Protocol (BACIP)*. arXiv.org. https://arxiv.org/abs/2406.15482

Mrgx. (2024, August 27). DMW explained: Department of Migrant Workers | Financial Assistance Philippines. Financial Assistance Philippines. https://financialassistance.ph/dmw-department-of-migrant-workers/

Mortensen, K. S. (2023, May 29). The Future of Digital Credentials: What to Expect - Diplomasafe. Diplomasafe. https://diplomasafe.com/the-future-of-digital-credentials/

Nama-Admin. (2023, May 10). PRC verification: Checking validity and status of my professional license. It's More Fun With Juan. https://www.morefunwithjuan.com/2023/05/prc-license-verification.html#google_vignette

Narayanan, A., Bonneau, J., Felten, E. W., Miller, A., & Goldfeder, S. (2016). *Bitcoin and cryptocurrency technologies: A comprehensive introduction*. Princeton University Press. https://bitcoinbook.cs.princeton.edu/

Narayanan, A., Bonneau, J., Felten, E., Miller, A., & Shasha, D. (2016). Bitcoin and Cryptocurrency Technologies. Princeton University Press.

NUS pilots first-in-market green bond reporting tokenisation initiative with Northern Trust and UOB. (n.d.). Northern Trust.

https://www.northerntrust.com/europe/pr/2025/nus-pilots-first-in-market-green-bond-reporting-tokenisation-initiative-with-northern-trust-and-uob

NUS to tokenise its green bonds using blockchain. (n.d.). https://govinsider.asia/intl-en/article/nus-to-tokenise-its-green-bonds-using-blockchain

Observatory of Public Sector Innovation. (2021, November 15). *OpenCerts - Observatory of Public Sector Innovation*. Observatory of Public Sector Innovation. https://oecd-opsi.org/innovations/opencerts/

OpenCerts. (2025). Smartnation.gov.sg. https://www.smartnation.gov.sg/initiatives/opencerts/

Osoka, A. (2023b, October 18). *Akowe's blockchain solution set to transform Africa's educational sector*. Mariblock. https://www.mariblock.com/akowes-blockchain-solution-set-to-transform-africas-educational-sector/

OWWA ECARD. (n.d.). https://ecard.owwa.gov.ph/

Owwamember. (2023, November 12). *How to become an OWWA member? | OWWA Member*. OWWA Member. https://owwamember.com/how-to-become-owwa-member/

Owwamember. (2023, November 12). How to process employment contract verification for OFWs | OWWA Member. OWWA Member. https://owwamember.com/contract-verification/#google_vignette

Pandya, N. S. (2022). Innovative blockchain solutions for enhanced security and verifiability of academic credentials. *International Journal of Science and Research Archive*, *6*(1), 347–357. https://doi.org/10.30574/ijsra.2022.6.1.0225

PDF-LIB · Create and modify PDF documents in any JavaScript environment. (n.d.). https://pdf-lib.js.org/

PDF.js - home. (n.d.). https://mozilla.github.io/pdf.js/

Pearson. (2025). *Pearson Acquires Digital Credentialing Leader Credly*. Pearsonvue.com. https://www.pearsonvue.com/us/en/about/news/2022/pearson-acquires-digital-credentialing-leader-cred.html

Philippine National Privacy Commission. (2021). *Data Privacy Act of 2012 (Republic Act No. 10173)*. Retrieved from https://www.privacy.gov.ph/

Philippine Overseas Employment Administration Pre-Employment Service Office-Landbased Center REQUIREMENTS FOR EVALUATION OF DIRECT HIRE APPLICATION PROFESSIONAL AND SKILLED WORKERS. (n.d.). Retrieved August 3, 2025, from https://dmw.gov.ph/archives/services/workers/Professional_Checklist.pdf

Philippines documents & compliance. (n.d.). https://recruit-nz.co.nz/philippines-documents-compliance/

PRC. (n.d.). LERIS | Professional Regulation Commission. https://online.prc.gov.ph/

Pulmano, C., Estuar, M. R. J., De Leon, M. M., Tan, H. C. L., Co, N. a. S., & Tamayo, L. P. V. (n.d.). *Towards the Development of a Blockchain-based Decentralized Digital Credential System using Hyperledger Fabric for Participatory Governance*. Archīum Ateneo. https://archium.ateneo.edu/discs-faculty-pubs/418/

Putri. (2024, December 18). The global rise of fake degrees: Six major scandals in 2024 - Education Verification. Education Verification. https://www.education-verification.com/the-global-rise-of-fake-degrees-six-major-scandals-in-2024/

Quigley, J. (2024, August 21). What Are Digital Credentials? *What Are Digital Credentials and How Are They Used?* https://www.accredible.com/blog/what-are-digital-credentials

R.A. 6715. (n.d.). https://lawphil.net/statutes/repacts/ra1989/ra_6715_1989.html

R.A. 8981. (n.d.). https://lawphil.net/statutes/repacts/ra2000/ra_8981_2000.html

Rahman, T., Mouno, S. I., Raatul, A. M., Azad, A. K. A., & Mansoor, N. (2023, July 11). *Verifi-Chain: A credentials verifier using blockchain and IPFS*. arXiv. https://doi.org/10.48550/arXiv.2307.05797

Ristenpart, T., Shacham, H., & Sharp, M. (2017). Security of Public Key Infrastructure: Attacks and Defenses. IEEE Transactions on Security and Privacy, 13(3), 87-98.

Ristenpart, T., Yilek, S., & Halderman, J. A. (2017). Security pitfalls in PKI-based digital signature systems. *IEEE Security & Privacy, 15*(6), 62–70. https://doi.org/10.1109/MSP.2017.4251114

Rohaidi, N. (2019, May 24). *Singapore fights education fraud with blockchain-powered platform*. Govinsider.asia. https://govinsider.asia/intl-en/article/singapore-education-fraud-blockchain-opencerts-steven-koh-govtech-patrice-choong-np

Romano, M. (2025, April 11). The Professional Regulation Commission (PRC) and its impact on the real estate industry. Housing Interactive. https://housinginteractive.com.ph/blog/the-professional-regulation-commission-prc-and-its-impact-on-the-real-estate-industry/

Rustemi, A., Dalipi, F., Atanasovski, V., & Risteski, A. (2023). A Systematic Literature Review on Blockchain-Based Systems for Academic Certificate Verification. IEEE Access, 11, 64679–64696. https://doi.org/10.1109/access.2023.3289598

Rustemi, A., Dalipi, F., Atanasovski, V., & Risteski, A. (2023). *A Systematic Literature Review on Blockchain-Based Systems for Academic Certificate Verification*. IEEE Access. https://doi.org/10.1109/access.2023.3289598

Rybkowski, Z. K., & Ballard, G. (2008, July). USING THE "FIVE WHYS" AS A DECISION-MAKING FRAMEWORK FOR EVIDENCE-BASED DESIGN. In Proceedings of the 16th annual conference on Lean Construction (IGLC-16), Manchester, UK (pp. 391-402).

Shen, X., Luo, J., Wang, H., Liu, M., Dustdar, S., & Wang, Z. (2024, December 2). A hybrid BPMN-DMN framework for secure inter-organizational processes and decisions collaboration on permissioned blockchain. arXiv.org. https://arxiv.org/abs/2412.01196#:~:text=The%20framework%20introduces%20three%20key%20innovations%3A%20%281%29%20a,through%20secure%20data%20transfer%20and%20external%20system%20integration.

Shoemaker, P. (2025, May 9). *What are decentralized identifiers (DIDs)?* Identity. https://www.identity.com/what-are-decentralized-identifiers-dids/

Skou, I. (2024, July 14). *How Digital Credentials Address Skill Gaps in the Identification Process*. Diplomasafe. https://diplomasafe.com/digital-transformation-education-student-perception/

Solat, S., Calvez, P., & Naït-Abdesselam, F. (2021). Permissioned vs. Permissionless Blockchain: How and Why There Is Only One Right Choice. Journal of Software, 95–106. https://doi.org/10.17706/jsw.16.3.95-106

Stefaniak, J., & Carey, K. (2019). Instilling purpose and value in the implementation of digital badges in higher education. *International Journal of Educational Technology in Higher Education*, *16*(1). https://doi.org/10.1186/s41239-019-0175-9

Tan, C. (2025, July 21). Two women jailed for submitting fake university certificates to MOM for employment passes. *The Straits Times*. https://www.straitstimes.com/singapore/courts-crime/two-women-jailed-for-submitting-fake-university-certificates-to-mom-for-employment-passes

Team, E. (n.d.-c). *What is Exonum - Exonum Documentation*. https://exonum.com/doc/version/latest/get-started/what-is-exonum/

TESDA eyes nationwide roll-out of e-certificates soon. (n.d.). https://www.tesda.gov.ph/Media/NewsDetail/20160

The Philippine Embassy in Berlin. (n.d.). https://philippine-embassy.de/overseas-employment-certificate-oec/

TruScholar. (2020). *TruScholar: Blockchain-Powered Digital Credential Issuance Platform*. Truscholar.io. https://www.truscholar.io/

TruScholar. (2025). *TruScholar vs DiplomaSafe: Comparing Digital Credentialing Platforms for Your Needs - Truscholar*. Truscholar.io. https://www.truscholar.io/blog/truscholar-vs-diplomasafe

TruScholar. (2025, April 16). *TruScholar's Patent Milestone: Pioneering Blockchain Innovation in India*. Linkedin.com. https://www.linkedin.com/pulse/truscholars-patent-milestone-pioneering-blockchain-innovation-qg1uc/

TruSkillsForce. (2024, November 12). *How Blockchain-Based Digital Credentials Enhance Employee Background Verification - TruSkillforce | Simplifying Work Credential Issuance & Skill Based Hiring*. Truskillsforce.io. https://www.truskillsforce.io/blog/blockchain-based-digital-credentials

UserTesting. (2025, April 18). *Sample size recommendations*. https://help.usertesting.com/hc/en-us/articles/14820712486941-Sample-size-recommendations

Varadarajan, S., Koh, J. H. L., & Daniel, B. K. (2023). A systematic review of the opportunities and challenges of micro-credentials for multiple stakeholders: learners, employers, higher education institutions and government. International Journal of Educational Technology in Higher Education, 20(1), 13. https://doi.org/10.1186/s41239-023-00381-x

Waterfall methodology — project management. (n.d.). https://business.adobe.com/blog/basics/waterfall

Web3Africa. (2023, October 2). *Akowe's Blockchain solution revolutionizing academic verification in Nigeria*. FurtherAfrica. https://furtherafrica.com/2023/10/02/akowes-blockchain-solution-revolutionizing-academic-verification-in-nigeria/

What are Smart Contracts & How Do They Work? | Kaspersky. (2024, October 15). /. https://www.kaspersky.com/resource-center/definitions/what-are-smart-contracts

What is Scrum? (n.d.-b). Scrum.org. https://www.scrum.org/resources/what-scrum-module

What's new, Department of Migrant Workers? | Article | Chambers and partners. (n.d.). https://chambers.com/articles/whats-new-department-of-migrant-workers

Wolniak, R., & Grebski, W. (2023). The usage of 5 Why in Industry 4.0 conditions. Scientific Papers of Silesian University of Technology. Organization and Management Series, 189, 735-745.

Yaga, D., Mell, P., Roby, N., & Scarfone, K. (2019). *Blockchain Technology Overview* (NISTIR 8202). National Institute of Standards and Technology. https://doi.org/10.6028/NIST.IR.8202

Yaga, D., Mell, P., Roby, N., & Scarfone, K. (2019). Blockchain technology overview. arXiv preprint arXiv:1906.11078. https://doi.org/10.6028/NIST.IR.8202

Zhao, H., Yang, Y., & Liu, B. (2020). Blockchain technology for secure and transparent academic credentials verification. Computers in Education, 78, 39-47.

Appendix A: Transcript of Interview with HR Recruiter of Avesco Marketing Corporation (June 28, 2025)

Interviewer: How do you currently verify applicant credentials?

Miss Tan: Through contacting the school that the new fresh graduate student studied at.

Interviewer: How do you deal with applicants who are suspicious?

Miss Tan: you could already checked/ Verify the credential or the truthfulness of the application applied by the applicant through initial screening of the hiring manager either phone call interview of the applicant or asking the applicant to go directly to the company that they applied for in person.

Interviewer: What are the key factors that contribute to your suspicion of an applicants credentials/credibility?

Miss Tan: One of which is having a phone interview talking to the applicant/ candidate for the position asking questions relating to what is written in there CV or resume.

Interviewer: How often do you encounter unverifiable or suspicious academic documents?

Miss Tan: so far none because each applicant that we hire we verify through call from there respective previous company they applied for

Interviewer: Would a QR code/blockchain-verification system be helpful for your recruitment?

Miss Tan: Yes

Interviewer: What features would make you trust such a system that verifies academic credentials?

Miss Tan: The accuracy of the time and date that the document has been verified by the block chain system. But do note that these answers are just purely base on the experience that I've seen in our recruitment process that I've worked at.

Appendix B: Transcript of Interview with HR Recruiter Intern of Sunlife Philippines (June 28, 2025)

Interviewer: : How do you currently verify applicant credentials?

Sir Lagunilla: The accuracy of the time and date that the document has been verified by the block chain system. But do note that these answers are just purely base on the experience that I've seen in our recruitment process that I've worked at. At present, we verify applicant credentials primarily through manual document review. This involves carefully checking submitted materials such as diplomas, transcripts, certificates, and licenses to ensure they appear legitimate and complete. We also follow up by asking the applicant specific questions about their academic and professional background during interviews. This helps us assess whether their answers align with the information provided in their documents. While this process is time-consuming, it allows us to evaluate both the content and the confidence with which they discuss their experiences.

Interviewer: How do you deal with applicants who are suspicious?

Sir Lagunilla: When an applicant seems suspicious, whether through inconsistent statements or questionable documents, we take extra steps to validate their claims. This might include contacting the issuing institutions such as universities or licensing boards, cross-checking credentials against available databases, or conducting online research. We also review the applicant’s interview responses closely to identify any contradictions. While we try to maintain fairness, we remain cautious and prioritize the integrity of the recruitment process.

Interviewer: What are the key factors that contribute to your suspicion of an applicants credentials/credibility?

Sir Lagunilla: Several indicators can raise red flags during the recruitment process. The most common are inconsistencies between what is written in the resume and what is shared during the interview. We also look for irregularities in the timeline of education or work history that seem unrealistic. Formatting issues in documents, such as strange fonts, outdated logos, or grammar mistakes in certificates, are also warning signs. Additionally, when an applicant provides vague or evasive answers about their qualifications or previous roles, it makes us more cautious.

Interviewer: How often do you encounter unverifiable or suspicious academic documents?

Sir Lagunilla: So far, during my time as an HR intern, I have not encountered any cases involving suspicious or unverifiable academic credentials. While the media often reports stories about fake diplomas and fraudulent documents, our experience has been different. These situations are not common in our workplace, but we still remain alert to the possibility.

Interviewer: In events where an employee was verified to have fake credentials, what were the consequences? (optional)

Sir Lagunilla: I have not personally experienced such a case. However, based on company policy and general practice, submitting fake credentials would be treated very seriously. Consequences could range from immediate disqualification during recruitment to termination if the person is already employed. In more serious cases, legal action could also be considered, especially if the credentials were essential to the employee's job responsibilities.

Interviewer: Would a QR code/blockchain-verification system be helpful for your recruitment?

Sir Lagunilla: Yes, I believe that a QR code or blockchain-based verification system would be a great help in recruitment. It could speed up the verification process, reduce the risk of human error, and improve the accuracy of the information we receive. Having a reliable and secure system for confirming academic and professional credentials would give HR staff greater confidence in their hiring decisions.

Interviewer: What features would make you trust such a system that verifies academic credentials?

Sir Lagunilla: To build trust in a digital verification system, it should be directly linked to official academic institutions and government-recognized bodies. Real-time verification, data integrity through a secure digital ledger, and a clear record of each credential's origin would be important. It would also be essential for the system to comply with data privacy regulations, protecting sensitive applicant information. A user-friendly interface and transparency in how the verification works would further support adoption and confidence in the tool.

Appendix C: Transcript of Interview #2 with HR Recruiter of Avesco Marketing Corporation (July 21, 2025)

Interviewer: What are the exact steps in interview, and what documents are required for each step?

Miss Tan: we check applicants skills and credentials, and appoint an interview with the applicant we find fit for the role via text or email, stating the address, date of interview, and the documents to be brought, which are copies of Diploma, TOR, and any valid id. After interview, a technical test is conducted, to further assess the skills of applicant. We announce the results either the day itself, or the day after. In the event that applicant is hired, we request the medical certifcate and NBI clearance.

Interviewer: Are there any renewable documents that are sought for by the recruiter?

Miss Tan: medical certificates which are renewed every 6 months and NBI clearance every year. These are strictly verified for hiring, and even when applicant is an employee already.

Appendix D: Survey Response From HR Recruiters Regarding Credential Verification Practices

The following responses were collected through Google Forms with purpose of gaining insight from HR recruiters and interns as part of the requirements analysis phase of the ChainCert system. These responses helped shape the Business Process Model and supported the identification of inefficiencies in traditional academic credential verification workflows.

Appendix D.1: Survey Response from Lagunilla, HR Recruiter Intern of Sunlife Philippines (July 21, 2025)

Survey Question: How do you currently verify academic credentials from job applicants?

Sir Lagunilla: For financial advisors, we request applicants to submit scans or photocopies of diplomas, TORs (Transcript of Records), or proof of ongoing studies. These are then manually cross-checked with the school or verified through alumni networks, HR references, or via request letters to the school registrar if necessary. For internship applicants, HR interns usually verify based on the submitted school endorsement letters, student IDs, and official email addresses. There's usually no direct verification with the school unless something seems inconsistent.

Survey Question: What are the main challenges you face in the credential verification process (e.g., time delays, manual errors, lack of standardization)?

Sir Lagunilla: One of the main challenges we face in the verification process is the lack of standardization in how schools issue academic documents. Different institutions have different formats, and not all have adopted digital systems, making manual verification time-consuming. Time delays are also a recurring issue, especially when waiting for school responses during peak periods or holidays. Additionally, the manual nature of the process increases the risk of clerical errors or missed inconsistencies. For HR interns, the limited access and authority to conduct direct verifications add another layer of difficulty in ensuring document authenticity.

Survey Question: How often do you encounter issues with fraudulent credentials? How do you handle these situations?

Sir Lagunilla: The time it takes to verify credentials also varies. For financial advisor applicants, it usually takes about five to ten working days, depending on how responsive they are, whether the in Metro Manila or a provincial area, and whether all submitted documents are complete and legible. For internship applicants, the process is much faster typically one to three days because most universities provide a standard endorsement letter and require students to complete documentation before applying. However, delays can still happen if documents are incomplete or inconsistent.

Survey Question: What problems do you face due to delays in verifying credentials (e.g., hiring delays, lost opportunities)?

Sir Lagunilla: Delays in verifying academic credentials often result in significant setbacks, particularly when it comes to hiring financial advisors. Since many candidates are eager to start, prolonged verification may cause them to accept other job offers, leading to lost opportunities for the company. This affects our recruitment targets and disrupts training schedules, especially when we're hiring in batches. For HR interns, the delays are less critical but can still hinder the onboarding process, especially when school documents like endorsement letters are incomplete or submitted late.

Survey Question: How does the lack of standardized credential verification systems impact your hiring decisions?

Sir Lagunilla: The lack of a standardized credential verification system across Philippine educational institutions further complicates the process. Each school has its own procedures, formats, and timelines, making it difficult to implement a uniform or automated system. As a result, hiring decisions are sometimes delayed or based on incomplete information, which can affect the fairness and consistency of our evaluations.

Survey Question: What data do you collect from candidates to verify their credentials (e.g., certificates, transcripts, personal information)?

Sir Lagunilla: We collect various documents from candidates to verify their academic credentials. These include diplomas, transcripts of records (TOR), certificates of enrollment or graduation, and in some cases, school endorsement letters for interns. We also collect personal information such as full name, date of birth, school name, course, and expected graduation date to cross-check with the submitted documents.

Survey Question: How do you currently store and manage this verification data? Are there any issues with data security or integrity?

Sir Lagunilla: Currently, we store and manage verification data through a combination of physical files and cloud-based storage, depending on the office handling the application. While access is restricted to authorized HR personnel, there are occasional

concerns regarding data organization, incomplete files, and the long-term integrity of physical documents. We also remain cautious about digital data security, especially with the risk of unauthorized access or accidental deletion in shared folders.

Survey Question: Are there any legal consequences for hiring individuals with fraudulent credentials in your industry?

Sir Lagunilla: Yes, there are legal consequences for hiring individuals with fraudulent credentials, especially in the financial services industry. Hiring someone under false qualifications may result in regulatory penalties, reputational damage, and in severe cases, legal action against both the individual and the company if negligence in due diligence is proven. This is particularly important when dealing with roles that involve financial advising, where trust and professional credibility are essential.

Survey Question: How much does the credential verification process currently cost your organization? Would blockchain technology help reduce these costs?

Sir Lagunilla: The current credential verification process does not have a fixed cost, but it does consume significant resources in terms of HR staff time, follow-up communications with schools, and administrative work. For financial advisor applicants, delays often lead to opportunity costs due to onboarding disruptions. In some cases, we also incur minor expenses for courier services or notarized document requests. Blockchain technology could help reduce these costs by eliminating manual follow-ups and providing instant verification, saving both time and administrative overhead.

Appendix D.2: Survey Response from Tan, HR Recruiter of Avesco Marketing Corporation (July 21, 2025)

Survey Question: How do you currently verify academic credentials from job applicants?

Miss Tan: Through there online resume or CV

Survey Question: What are the main challenges you face in the credential verification process (e.g., time delays, manual errors, lack of standardization)?

Miss Tan: None

Survey Question: How often do you encounter issues with fraudulent credentials? How do you handle these situations?

Miss Tan: 1-2 weeks

Survey Question: What problems do you face due to delays in verifying credentials (e.g., hiring delays, lost opportunities)?

Miss Tan: Outsource

Survey Question: How does the lack of standardized credential verification systems impact your hiring decisions?

Miss Tan: Delays for deployment for vacant positions that is needed in the org.

Survey Question: What data do you collect from candidates to verify their credentials (e.g., certificates, transcripts, personal information)?

Miss Tan:

Appendix E: Survey Response from the Human Resources Development and Management Department Department Head 1 of the City Government of Bacoor, Cavite (July 12, 2025)

Question: How do you currently verify applicant credentials?

Answer: We verify applications credentials by:

- Requiring applicants to submit a notarized CSC Personal Description sheet/pds (Revised 2017 form). The form has a subscribed and sworn part which is considered a public document. If the contents are found to be erroneous or falsified or perjured, applicant is liable under the law;
- Human Resource Management Practitioners must do background checks, calling previous employers, schools etc. PRC licenses or CSC eligibilities are also checked through PRC and CSC websites dedicated to checking the veracity of the document submitted,

- Transcript of records and diploma must be certified true copy as per requirements of the Omnibus Rules on Appointments and Other Human Resource Actions (ORA-OHRA);
- All movements and actions of the HRMO on handling appointments (Recruitment, Selection and Placement are all provided for under the said law);
- We administer and interpret psychological tests, intelligence tests, etc. If client is found to be unfit, his application does not proceed to the next step since he is unfit; if applicant is applying for a position that involves security, peace and order related jobs, we require them to submit neuro-psych exams administered and interpreted by an accredited clinic.

Question: How do you deal with suspicious applicants?
Answer: Our recruitment and selection process is stringent and rarely do we get or experience dealing with a suspicious client reaching the point of dealing with them face to face because the initial screening and submission of the pds is done over the window transaction or via e-mail, hence weeding out suspicious or unwanted applicants. Further as per a recent memo, offices are strictly prohibited from allowing the non-employees from transactions inside the office premises. All are strictly required to do window transactions. Offices have cctv and guards on duty to monitor and provide security.

Question: What are the key factors that contribute to your suspicion of an applicant's credentials/credibility?
Answer: Inconsistencies in the PDS submitted- mismatched employment details, claims of positions that don't align with the education, work experience and training:

- Unverifiable educational background- unable to provide documents supporting it like TORs or diplomas or schools/colleges or universities that are not accredited;
- Employment history red flags- especially for frequent changes of job for less than a year and long unexplained gaps in work history;,
- Reference check irregularities;,
- Inconsistencies during the actual interview vis a vis with the documents submitted
- Too good to be true
- Additional answer to what are the key factors that contribute...? If applicants seems to have no digital footprint

Question: How often do you encounter unverifiable or suspicious academic documents?

Answer: My experience as head of the HRDMD, we seldom experience it. See my answers in the previous questions. Submission of falsified or perjured documents would mean a criminal and administrative liability. It would also carry with it a perpetual disqualification in government service. A heavy penalty indeed. Hence, it serves a deterrent.

Question: Would a blockchain-based verification system be useful in your recruitment process?
Answer: In the government sector seemingly yes, because for high integrity, large scale hiring as it reduces credential fraud, builds trust and enhances recruitment efficiency. However, with our current set up, no as this would entail large public funds to set up linking to the HRIS (cost and infra), needs training of staff which would also mean additional funding. Consider also the Data Privacy Act - the design should definitely comply with the requirements set forth by the law. Call me conservative but I would not risk the data of my employees and be subjected to administrative and criminal liabilities that entails the violation of the provisions of the said Act.
Maybe more appropriate for CHED, TESDA and PRC to issue blockchained verified certificates and licenses first before other smaller offices of the LGU can follow suit.

Question: What features would make you trust a blockchain system designed to verify academic credentials?
Answer:
Once uploaded data cannot be altered;,

- Only those verified institutions like CHED etc should have permission to issue credentials
- there is transparency in the audit trail,
- Must be user friendly and;
- Must comply with the Data Privacy Act of 2012. As this is really a big issue

Appendix F: Survey Response Summary – Measuring the Feasibility and Impact of Blockchain-Based Credential Verification Systems

1. Company

The survey was answered by individuals from a diverse set of organizations, indicating broad interest in credential verification systems across sectors. Each of the seven resp ondents represented a different entity, including Avesco Marketing Corp, Sun Life Financial, Interpublic Group, Notified, Toyota, St.

Thomas Aquinas Academy of Bacoor Cavite, Inc., and a private school in Bacoor. This spread highlights perspectives from both private corporations and educational institutions.

2. What is your current occupation or position?

Three respondents currently hold roles as HR professionals, recruiters, or employers—signifying their direct involvement in applicant screening and document verification. The remaining respondents hold various roles across departments, such as a Web Development Manager, Sales staff, LIS (Learner Information System) Coordinator, and School Principal. These varied roles provide a cross-functional view of how academic credentials are handled across organizations.

3. Type of organization

A majority of the respondents (five out of seven) come from corporate or employer organizations, while the remaining two work in academic institutions. This mix reflects how both credential issuers (schools) and credential consumers (companies) view and handle the verification process.

4. How do you currently issue and manage academic credentials?

From the educational side, credential issuance practices vary slightly. One institution issues credentials during graduation, with transcript records processed through SF10 forms during school-to-school transfers. Another noted that their credentials are issued only upon student request. These practices reflect typical procedures in Philippine academic institutions, where document handling is largely administrative and manually triggered.

5. Verification process when requested by employers or third parties

Verification is conducted either by reviewing internal academic records or by issuing official certifications. This manual approach, while functional, is resource-dependent and susceptible to institutional delays, especially when employers require quick validation of an applicant's credentials.

6. Duration to verify a credential

Credential verification durations varied among respondents. One noted it typically takes seven working days to complete verification, while another estimated a faster turnaround of two to three days. This range underscores inconsistencies in processing times, which can delay hiring or admissions processes.

7. Managing updates to credentials

Credential updates are handled either by storing soft copies for internal access or by requiring an affidavit

of loss before reissuance. This reveals how institutions balance convenience and security, although both methods are limited by manual processing and document recovery challenges.

8. Challenges faced in current verification process

Respondents cited multiple obstacles in the existing system. Manual processing and human error were major concerns, as was the accuracy of credential data. These factors can compromise the efficiency and trustworthiness of verification, especially under time-sensitive or high-volume conditions.

9. Issues with fraudulent credentials

Instances of credential fraud were reported, including students with unpaid obligations continuing their education and the submission of falsified documents like birth certificates. Such cases reveal vulnerabilities in existing systems that are not robust enough to consistently detect or prevent fraud.

10. Tracking issued credentials

Tracking mechanisms for issued credentials were inconsistent. One respondent admitted there is no way to track credentials once they are transferred to another institution, though internal copies are retained. Another claimed they rely on a proper filing system for record-keeping. These methods highlight the lack of centralized or real-time tracking of issued documents.

11. Data collected when issuing credentials

The credential issuance process requires different supporting documents depending on the institution. One school collects SF9 Report Cards and SF10 Form 137, while another also requires request forms, school IDs, and birth certificates. These varying document sets suggest the absence of a standardized data collection framework across institutions.

12. Storage and protection of student data

Data storage practices also differ in terms of reliability and risk exposure. One respondent acknowledged that hardcopy records are vulnerable to natural disasters like floods, whereas soft copies risk digital corruption. This indicates a need for more secure, disaster-resilient, and tamper-proof storage systems—possibly through modern technologies like blockchain.

Appendix G: Survey Response from Roland Talagtag from Reliance Producers Cooperative (July 12, 2025)

Question: How do you currently verify applicant credentials?

Answer: We ask applicants to submit the original copies of documents with expiration dates, such as police clearance, medical results, etc. For documents like birth certificates, diplomas, and transcripts of records, they are required to present the original copies for validation. As for government-issued IDs related to mandatory deductions such as SSS, Pag-IBIG, PhilHealth, and TIN, applicants must present the actual cards to verify that the name and number on the card match our records.

Question: How do you deal with suspicious applicants?
Answer: If the applicant is not applying for a skilled, technical, or hard-to-fill position, we reject the application and look for another qualified candidate. For critical positions, we contact the previous or most recent employer to inquire about the applicant's performance records.

Question: What are the key factors that contribute to your suspicion of an applicant's credentials/credibility?
Answer: If there is a gap in the applicant's employment history that they cannot clearly explain, or if the applicants suddenly resigned from a recent job where they had worked for years, especially if the company is known to be a good employer, these may raise concerns during the evaluation process.

Question: How often do you encounter unverifiable or suspicious academic documents?
Answer: Not very often, let's say around 1 to 2 times a month and there are month/s that we don't have. This usually involves not declaring previous employer, birth certificates, or school records.

Question: Would a blockchain-based verification system be useful in your recruitment process?
Answer: Yes, definitely. The verification process will be faster and will provide a secure and transparent method for verifying the required information.

Question: What features would make you trust a blockchain system designed to verify academic credentials?
Answer: Something like a document with a QR code that, when scanned, links to the issuer and allows us to verify the authenticity of the submitted document.

Question: In events when you an employee was verified to have fake credentials, what were the consequences?

Answer: If the applicant has not yet been onboarded, we will no longer proceed with the hiring. If dishonesty is discovered after the applicant has already been employed, disciplinary action will be taken. This may include penalties ranging from suspension to termination of employment, depending on the severity and circumstances of the violation.